

8.1 Magnetism

Magnet:

The material which has directional, attractive and repulsive properties and the material by which magnetic properties can be given to other materials are called magnets. Its chemical formula is Fe_3O_4 .

Magnetism:

The directional and attractive property of a magnet is called its magnetism.

Magnetic field:

The region around a magnet up to which it can show its magnetic properties is called magnetic field.

Curie point or temperature a magnet:

A particular temperature at which the magnetism of a magnet is destroyed completely is called the Curie point.

Magnetic permeability of a medium:

The characteristic of a material for which magnetic field lines can pass more easily through a material compared to another material is called magnetic permeability.

In other word, the ratio of the magnetic flux density created in a medium to the magnetic intensity is called absolute permeability of that medium. It is denoted by μ and its unit is Henry/meter (H/m).

$$\text{i.e. Magnetic permeability} = \frac{\text{Magnetic flux density}}{\text{Magnetic intensity}}; \therefore \mu = \frac{B}{H}$$

Magnetic induction:

The process by which a magnetic material becomes a magnet is called magnetic induction.

Magnetic materials and Types of them:

The materials which can be attracted and which can be transformed into magnets are called magnetic materials. For example, Iron, nickel, cobalt etc.

There are three types of magnetic materials those are given below,

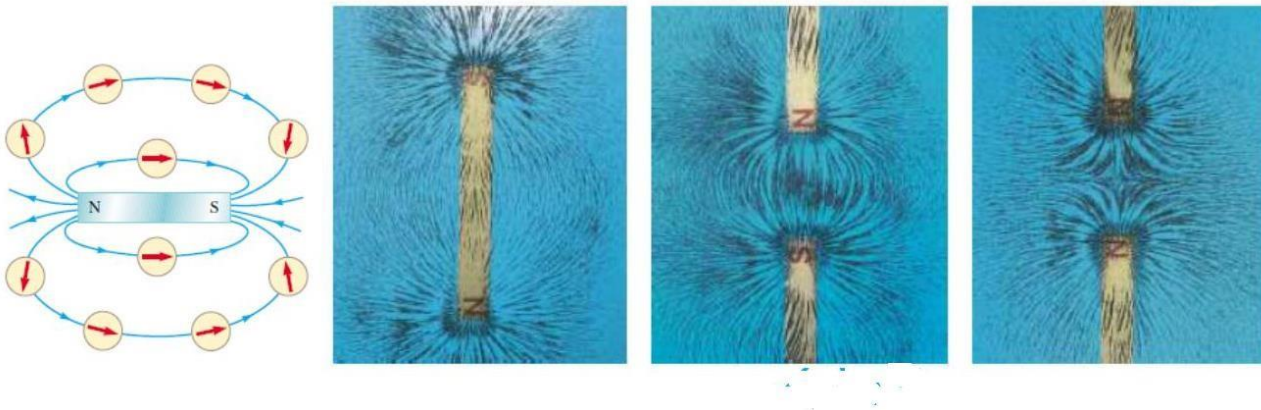
- (i) Diamagnetic substance ($\mu < 1$); Gold, Copper etc.
- (ii) Paramagnetic substance ($\mu > 1$); Platinum, Manganese etc.
- (iii) Ferromagnetic substance ($\mu \gg 1$); Iron, Cobalt etc.

MAGNETIC FIELD EFFECTS

8.1.2 Magnetic Field

Magnetic field is the region or space around any charge within which its influence can be felt by other magnetic substances.

The magnetic field around any magnet is considered as closely spaced magnetic field lines. The magnetic field lines of a bar magnet can be traced with the aid of a compass as shown in the figure below:



In addition to a bar magnet, a moving charge or a current creates a magnetic field in the surrounding space (in addition to its electric field). The magnetic field exerts a force on any other moving charge or current that is present in the field.

8.2 Magnetic Force on a Charged Particle

Consider a charged object having charge q is projected a uniform magnetic field of flux density with velocity. The magnetic force acting on the object can be expressed as:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

$$F = qvB \sin \theta \quad \hat{n} \text{ --- (1)}$$

Here $\hat{n}$ is the unit vector that is perpendicular to the plane of $\mathbf{v}$ and $\mathbf{B}$, and is used to describe the direction of magnetic force on the charged object.

It is clear from equation (1) that the maximum magnetic force will act on the charged object when it will be projected perpendicular to the magnetic field. The maximum magnetic force on the charged object will be

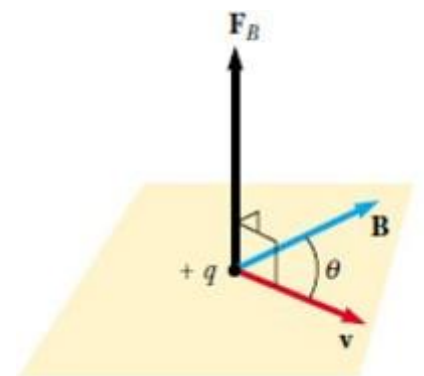
$$F = qvB$$

$$B = \frac{F}{qv}$$

$\mathbf{B}$ is also called the magnetic field induction which can be defined as:

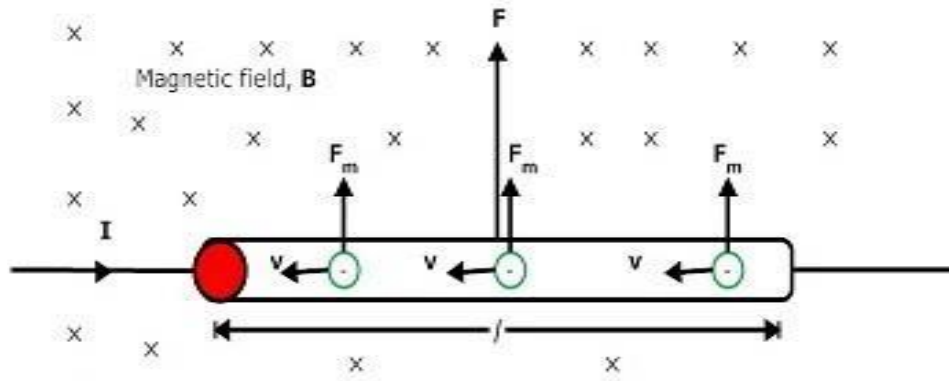
“The force on a unit positive charge moving perpendicular to the magnetic field with uniform velocity”. The SI unit of magnetic field induction $\mathbf{B}$ is tesla, while in cgs system of units $\mathbf{B}$ is measured in gauss.

$$1 \text{ tesla} = 10^4 \text{ gauss}$$



8.3 Magnetic Force on a Current Carrying Conductor

Consider a current carrying wire of length L and cross-sectional area A , is placed in a uniform magnetic field of flux density $\mathbf{B}$ as shown in the figure below:



If n is the number of free charges per unit volume of a conductor (each having charge e), then the total charge flowing through the wire is

$$q = nAle$$

Suppose the charges are moving with drift velocity v_d and cover length l in t seconds, then

$$t = \frac{l}{v_d}$$

If I is the current flowing through the conductor, then

$$I = \frac{q}{t}$$

$$\Rightarrow I = \frac{nAle}{\frac{l}{v_d}}$$

$$\therefore v_d = \frac{I}{nAe}$$

Now

$$\text{Force on one charge } F' = ev_d B_{\perp}$$

$$\text{Force on } nAl \text{ charges, } F = nAlF' = nAle v_d B_{\perp}$$

Putting value of v_d we get,

$$F = nAle \left(\frac{I}{nAe} \right) B_{\perp} = IB_{\perp} l$$

In vector form:

$$\vec{F} = I(\vec{l} \times \vec{B})$$

The direction of F will be perpendicular to the plane of $\vec{l}$ and $\vec{B}$.

8.3.1 Torque on a Current Loop in Magnetic Field

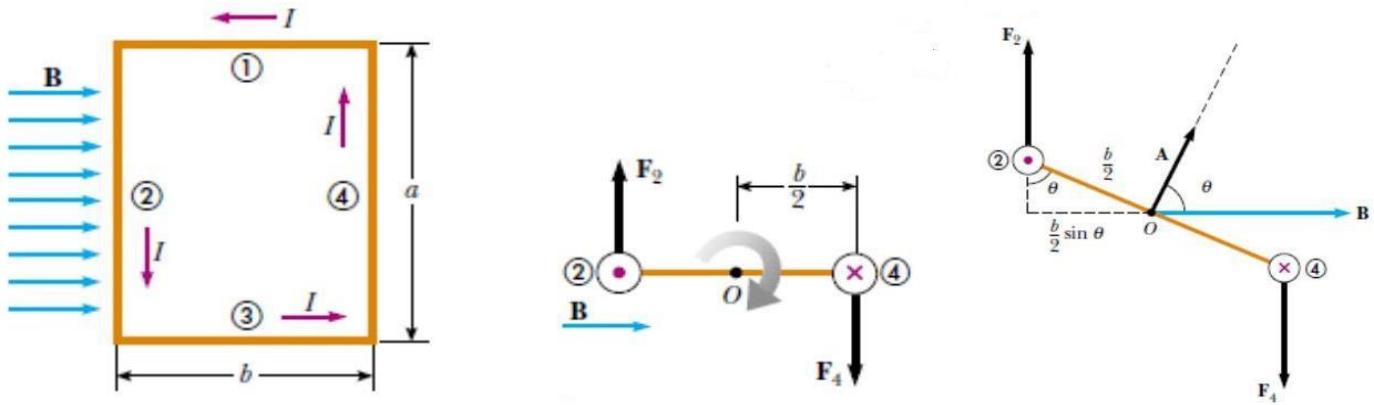
Consider a loop of wire carrying current suspended in a uniform magnetic field of flux density $\mathbf{B}$ as shown in the figure below:

The sides 1 and 3 of the coils are oriented parallel to the magnetic field. Thus,

$$F_1 = IbB \sin 180^\circ = 0 \text{ and}$$

$$F_3 = IbB \sin 0^\circ = 0$$

The sides 2 and 4 are held perpendicular to the magnetic field. Thus,



$$F_2 = F_4 = F = IlB\sin\theta = IaB\sin 90^\circ = IaB \quad \because l = a$$

The direction of F_2 , is out of the page and that of F_4 , is into the page. These two anti-parallel forces separated by a small distance form a couple. The couple tends to rotate the loop about point O. The magnitude of torque of this couple will be:

$$\text{Torque} = (\text{Force})(\text{Perpendicular distance bet}^n \text{ line of action of Force})$$

$$\tau = Fb\sin\theta = IaB b\sin\theta = IabB\sin\theta$$

Where θ is the angle between the magnetic field $\mathbf{B}$ and vector area $\mathbf{A}$ of the loop.

$$\tau = IAB\sin\theta \because A = ab = \text{area of loop}$$

This is the expression of torque produced in a loop with single turn. For a loop with N turns, the torque will be:

$$\tau = NIAB\sin\theta$$

In vector form:

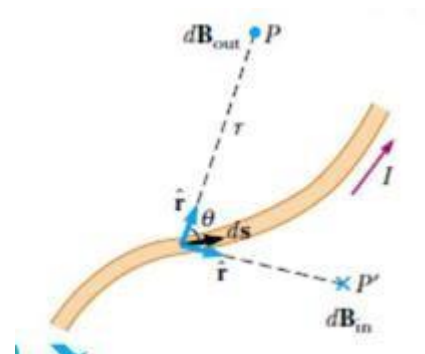
$$\boldsymbol{\tau} = NI(\mathbf{A} \times \mathbf{B})$$

This is the expression of the torque on a current carrying loop in a magnetic field.

8.4 Biot-Savart's law:

Statement: When current flows through a conductor, magnetic field is generated around it. According to this law, the magnitude of magnetic field produced around a conductor of small length due to the flow of current through it is –

- Directly proportional to the amount of **current**,
- Directly proportional to the **length** of the conductor,
- Directly proportional to the **sine of the angle** between the lines connecting the point from the mid-point of the conductor and the conductor itself,
- Inversely proportional to the **square of the distance** between the point and the mid-point of the conductor.



Explanation:

According to this law, the magnetic flux density dB at a point P due to a small element dl of a conductor carrying a current I is given by,

$$dB \propto \frac{I dl \sin\theta}{r^2} \dots \dots \dots (1)$$

Where, r is the distance from the point P to the mid-point of the element dl and θ is the angle between the element and the line joining it to P as shown in Fig-3.

From equation (1) we can write,

$$dB = K \frac{I dl \sin\theta}{r^2} \dots \dots \dots (2)$$

Where, K is a proportionality constant whose value depends on nature of the medium and the unit of the quantities.

For air medium and in S.I. unit

$$K = \frac{\mu_0}{4\pi}$$

Where, $\mu_0 = 4\pi \times 10^{-7} \text{H/m}$.

So, putting this value in (2) we get,

$$dB = \frac{\mu_0 I \cdot dl \cdot \sin\theta}{4\pi r^2}$$

For other medium we get,

$$dB = \frac{\mu}{4\pi} \frac{I \cdot dl \cdot \sin\theta}{r^2}$$

where, μ is the permeability of that medium.

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

9.1 Magnetic Flux

The number of magnetic lines of force passing normally through certain area is called magnetic flux. It is denoted by ϕ_B . It is a scalar quantity and its SI unit is weber (Wb). It is measured by the product of magnetic field strength and the component of vector area parallel to magnetic field.

If $d\mathbf{A}$ is the vector area element of the surface placed in uniform magnetic field of magnetic field strength B as shown figure below.

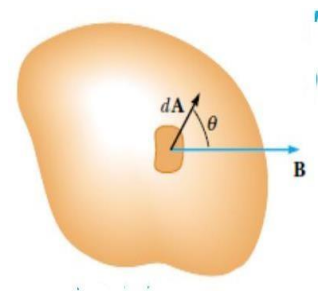
The magnetic flux $d\phi_B$ through $d\mathbf{A}$ is given by:

$$d\phi_B = \mathbf{B} \cdot d\mathbf{A}$$

$$\phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

$$\phi_B = \int B dA \cos\theta$$

Where θ is the angle between magnetic field strength and vector the area element.



9.2 Faraday's Law of Electromagnetic Induction

Statement:

The induced emf in a circuit is equal to the negative of rate at which the magnetic flux through the circuit is changing with time.

Or

The magnitude of induced emf in a circuit is directly proportional to the rate of change of magnetic flux.

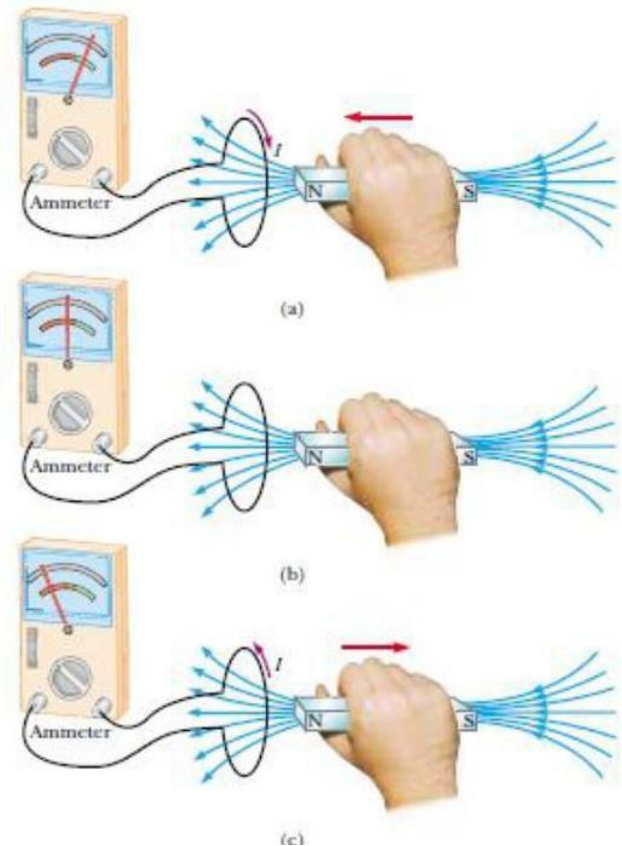
*Explanation

When a magnet is moved toward the loop, the ammeter needle deflects in one direction, as shown in the figure (a). When the magnet is brought to rest and held stationary relative to the loop figure (b), no deflection is observed. When the magnet is moved away from the loop, the needle deflects in the opposite direction, as shown in figure

(c). Finally, if the magnet is held stationary and the loop is moved either toward or away from it, the needle deflects.

From these observations, we conclude that the loop detects that the magnet is moving relative to it and we relate this detection to a change in magnetic field. Thus, it seems that a relationship exists between current and changing magnetic field.

These results are quite remarkable in view of the fact that a current is set up even though no batteries are present in the circuit. We call such a current the induced current which is produced by an induced emf . This phenomenon is called electromagnetic induction.



Mathematical form of Faraday's Law

If $\mathcal{E}$ is the induced emf due to change in flux $d\phi_B$ in time dt , then we can describe the Faraday's law of electromagnetic induction as:

$$\mathcal{E} = - \frac{d\phi_B}{dt}$$

If there are N number of turns in a coil, then,

$$\mathcal{E} = - \frac{d(N\phi_B)}{dt}$$

$$\mathcal{E} = -N \frac{d\phi_B}{dt}$$

9.3 Lenz law

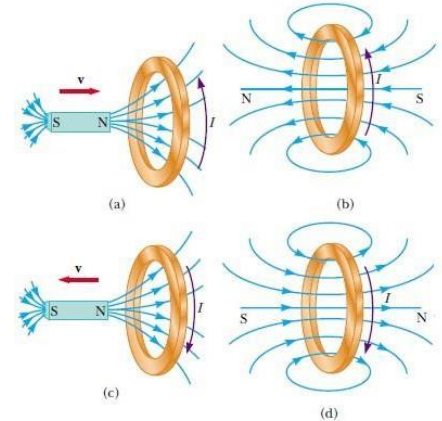
The direction of induced current is such that it opposes its own cause.

Explanation:

Consider a bar magnet moves toward a stationary metal loop, as in figure (a). As the magnet moves to the right toward the loop, the external magnetic flux through the loop increases with time. As the result, the induced current set up in the loop which produces magnetic field, as illustrated in figure (b).

Knowing that like magnetic poles repel each other, we conclude that the left face of the current loop acts like a north pole and that the right face acts like a south pole.

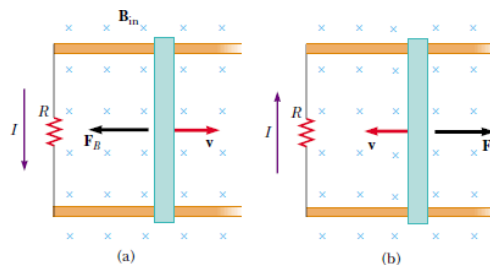
If the magnet moves to the left, as in figure (c), its flux through the area enclosed by the loop decreases in time. Now the induced current in the loop produces the magnetic field as shown in figure (d). In this case, the left face of the loop is a south pole and the right face is a north pole.



9.3.1 Lenz Law and Conservation of Energy

*“Lenz law is the statement of law of conservation of energy
for the circuit involving induced current”*

To understand this statement, consider a conducting bar moving to the right on two parallel rails in the presence of a uniform magnetic field as shown in the figure below. As the bar moves to the right, the magnetic flux through the area enclosed by the circuit increases with time because the area increases. As the result, the induced current must be directed counterclockwise when the bar moves to the right. Since the current carrying bar is moving in the magnetic field, it will experience a magnetic force. By using right hand rule, the direction of is opposite to that of, that tends to stop the rod. An external dragging force must be applied to keep the rod moving in the magnetic field.

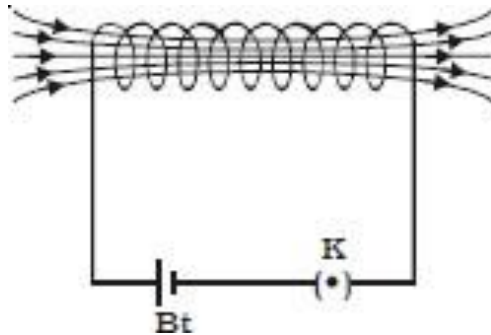


This dragging force provides the energy for the induced currents to flow. This energy is the source of induced current. Thus, the electromagnetic induction is exactly according to the law of conservation of energy.

If the bar is moving to the left, as in figure (b), the external magnetic flux through the area enclosed by the loop decreases with time. Because the field is directed into the page, the direction of the induced current must be clockwise.

9.4 Self-Induction

The property of a coil which enables to produce an opposing induced *emf* in it when the current in the coil changes is called self-induction.



Coefficient of self induction

When a current I flows through a coil, the magnetic flux (ϕ) linked with the coil is proportional to the current.

$$\phi \propto I$$

$$\phi = LI$$

Where L is a constant of proportionality and is called coefficient of self induction or self inductance.

From Faraday's law of induction.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

So,

$$\mathcal{E} = -\frac{d(LI)}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

If there are N no. of turns in the coil then,

$$\mathcal{E} = -N \frac{d\phi}{dt}$$

The unit of self inductance is henry.

9.5 Mutual induction

The phenomenon of producing an induced *emf* in a coil due to the change in current in the other coil is known as mutual induction.

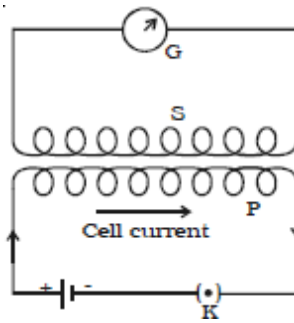


Figure: Mutual Induction

Coefficient of mutual induction

When a current I flows through a coil, the magnetic flux (ϕ) linked with the other coil is proportional to the current.

$$\phi \propto I$$

$$\phi = MI$$

Where M is a constant of proportionality and is called coefficient of mutual induction or mutual inductance.

From Faraday's law of induction.

$$\mathcal{E} = -\frac{d\phi}{dt}$$

So,

$$\mathcal{E} = -\frac{d(MI)}{dt}$$

$$\mathcal{E} = M \frac{dI}{dt}$$

If there are N no. of turns in the coil then,

$$\mathcal{E} = -N \frac{d\phi}{dt}$$

The unit of mutual inductance is henry.

Magnetic Properties of Matter

Factors Determining Magnetic Properties

The most basic factor is the configuration of the electrons in the material's atoms. The ability of the atoms or molecules in the material to align magnetically is important in determining whether the material responds to a magnetic field. The orientation of the spin of the electrons in an atom, the orientation of the atoms in a molecule or alloy, and the ability of domains of atoms or molecules to line up are the factors that determine how a material responds to a magnetic field.

Electron orientation

Electrons can behave as tiny magnets, each with north (N) and south (S) poles. When an atom's electrons are lined up in the same orientation, with most having their N pole facing one direction, the atom becomes like a magnet, with N and S poles. It is also possible for the electrons to be in various directions, making the atom not magnetic.

Moving electrons create magnetic field

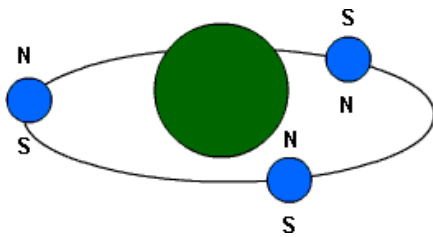
The reason that electrons can behave like tiny magnets is the fact that when electrons move, they create a magnetic field. Placing a compass near a wire carrying DC electrical current can show that a magnetic field is created due to the electrons moving through the wire.

A magnetic field is also created when electrons rotate around a nucleus and when they spin while in orbit.

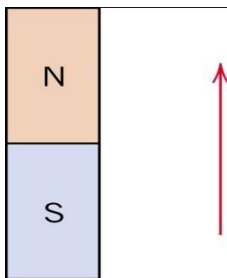
Spinning electrons

Electrons have a property called *spin*. This spinning creates a magnetic field with N and S poles; just as the spinning Earth has magnetic poles. Note that the N pole on an electron is really a North-seeking pole, just as in a magnet.

If electrons in the shells of an atom spin in the same direction, the atom will exhibit a magnetic field and will respond to the forces of a magnet. If half of the electrons spin one way and the rest spin the other way, they will neutralize each other and the material will not be affected by a magnetic field. This atom is barely magnetic because all its electrons are not aligned.



The magnetic moment represented by a vector



Strong and weak electron alignments

Atoms such as iron have most of their electrons aligned in the same direction. Thus, iron or nickel would be attracted to a magnet. Aluminum only has a few electrons aligned, and thus it is only weakly magnetic. An element with half of its electrons oriented one way would not be attracted to a magnet.

Atomic and molecular alignment

Although some atoms may be highly magnetic, they really need to be aligned to make a material magnetic. If magnetic atoms are facing different directions, their fields will cancel out each other.

Solids and fluids

Since the atoms or molecules in a solid are fixed in place, most magnetic materials are solids. This is because once the atoms or molecules become aligned, they tend to stay in place. An example is seen when you magnetize a piece of iron.

As a material becomes heated or when it is in its liquid or gaseous state, the atoms or molecules are in rapid motion and are not aligned. Thus, fluids are seldom magnetic.

An exception is when a magnetic material such as iron is in its liquid state and is continually rotating around an axis. In such a situation, the atoms can be aligned in one direction, even though they are in rapid motion.

For example, the core of the Earth is made of liquid iron. Since the Earth rotates on its axis, the liquid iron is rotating, thus creating the Earth's magnetic field. Also, the Sun rotates on its axis, and the material in its plasma state creates the Sun's magnetic field.

Molecules

If two or more elements are chemically combined to form a molecule, it is quite possible that the compound is not very magnetic because the orientations of the atoms in the molecule work against each other.

A good example of this is to compare the magnetic properties of iron as compared to its compounds if iron oxide (rust) and iron sulfide. A piece of iron is highly magnetic, but a hunk of rusty iron is not.

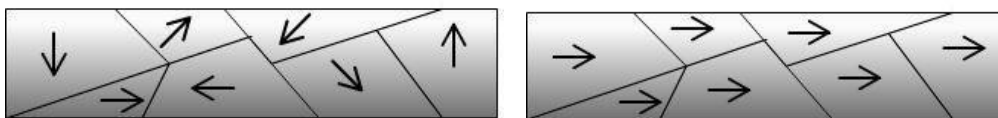
Alloys

Metals of different elements can be mixed when they are in the molten or liquid state to form alloys. These combinations result in materials with slightly different physical and chemical properties than the elements by themselves.

If the metals typically respond well to a magnetic field--such as iron and nickel--then their alloy has even a stronger reaction to magnetism. On the other hand, there are some alloys of iron--such as forms of stainless steel--that do not respond well at all to a magnet.

Domains

The final factor is the orientation of domain in a solid. A group of atoms in a metal may become aligned, but the various groups may be misaligned. These groups are called domains. It is necessary to line up many of the domains in a material like iron in order for it to become a magnet.



Magnetic material with domains misaligned, aligned domains make material highly magnetic

Summary

Alignment of electrons, atoms and domains are important in determining the magnetic response of a material and whether it is a magnet. Since the atoms or molecules need to be aligned, gases and liquids are typically not magnetic, and most magnets are solid metals. An exception is in the rotating liquid iron core of the Earth and the rotating plasmas of the Sun.

Magnetic susceptibility

Magnetic susceptibility is a dimensionless proportionality constant that indicates the degree of magnetization of a material in response to an applied **magnetic field**.

Magnetic permeability

In electromagnetism, permeability is the measure of the ability of a material to support the formation of a magnetic field within itself. Hence, it is the degree of magnetization that a material obtains in response to an applied magnetic field.

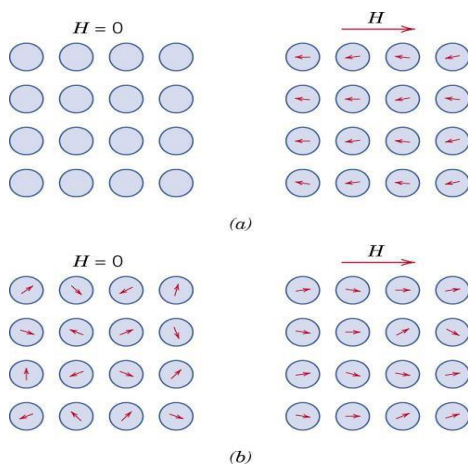
Diamagnetic, Paramagnetic, and Ferromagnetic Materials

When a material is placed within a magnetic field, the magnetic forces of the material's electrons will be affected. This effect is known as Faraday's Law of Magnetic Induction. However, materials can react quite differently to the presence of an external magnetic field. This reaction is dependent on a number of factors, such as the atomic and molecular structure of the material, and the net magnetic field associated with the atoms. The magnetic moments associated with atoms have three origins. These are the electron motion, the change in motion caused by an external magnetic field, and the spin of the electrons.

In most atoms, electrons occur in pairs. Electrons in a pair spin in opposite directions. So, when electrons are paired together, their opposite spins cause their magnetic fields to cancel each other. Therefore, no net magnetic field exists. Alternately, materials with some unpaired electrons will have a net magnetic field and will react more to an external field. Most materials can be classified as diamagnetic, paramagnetic or ferromagnetic.

Diamagnetic

- Materials have a weak, negative susceptibility to magnetic fields.
- Repelled by a magnetic field
- The material does not retain the magnetic properties when the external field is removed.
- No hysteresis properties, curie points and retentivity.
- In diamagnetic materials all the electron are paired so there is no net permanent magnetic moment per atom.
- Diamagnetic properties arise from the realignment of the electron paths under the influence of an external magnetic field.
- The value of diamagnetic susceptibility is independent of temperature and applied magnetic field.
- Most elements in the periodic table, including copper, silver, and gold, Bismuth, antimony, Zn, organic materials, Alkali earth metals are diamagnetic. They can be solids, liquids and gases.



Paramagnetic

- Materials have a small, positive susceptibility but of low order to magnetic fields.

- Slightly attracted by a magnetic field and the material do not retain the magnetic properties when the external field is removed.
- Paramagnetic properties are due to the presence of some unpaired electrons, and from the realignment of the electron paths caused by the external magnetic field
- No hysteresis properties, curie points and retentivity.
- Paramagnetic substance tends to go slowly to the stronger region of the field from the weaker region.
- Paramagnetic materials include magnesium, molybdenum, lithium, and tantalum. They can be solids, liquids and gases.
- Paramagnetic materials like aluminum, uranium and platinum become more magnetic when they are very cold.

Ferromagnetic

- Materials have a large, positive susceptibility to an external magnetic field.
- Exhibit a strong attraction to magnetic fields and are able to retain their magnetic properties after the external field has been removed.
- They have fixed curie points
- Show hysteresis properties.
- Magnetic permeability is of high order.
- Their susceptibility depends on temperature i.e., $K \propto \frac{1}{T}$
- Ferromagnetic substance tends to go quickly to stronger region of the field from weaker region.
- Ferromagnetic materials have some unpaired electrons so their atoms have a net magnetic moment.
- They get their strong magnetic properties due to the presence of magnetic domains. In these domains, large numbers of atom's moments (10^{12} to 10^{15}) are aligned parallel so that the magnetic force within the domain is strong.
- When a ferromagnetic material is in the unmagnetized state, the domains are nearly randomly organized and the net magnetic field for the part as a whole is zero. When a magnetizing force is applied, the domains become aligned to produce a strong magnetic field within the part.
- Iron, nickel, and cobalt are examples of ferromagnetic materials. They can be crystalline and solids.

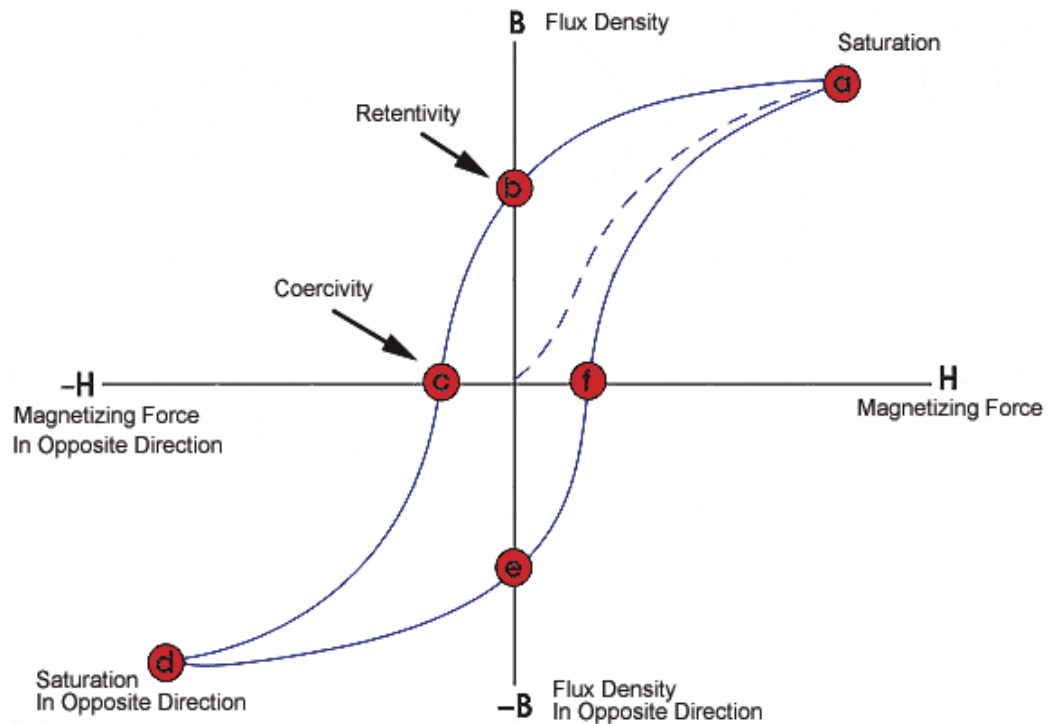
Temperature effect

Strongly magnetic ferromagnetic materials like nickel or steel lose all their magnetic properties if they are heated to a high enough temperature. The atoms become too excited by the heat to remain pointing in one direction for long.

The temperature at which a metal loses its magnetism is called the Curie temperature, and it is different for every metal. The Curie temperature for nickel, for example, is about 350°C.

The Hysteresis Loop and Magnetic Properties

“The striking property of Ferro Magnetic materials is the relation between Magnetization and the strength of Magnetic field. This property is called Hysteresis”



The loop is generated by measuring the magnetic flux of a ferromagnetic material while the magnetizing force is changed. A ferromagnetic material that has never been previously magnetized or has been thoroughly demagnetized will follow the dashed line as H is increased.

As the line demonstrates, the greater the amount of current applied ($H+$), the stronger the magnetic field in the component ($B+$). At point "a" almost all of the magnetic domains are aligned and an additional increase in the magnetizing force will produce very little increase in magnetic flux. The material has reached the point of magnetic saturation.

When H is reduced to zero, the curve will move from point "a" to point "b." At this point, it can be seen that some magnetic flux remains in the material even though the magnetizing force is zero. This is referred to as the point of retentivity on the graph and indicates the remanence or level of residual magnetism in the material. (Some of the magnetic domains remain aligned but some have lost their alignment.)

As the magnetizing force is reversed, the curve moves to point "c", where the flux has been reduced to zero. This is called the point of coercivity on the curve. (The reversed magnetizing force has flipped enough of the domains so that the net flux within the material is zero.) The force required to remove the residual magnetism from the material is called the coercive force or coercivity of the material.

As the magnetizing force is increased in the negative direction, the material will again become magnetically saturated but in the opposite direction (point "d"). Reducing H to zero brings the curve to point "e." It will have a level of residual magnetism equal to that achieved in the other direction. Increasing H back in the positive direction will return B to zero. Notice that the curve did not return to the origin of the graph because some force is required to remove the residual magnetism. The curve will take a different path from point "f" back to the saturation point where it will complete the loop. The area of loop indicates the amount of energy wasted in one cycle of operation.

From the hysteresis loop, a number of primary magnetic properties of a material can be determined.

1. **Retentivity** - A measure of the residual flux density corresponding to the saturation induction of a magnetic material. In other words, it is a material's ability to retain a certain amount of residual magnetic field when the magnetizing force is removed after achieving saturation. (The value of B at point b on the hysteresis curve.)
2. **Residual Magnetism** or **Residual Flux** - the magnetic flux density that remains in a material when the magnetizing force is zero. Note that residual magnetism and retentivity are the same when the material has been magnetized to

the saturation point. However, the level of residual magnetism may be lower than the retentivity value when the magnetizing force did not reach the saturation level.

3. **Coercive Force** - The amount of reverse magnetic field which must be applied to a magnetic material to make the magnetic flux return to zero. (The value of H at point c on the hysteresis curve.)
4. **Permeability, μ** - A property of a material that describes the ease with which a magnetic flux is established in the component.
5. **Reluctance** - Is the opposition that a ferromagnetic material shows to the establishment of a magnetic field. Reluctance is analogous to the resistance in an electrical circuit.

Hard Magnetic Materials

- ✓ Hard magnetic materials have large hysteresis loss due to large hysteresis loop area
- ✓ The Coercivity and retentivity are large hence these materials cannot be easily magnetized and demagnetized.
- ✓ These materials have small values of permeability and susceptibility.
- ✓ These are used to make permanent magnets.
- ✓ Ex. Alnico alloy

Soft Magnetic Materials

- ✓ Soft Magnetic materials have low hysteresis loss due to small hysteresis loop area.
- ✓ The coercivity and retentivity are small; hence these materials can be easily magnetized and demagnetized.
- ✓ These materials have large values of permeability and susceptibility.
- ✓ These are used to make electromagnets.
- ✓ Ex: Iron silicon alloys, Ferrous nickel alloy

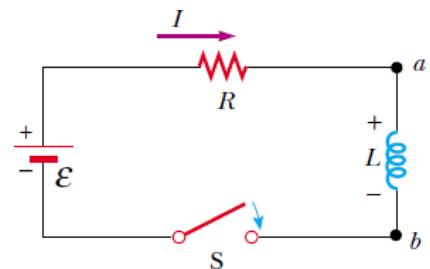
10.1 Energy Stored in the Magnetic Field

Consider a resistor and an inductor is connected in series with a battery of emf $\mathcal{E}$. According to the Kirchhoff's 2nd rule;

$$\mathcal{E} - V_R - V_L =$$

$$0 \quad \mathcal{E} = V_R + V_L$$

$$\mathcal{E} = IR + L \frac{dI}{dt}$$



Multiplying both sides by I , we have:

$$\mathcal{E}I = I^2R + LI \frac{dI}{dt}$$

Here $\mathcal{E}I$ is the power supplied by the source, I^2R is the power dissipated in resistance and $LI \frac{dI}{dt}$ is the energy supplied per time in the inductance coil, where the magnetic field also exists.

If U_B is the total energy stored in the magnetic field of inductance coil, then the energy stored per unit time is:

$$\frac{dU_B}{dt} = LI \frac{dI}{dt}$$

$$dU_B = LI dI$$

Integrating both sides, we get:

$$\int_0^{U_B} dU_B = \int_0^I LI \, dI$$

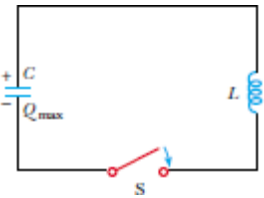
$$U_B = L \left[\frac{I^2}{2} \right]_0^I$$

$$U_B = \frac{1}{2} LI^2$$

This is the expression of energy stored in the magnetic field of a current carrying inductor.

10.2.1 Electromagnetic Oscillations (Qualitative discussion)

Consider an oscillating circuit which consists of a capacitor and an inductor. If the capacitor is initially charged and the switch is then closed, we find that both the current in the circuit and the charge on the capacitor oscillate between maximum positive and negative values. If the resistance of the circuit is zero, no energy is transformed to internal energy. In the following analysis, we neglect the resistance in the circuit. We also assume an idealized situation in which energy is not radiated away from the circuit.



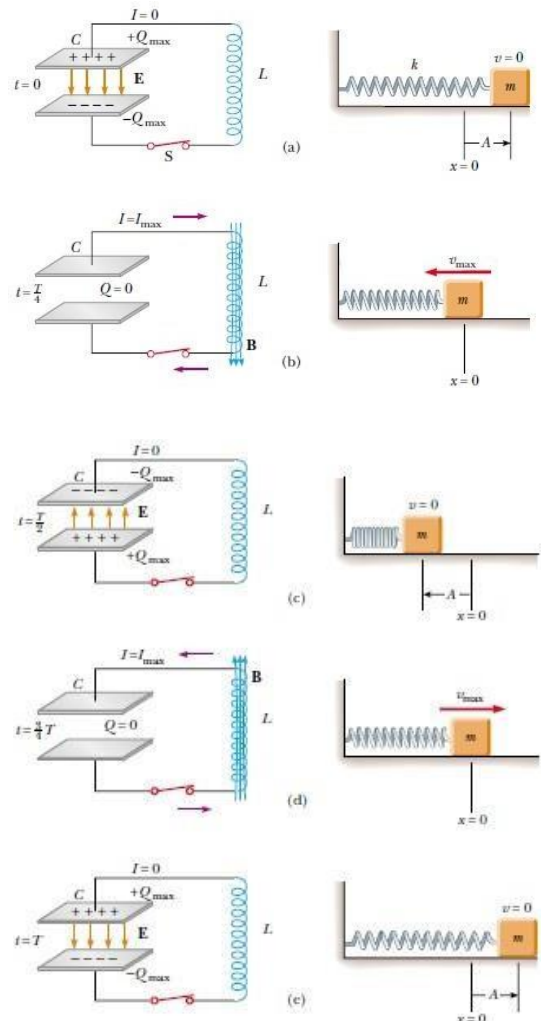
When the capacitor is fully charged, the energy in the circuit is stored in the electric field of the capacitor and is equal to

$$U_E = \frac{1}{2} \frac{Q_{\max}^2}{C}$$

At this time, the current in the circuit is zero, and therefore no energy is stored in the inductor.

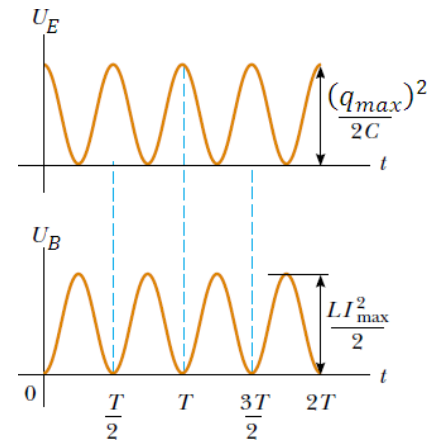
$$U_B = 0$$

As the switch S is closed, the capacitor begins to discharge and the energy stored in its electric field decreases. The discharge of the capacitor represents a current in the circuit, and hence some energy is now stored in the magnetic field of the inductor. Thus, energy is transferred from the electric field of the capacitor to the magnetic field of the inductor. When the capacitor is fully discharged, it stores no energy. At this time, the current reaches its maximum value, and all of the energy is stored in the inductor.



$$(U)_E \min = 0, (U)_B \max = \frac{1}{2} L I_{\max}^2$$

The current continues in the same direction, decreasing in magnitude, with the capacitor eventually becoming fully charged again but with the polarity of its plates now opposite the initial polarity. This is followed by another discharge until the circuit returns to its original state of maximum charge $q_{\max}$. The energy continues to oscillate between inductor and capacitor. When the energy stored in the capacitor is maximum, the energy stored in inductor is minimum and vice versa. But the total energy $U = U_E + U_B$ remains constant.



As the energy in such circuit is used to oscillate between capacitor and inductor, therefore, it is also called ‘Oscillating Circuit’.

10.2.2 LC Circuit: Electromagnetic Oscillations (Quantitative analysis)

Let, U_E and U_B are the values of energy stored in the electric field of capacitor and magnetic field of an inductor, respectively.

So, the total energy U stored in the circuit is:

$$U = U_E + U_B$$

$$U = \frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L I^2$$

Differentiating the above equation we get,

$$\frac{dU}{dt} = \frac{d}{dt} \left[\frac{1}{2} \frac{q^2}{C} + \frac{1}{2} L I^2 \right]$$

as, $U = \text{constant}$, we have,

$$0 = \frac{q}{C} \frac{dq}{dt} + L I \frac{dI}{dt}$$

$$\Rightarrow \frac{q}{C} \frac{dq}{dt} + L \frac{dq}{dt} \frac{d}{dt} \left(\frac{dq}{dt} \right) \quad [\because I = \frac{dq}{dt}]$$

$$\Rightarrow \frac{dq}{dt} \left[\frac{q}{C} + L \frac{d^2 q}{dt^2} \right] = 0$$

$$\Rightarrow \frac{q}{C} + L \frac{d^2 q}{dt^2} = 0$$

$$\therefore \frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0 \dots \dots \dots (1)$$

Solution of equation (1) will be,

$$q = q_m \cos(\omega t + \phi) \dots \dots \dots (2)$$

Where,

$$\omega = \frac{1}{\sqrt{LC}}$$

Thus, time period, $T = \frac{2\pi}{\omega}$

$$T = 2\pi\sqrt{LC}$$

And frequency,

$$f = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

Total Energy in LC Oscillations:

The energy stored in the electric field of a capacitor is,

$$U_E = \frac{1}{2} \frac{q^2}{C}$$

$$U_E = \frac{1}{2C} q_m^2 \cos^2(\omega t + \phi) \dots \dots \dots (3)$$

The energy stored in the magnetic field of an inductor will be,

$$U_B = \frac{1}{2} LI^2$$

$$U_B = \frac{1}{2} L \left(\frac{dq}{dt} \right)^2$$

$$U_B = \frac{1}{2} L [\omega^2 q_m^2 \sin^2(\omega t + \phi)]$$

$$U_B = \frac{1}{2} L \left[\frac{1}{LC} q_m^2 \sin^2(\omega t + \phi) \right]$$

$$U_B = \frac{1}{2C} q_m^2 \sin^2(\omega t + \phi) \dots \dots \dots (4)$$

The total energy stored in the LC circuit is:

$$U = U_E + U_B$$

$$U = \frac{1}{2C} q_m^2 \cos^2(\omega t + \phi) + \frac{1}{2C} q_m^2 \sin^2(\omega t + \phi)$$

$$U = \frac{1}{2C} q_m^2 [\cos^2(\omega t + \phi) + \sin^2(\omega t + \phi)]$$

$$U = \frac{1}{2C} q_m^2$$

As, q_m is the maximum charge stored in the capacitor, so the total energy stored in LC circuit is constant.

B.Sc. II Semester (NEP)**DSC 2: Electricity and Magnetism****Unit 3 (11 hrs): Magnetism:**

Statement of Biot-Savart law, Applications – (a) Magnetic field due to steady current in a long straight wire (b) Expression for force between two long straight conductors, Ampere's circuit law, Applications – (a) Magnetic field near a long wire carrying current (b) Magnetic field due to a solenoid, Current loop as a dipole, Torque on a dipole.

Electromagnetic induction:

Review of Faradays laws of electromagnetic induction, Deduction of Faradays laws from Lorentz force. Induction and Relative Motion: A Conducting rod moving in a uniform magnetic field, Concept of self-induction and mutual induction, Energy stored in a magnetic field (derivation).

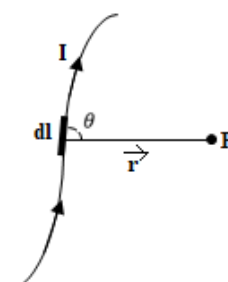
Magnetism**Biot- Savart law:**

This is an empirical law which was formulated based of the experimental results and gives the relation for magnetic field produced by a current element.

Consider a current element of length dl carrying current I . P is a point at a distance r from the current element. Let θ be the angle between the current element and the line joining the point to the element.

The law states that the magnetic field dB at P due to current element is

1. Directly proportion to the strength of the current, I
2. Directly proportional to the length of the current element, dl
3. Directly proportional to the sine of the angle between the element and the line joining the point to the element, $\sin\theta$
4. Inversely proportional to the square of the distance r of the point P from the element dl .



Combining all these factors

$$dB \propto \frac{I \cdot dl \cdot \sin\theta}{r^2}$$

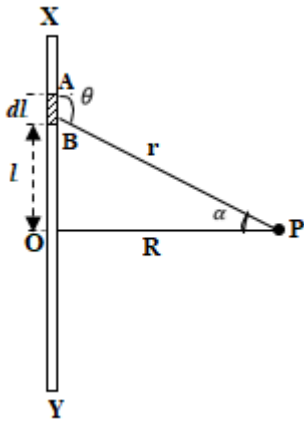
$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot dl \cdot \sin\theta}{r^2}$$

Where $\frac{\mu_0}{4\pi}$ is the proportionality constant and μ_0 is the permeability of free space.

The value of $\mu_0 = 4\pi \times 10^{-7} \frac{Wb}{Amp-m}$ or H/m .

In vector form Biot –Savart's is given by

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot \vec{dl} \times \vec{r}}{r^3}$$

Magnetic field due to steady current in a long straight wire:

Consider an infinitely long conductor placed in vacuum and carrying a current I amperes. Let P be a point at a distance R from the conductor. O is the foot of the perpendicular from P to the conductor. Consider a small element AB of length dl of conductor at a distance l from O . Let r be the distance of the element from the point P . Let θ be the angle between the current element and the line joining the point to the current element.

The magnitude of the field dB due to element AB at P is given by

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin\theta}{r^2} \dots \dots \dots (1)$$

The field due to the whole conductor is given by

$$B = \int_{-\infty}^{\infty} dB = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{\sin\theta dl}{r^2} \dots \dots \dots (2)$$

From figure,

$$r = (l^2 + R^2)^{1/2}$$

$$\sin\theta = \sin(\pi - \theta) = \frac{R}{r}$$

$$\therefore \sin\theta = \frac{R}{(l^2 + R^2)^{1/2}}$$

Substituting the value of r and $\sin\theta$ in equation (2), we get,

$$B = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{R dl}{(l^2 + R^2)^{3/2}}$$

In order to evaluate the integral, consider

$$l = R \tan\alpha$$

$$\therefore dl = R \sec^2\alpha d\alpha$$

$$\therefore B = \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{R R \sec^2\alpha d\alpha}{(R^2 \tan^2\alpha + R^2)^{3/2}}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{R^2 \sec^2\alpha d\alpha}{R^3 (1 + \tan^2\alpha)^{3/2}}$$

$$B = \frac{\mu_0 I}{4\pi} \int_{-\pi/2}^{\pi/2} \frac{d\alpha}{R \sec\alpha}$$

$$B = \frac{\mu_0 I}{4\pi R} \int_{-\pi/2}^{\pi/2} \cos\alpha d\alpha$$

$$B = \frac{\mu_0 I}{4\pi R} [\sin\alpha]_{-\pi/2}^{\pi/2}$$

$$B = \frac{\mu_0 I}{4\pi R} (1 + 1)$$

$$B = \frac{\mu_0 I}{2\pi R}$$

This is the expression for the magnetic field near a long straight conductor.

Expression for force between two long straight conductors:

Consider two infinitely long, straight parallel conductors X and Y separated by a distance r and carrying currents I_1 and I_2 respectively in the same direction as shown in figure. Magnetic fields are produced due to currents flowing in each conductor. Since each conductor is in the magnetic field produced by the other, each conductor experiences a force.

Consider a point P on the conductor. The magnitude of the magnetic field at any point P on the conductor Y due to current I_1 in conductor X is

$$B_1 = \frac{\mu_0 I_1}{2\pi r} \dots \dots \dots (1)$$

According to the right-hand rule (fore figure-current, middle figure-magnetic field, thumb-direction of force), the force on the conductor Y is inwards (Fig 1), that is towards conductor X. thus the magnitude of force experienced by length l of conductor Y is given by

$$F_2 = B_1 I_2 l \dots \dots \dots (2)$$

Substituting the value of B_1 from equation (1), we get

$$F_2 = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2 l}{r}$$

$\therefore$ Force per unit length of conductor,

$$F_2 = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r} \dots \dots \dots (3)$$

Similarly, it can be shown that the conductor X also experiences a force and is directed towards the conductor Y as shown in Fig (2). The force on the conductor X per unit length is

$$F_1 = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r} \dots \dots \dots (4)$$

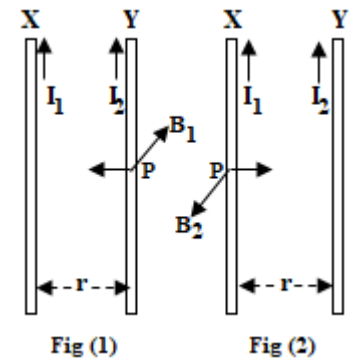
Thus, the force between two long straight conductors is

$$F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r}$$

From the above expression it is clear that the parallel conductors carrying currents in the same direction attract each other, and parallel conductors carrying currents in opposite directions repel each other.

Ampere's circuit law:

Oersted in 1820 discovered the existence of the magnetic field due to the current flowing in a conductor. Ampere derived a relationship between current I and the magnetic field B . The relationship is known as Ampere's law.



According to the law, “The line integral $\oint B \, dl$ for a closed curve is equal to μ_0 times the net current I through the area bounded by the curve”. Thus

$$\oint B \, dl = \mu_0 I$$

Proof:

Consider a long straight conductor carrying a current I as shown in figure. Let the conductor be perpendicular to the plane of the paper, directed outwards. According to the Biot-Savart's law the magnitude of the magnetic induction at a distance R from the conductor is given by

$$B = \frac{\mu_0 I}{2\pi R} \dots \dots \dots (1)$$

At each point on this circle, B has a constant magnitude. Further, the direction of B at every point is along the tangent to the circle. That is, the angle between dl and B at every point on this path is zero. The line integral of magnetic field around this closed path is given by

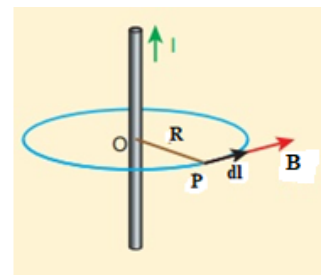
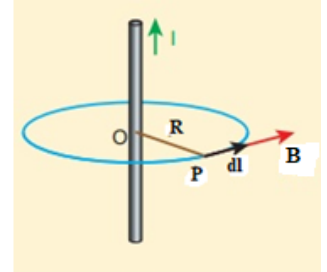
$$\begin{aligned} \oint B \, dl \cos \theta &= \oint B \, dl = \frac{\mu_0 I}{2\pi R} \oint dl \\ \oint B \, dl &= \frac{\mu_0 I}{2\pi R} \cdot 2\pi R \quad \because \oint dl = 2\pi R \\ \therefore \oint B \, dl &= \mu_0 I, \text{ Hence the proof} \end{aligned}$$

Applications of Ampere's circuit law:

1. Magnetic field due to a straight conductor carrying current:

Consider a long straight conductor carrying current I in the direction shown in figure. The magnetic lines of force are concentric circles all centered at the conductor. Choose a circle of radius R as the closed path. It is desired to find the magnetic field B at a point P . the magnitude of B is the same everywhere on this closed path. The angle between B and dl is 0° everywhere on this path. Therefore, applying Ampere's law to this closed path, we have

$$\begin{aligned} \oint B \, dl &= \mu_0 I \\ \oint B \, dl \cos 0 &= \mu_0 I \\ B \oint dl &= \mu_0 I \\ B \cdot 2\pi R &= \mu_0 I \\ B &= \frac{\mu_0 I}{2\pi R} \end{aligned}$$



2. Magnetic field due to a long solenoid carrying current:

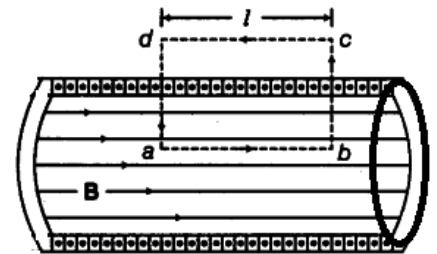
Consider a long air-cored solenoid having closely packed coils. Let I be the current flowing through the solenoid, n the number of turns per unit length. Choose a rectangular path $abcd$ in a long solenoid as shown in figure. Let $ab = l$. The line integral of B over the closed path $abcd$ is given by

$$\oint \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} + \int_b^c \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} + \int_d^a \vec{B} \cdot d\vec{l}$$

$$\int_a^b \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} \cos 0 = B \int_a^b dl = Bl$$

$$\int_b^c \vec{B} \cdot d\vec{l} = \int_c^d \vec{B} \cdot d\vec{l} = 0 \quad \because \text{angle between } \vec{B} \text{ and } d\vec{l} \text{ is } 90^\circ$$

$$\int_c^d \vec{B} \cdot d\vec{l} = 0 \quad \because \text{Field outside the solenoid is zero.}$$



$$\oint \vec{B} \cdot d\vec{l} = Bl \dots \dots (1)$$

According to Ampere's law we have

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \times \text{Current enclosed by } a \text{ } b \text{ } c \text{ } d \text{ } a$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \times \text{Number of turns in } abcd \times I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \times nl \times I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \cdot n \cdot l \cdot I \dots \dots (2)$$

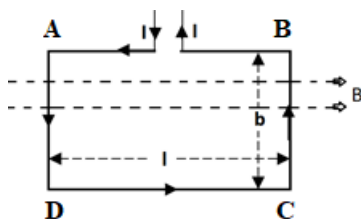
From equations (1) and (2), we have

$$Bl = \mu_0 \cdot n \cdot l \cdot I$$

$$\therefore B = \mu_0 nI$$

Concept of dipole & Current loop as a dipole:

According to the atomic theory, an atom consists of electrons rotating along closed orbital paths round the nucleus. The nucleus is a central heavy core consisting of neutrons and protons. The electron and the proton would appear as an electric dipole. The electrons which are revolving round the nucleus thus behave as a current loop. According to the explanation by Oersted, whenever current flows in a conductor, it gives rise to a magnetic field. Thus the current loop can be treated as a magnetic dipole. Further ampere explained that there are no such entities as individual independent magnetic north and south poles. He suggested that all observed magnetic phenomena are likely to be due to circulating currents, because of these circulating currents atoms and molecules can be treated as tiny magnets or magnetic dipoles.

Torque on a dipole:

Consider one turn of a rectangular coil $ABCD$ of length l and breadth b suspended in uniform magnetic field B which is parallel to the plane of the coil as shown in figure. Let a current I flow through the coil. Since the sides AB and DC are parallel to the magnetic field, the force experienced by them will be zero.

The magnitude of the force on the side AD , is

$$F_{AD} = B I b \sin\theta$$

$$F_{AD} = B I b$$

This force will be acting perpendicular to the plane of the coil and

Similarly, the magnitude of the force on the side BC is

$$F_{BC} = B I b$$

This force will be acting perpendicular to the plane of the coil and away from the observer. These two equal unlike forces constitute a couple which tends to rotate the coil about an axis through the suspension. The torque exerted by this couple in the coil, is

$$\tau = B I b \times l$$

$$\tau = B I A$$

$$\tau = M \times B$$

This is the expression for Torque on a dipole.

Electromagnetic induction:**Faradays laws of electromagnetic induction:**

I law: Whenever there is change in the magnetic flux linked with a circuit an emf and hence a current is induced in the circuit. The emf lasts as long as the change in flux persists.

II law: The magnitude of the induced emf is directly proportional to the rate of change of magnetic flux linked with the circuit. $e = -\frac{d\phi}{dt}$

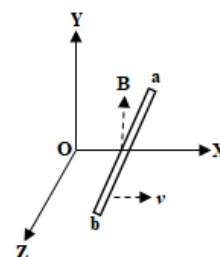
Deduction of Faradays laws from Lorentz force:

Consider a straight conducting rod ab lying with its length along the Z -axis in a uniform magnetic field $\vec{B}$ directed along the Y -axis and moving with a velocity $\vec{v}$ along the X -axis.

According to Lorentz force equation a charged particle on the conductor carrying a charge q will experience a force $\vec{F}$ given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\frac{\vec{F}}{q} = (\vec{v} \times \vec{B})$$



But $\frac{\vec{F}}{q} = \vec{E}$, the induced electric field.

$$\therefore \vec{E} = (\vec{v} \times \vec{B})$$

Consider an elemental length $\vec{dl}$ of the rod ab . The induced e.m.f. across the length $\vec{dl}$ is $\vec{E} \cdot \vec{dl}$. Thus, the induced e.m.f. across the ends of the conductor due to the electric field is given by

$$\begin{aligned} e &= \int_a^b \vec{E} \cdot \vec{dl} = \int_a^b (\vec{v} \times \vec{B}) \cdot \vec{dl} \\ e &= \int_a^b -\vec{B} \cdot (\vec{v} \times \vec{dl}) \\ e &= -\vec{B} \int_a^b (\vec{v} \times \vec{dl}) \dots (1) \end{aligned} \quad (A \times B) \cdot C = -B \cdot (A \times C)$$

The conductor ab moves through a distance $\vec{v} dt$ along the X-axis in a time dt and in this time the area swept by the element $\vec{dl}$ is

$$\vec{v} dt \times \vec{dl} \quad \text{area} = l \times b$$

Therefore the total area $\vec{ds}$ swept by the conductor ab in time dt is given by

$$\begin{aligned} \vec{ds} &= \int_a^b (\vec{v} dt \times \vec{dl}) \\ \frac{d\vec{s}}{dt} &= \int_a^b (\vec{v} \times \vec{dl}) \end{aligned}$$

Substituting this value in eqn (1), we have

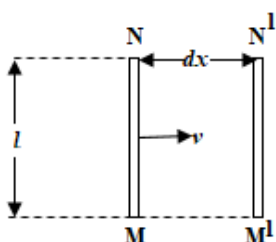
$$\begin{aligned} e &= -\vec{B} \left[\frac{d\vec{s}}{dt} \right] \\ e &= -\frac{\vec{B} \cdot d\vec{s}}{dt} \end{aligned}$$

But, $\vec{B} \cdot d\vec{s} = d\phi$ a change in magnetic flux

$$\therefore e = -\frac{d\phi}{dt}$$

This is the mathematical form of Faraday's II law of electromagnetic induction.

A Conducting rod moving in a uniform magnetic field:



Consider a straight conducting rod MN of length l moving at right angles to a uniform magnetic field B with a constant speed v . The magnetic field is perpendicular to the plane of the diagram and away from the observer as shown in figure.

Let the conductor move from MN to $M'N'$ through a distance dx in time dt . Then area swept by the conductor $= MNM'N' = l dx$.

Therefore, the change in magnetic flux linked with the conductor in time dt is

$$d\phi = B(l dx)$$

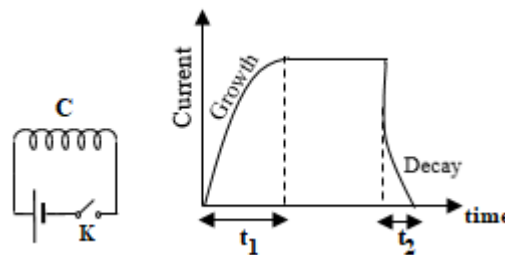
Magnitude of induced e.m.f. is

$$e = \left| \frac{d\phi}{dt} \right| = \left| Bl \frac{dx}{dt} \right| = B l v$$

$$\therefore \text{motional emf, } e = B l v$$

Concept of self-induction and mutual induction:

The phenomenon of self-induction was discovered by J.Henry in 1832. To understand the concept of self-inductance consider a coil C connected in series with a cell and a tap key as shown in figure. When the key is pressed the current in the circuit gradually increases and takes finite time t_1 . As the current changes the magnetic flux linked with the coil also changes zero to a certain maximum value. In accordance with Faraday's I law an emf is induced in the coil. The emf induced opposes the growth of the current in the coil and is called the back emf. When the key is released the current in the coil decays from maximum to zero in time t_2 . The flux linked with the coil also decreases from maximum to zero. This results in an induced emf but in the opposite direction. This emf is called the forward emf. It is observed that the time of growth t_1 is greater than the time of decay t_2 (fig). Hence the forward emf is greater than the back emf.



The phenomenon in which an emf is induced in a coil due to the change of current through the coil itself is known as self-induction. The property of a coil to oppose any change of current through it is known as self-inductance.

Coefficient of self-induction:

The magnetic flux ϕ linked with a coil at any instant is directly proportional to the current I through it at that instant. That is $\phi \propto I$ or $\phi = LI$ where L is a constant called the self-inductance or coefficient of self-conduction. According to Faraday's II law

$$e = \frac{d\phi}{dt} = \frac{d}{dt}(LI) = L \frac{dI}{dt}$$

Hence, self-inductance of a coil is numerically equal to the emf induced in it when the rate of change of current through it is unity. The SI unit of self-inductance is henry.

Mutual induction: Consider a pair of coils placed side by side without touching each other. A varying current flowing through one coil induces an emf in the other. This phenomenon in which a change of current in one coil induces an emf in another neighbouring coil is called mutual induction.

The emf produced in the secondary coil is directly proportional to the rate of change of current through the primary coil.

$$e \propto \frac{dI}{dt} \text{ or } e = M \frac{dI}{dt}$$

The constant M is called the coefficient of mutual induction or the mutual inductance between the coils.

Hence, mutual inductance between a pair of coils is numerically equal to the emf induced in the secondary coil when the current through the primary coil changes at unit rate.

Energy stored in a magnetic field:

Consider the circuit in which a source of emf E is connected to a resistance R and inductance L. From Kirchhoff's law,

$$E = IR + L \frac{dI}{dt}$$

$$EI = I^2 R + LI \frac{dI}{dt}$$

The first term on R.H.S is the rate at which energy appears as heat in the resistance R. The energy that does not appear as heat must according to the rate of conservation of energy in the magnetic field. Therefore $LI \frac{dI}{dt}$ represents the rate at which the energy is stored in the magnetic field.

$$\frac{dU}{dt} = LI \frac{dI}{dt}$$

$$dU = LI dI$$

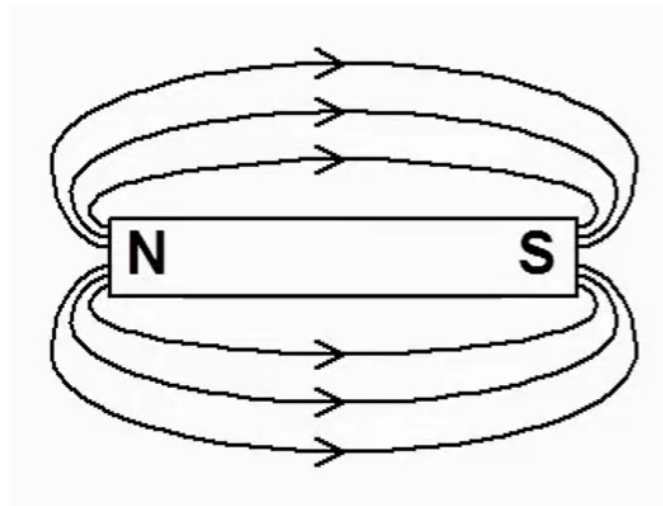
Therefore total energy stored in the magnetic field is

$$U = \int_0^{I_0} LI dI$$

$$U = \frac{1}{2} LI_0^2$$

Magnetic field, Magnetic field intensity, Magnetic flux and flux density

The magnetic field is defined as the region around the magnet where its poles and the electrical charges experience the force of attraction or repulsion. The presence of the field is determined through the needle. In actual practice, the magnetic field has no real existence, and they are purely imaginary. The magnetic field is created because of the magnetic line of force which possesses the following properties.

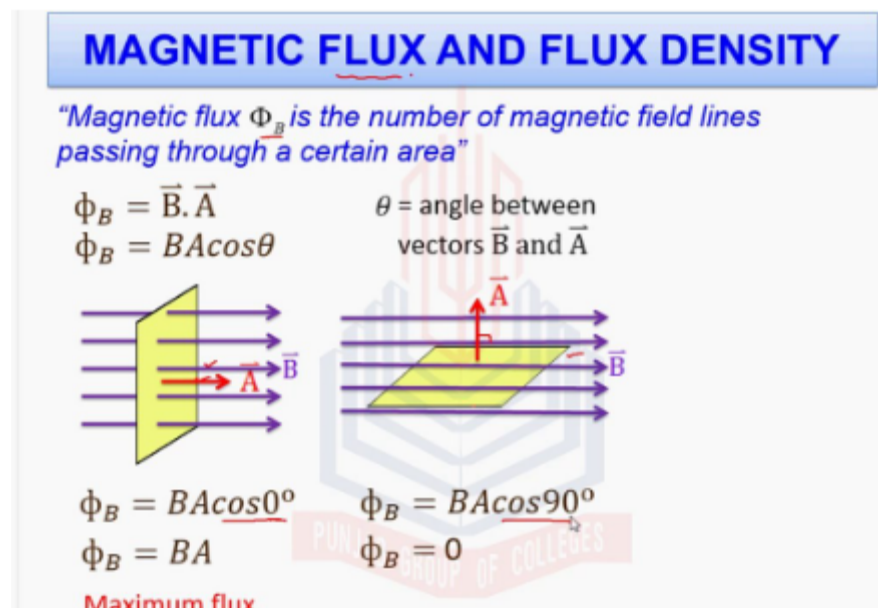


1. The magnetic line of force forms the closed loop.
2. The direction of magnetic line force is from north to south.
3. The magnetic lines never intersect each other.
4. The magnetic lines repel each other when they are parallel and in the same direction.
5. They are not affected by the non-magnetic materials.

Definition of Magnetic Flux

The magnetic flux is defined as the total number of magnetic lines of force produces by the magnet. It is measured in Weber. The one Weber is equal to the 10^8 line of forces.

Magnetic flux density (B) is a measure of the concentration or strength of a magnetic field at a specific point. It is defined as the number of magnetic field lines passing through a unit area perpendicular to the direction of the field. In more practical terms, it is the force exerted per unit current per unit length on a conductor placed perpendicular to the magnetic field.



Magnetic force on a moving charge

Magnitude of the Force The magnitude of the magnetic force, F , on a particle with charge q moving at a velocity v in a magnetic field B is calculated using the formula: $F = qvB \sin \theta$. In this equation, θ represents the angle between the velocity vector and the magnetic field lines.

Maximum Force: The force is strongest when the charge moves perpendicularly to the field lines $\theta = 90^\circ$.

Zero Force: The force is zero if the charge moves parallel or anti-parallel to the field lines ($\theta = 0$ or 180°).

Velocity Dependence: The force increases with higher velocity, greater charge, or stronger magnetic field strength. Direction of the Force The magnetic force is always perpendicular to both the velocity of the charge and the magnetic field. To determine this direction for a positive charge, physicists use Right Hand Rule

Lorentz force

The Lorentz force is the combined electric (qE) and magnetic ($qv \times B$) force on a charged particle moving through electromagnetic fields, defined by $F = qE + qv \times B$

It dictates that a current-carrying conductor in a magnetic field experiences a force perpendicular to both current and field.

1. Force on a Wire in an External Magnetic Field

When a straight wire of length l carrying a current I is placed in a uniform external magnetic field B , it experiences a force. This occurs because the moving charge carriers (electrons) within the wire experience the Lorentz force, which is then transmitted to the conductor itself.

Magnitude of the Force

The magnitude of this force is calculated using the formula:

$$F = IlB \sin \theta$$

- F : Magnetic force (measured in Newtons, N).
- I : Current flowing through the conductor (Amperes, A).
- l : Length of the conductor within the magnetic field (meters, m).
- B : Magnetic flux density of the external field (Teslas, T).
- θ : The angle between the direction of the current and the magnetic field lines.

Key Effects of Orientation:

- **Maximum Force:** Occurs when the wire is perpendicular to the field ($\theta = 90^\circ$), resulting in $F = IlB$.
- **Zero Force:** Occurs when the wire is parallel to the magnetic field ($\theta = 0^\circ$ or 180°).

What are Magnetic Circuits?

Magnetic circuits are similar to normal electrical circuits that have a closed path followed by magnetic lines of force. It is important to know that in a magnetic circuit, the magnetic lines of force originate from a point and end at the same point after completing the full path. Despite being a circuit, it is important to note that nothing flows in a magnetic circuit like current that flows in a standard electrical circuit.

As the name suggests, a magnetic circuit consists of magnetic materials which have high permeability, these materials are usually steel or iron. Magnetic circuits also include electric motors, transformers, generators galvanometers, etc.

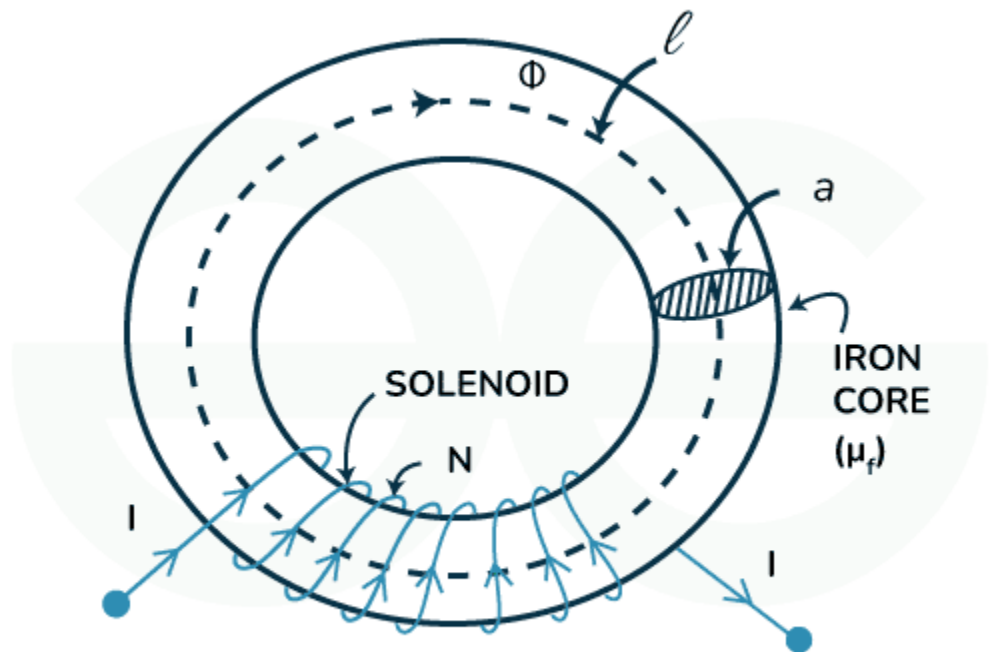
Applications of Magnetic Circuits

Magnetic circuits remain indispensable in electrical engineering thereby having various applications in real-day life like.

- Magnetic circuits are majorly used in transformers. This is mainly because of their ability to transfer energy efficiently. These circuits result in efficient power transfer by giving direct control over power supply in circuit.
- The circuits are used in inductors for storage of energy efficiently. The magnetic circuits generate a magnetic field of electrons and the energy of these electrons is stored in inductors. Using magnetic circuits also reduces electrical noise.
- Other appliances using magnetic circuits includes electric motors and generators. They help to operate motors by converting electrical energy to mechanical energy. The generators use these circuits for performing the exact opposite task i.e. converting mechanical energy to electrical energy.

Circuit Diagram of Magnetic Circuits

The magnetic circuit diagram shown below consist of a magnetic material, generally an iron core which is used for making the solenoid part of the circuit. The solenoid with N turns creates a magnetic flux which generates a current I in the solenoid.



Circuit Diagram

Magnetomotive force (mmf)

The magnetic pressure that sets up or tends to set up in a magnetic field is known as magnetomotive force.

A coil of N turns carrying a current I generates a magnetomotive force equal to $MMF = NI$

Unit: Ampere turns

Magnetic flux: The amount of magnetic lines of force set up in a magnetic circuit is called magnetic flux

Magnetic flux = MMF/S

Here S is the reluctance.

The unit of magnetic flux is weber

Reluctance: The opposition offers to the flow of magnetic flux in a magnetic circuit is known as reluctance. It is the ratio of MMF to the magnetic flux in the circuit.

Reluctance depends upon the length, cross-sectional area and permeability of the material.

$$S = \frac{l}{\mu A}, \text{ l is the length and A is the area of cross section.}$$

The unit of reluctance is AT/Weber

Magnetic field strength (H): If the magnetic circuit is uniform or the area of cross-section is uniform then the magnetomotive force per unit length is called magnetic field strength.

$$H = \text{MMF/Length}$$

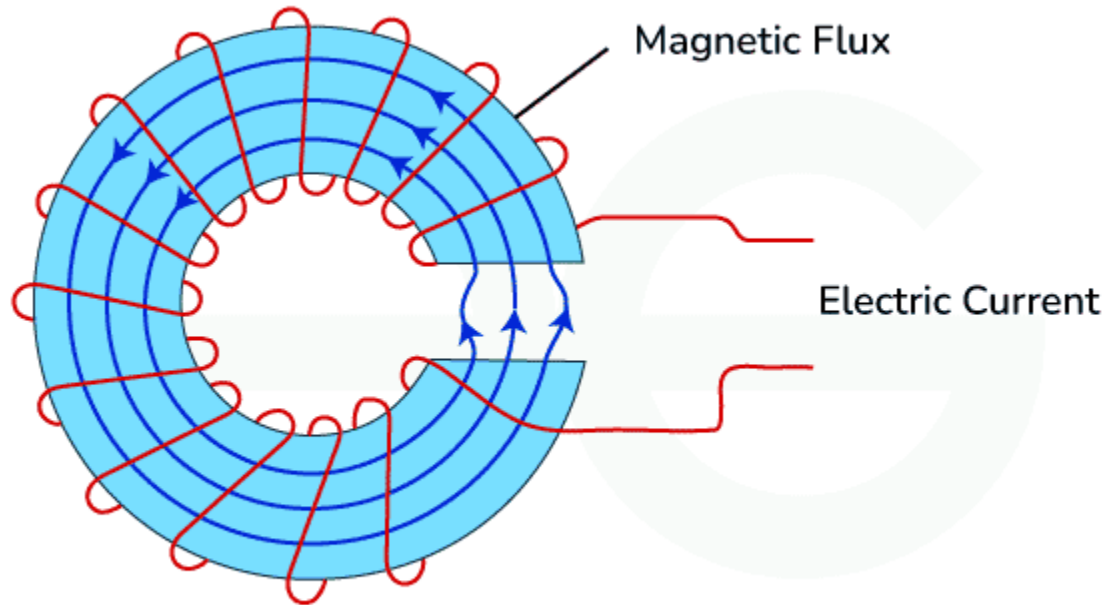
Unit is AT/m

Magnetic flux density: The magnetic flux established per unit area is called magnetic flux density.

$$B = \phi/A, \text{ Its unit is wb/m}^2 \text{ or tesla}$$

Ohm's Law For Magnetic Circuits

Consider the magnetic circuit given below.



Magnetic Circuits

Magnetic Circuit

We have already seen that magnetic circuits are analogous to electric circuits i.e. we can compare magnetic circuits to electric circuits in certain respects. That being said, we can say that there must be a law analogous to ohm's law for magnetic circuits. This indeed is true that there is a law called **Rowland's law** which is very similar to ohm's law. The statement of this law is:

Rowland's law states that the number of magnetic lines of force (Φ) is proportional to the magnetomotive force (F_m) and inversely proportional to the circuit's reluctance (S).

Let us see the analogy, According to ohm's law

$$V = IR$$

Here, V is the voltage I is the current in circuit R is circuit resistance. Similarly for magnetic circuits

$$F = \Phi S$$

Here, F is the magnetomotive force Φ is the magnetic flux S is the reluctance of the material.

The magnetomotive force can be compared to the voltage of the source while the magnetic flux can be compared to the current flow in electric circuit. The magnetic reluctance is synonymous to the resistance in electric circuit. This makes Rowland's law an equivalent version of ohm's law in magnetic circuit.

Properties of Magnetic Circuits

There are two major properties of magnetic circuits namely magnetic flux and magnetomotive force . We will discuss them here.

1. Magnetic Flux

Magnetic Flux is the term used for defining magnetic induction in magnetic circuits. It is often referred to when discussion is about strength of magnetic field in a particular region. Formally magnetic flux can be defined as a quantitative measure of magnetic field lines passing through a given region. It is generally represented by symbol Φ and can be written as

$$\Phi = B \cdot A \cdot \cos(\theta)$$

Φ is the magnetic flux measured in Weber (Wb), B is the magnetic field strength measured in Tesla (T), A is the area of magnetic field lines measured in square meters (m^2), θ is the angle between the magnetic field lines and the normal to the surface.

2. Magnetomotive Force (MMF)

This concept comes into picture when we talk about maintaining magnetic flux in a magnetic circuit. Similar to electromotive force in electrical circuits we talk about MMF as an analogous force. This is an important force that helps in sustaining the magnetic circuit. Formally, It is the total magnetic potential difference in a magnetic circuit and which performs the task of maintaining magnetic flux in a magnetic circuit. It can be written as

$$F = N \cdot I$$

F is the magnetomotive force in ampere-turns, N is the number of turns of the coil or winding, I is the current flowing through the coil in amperes.

Difference Between Electric Circuit and Magnetic Circuit

Lets see the difference between electric and magnetic circuits.

Electric Circuits	Magnetic Circuits
It is the path for electric current to flow.	It is the path traced by magnetic flux.
It follows ohms law $V=IR$	It follows law analogous to ohms law $F=\Phi R$
The force involved in maintaining circuits is electromotive force.	The force involved in maintaining circuits is magnetomotive force.
Resistance opposes the flow of current.	Reluctance is opposed by magnetic path to flux.
The flow of electrons decides current in conductor.	The magnetic lines of force decide the flux in core.

Magnetic circuits with ac excitations

Contents

1	GAUSS'S LAW	1
1.1	The Electric Field	1
1.2	Electric Field Lines	1
1.3	The Flux of the Electric Field	3
1.4	Gauss's Law	5
1.5	Coulomb's Law and Gauss's Law	5
1.6	Applications of Gauss's Law	6
1.6.1	Infinite line of charge	6
1.6.2	Infinite sheet of charge	7
1.6.3	A spherical shell of charge	8
1.6.4	Spherical symmetric charge distribution	9
1.7	Gauss Law for Dielectric Materials	10
1.7.1	Three electric vectors	11
1.8	Examples	12
2	ELECTRIC POTENTIAL	16
2.1	Potential Difference and Potential	16
2.2	Relation Between E and V	17
2.2.1	Potential when E is uniform	18
2.2.2	Potential when E is not uniform	18
2.3	Calculation of Potential	19
2.3.1	Potential due to a point charge	19
2.3.2	Potential due to a collection of point charges	20
2.3.3	Potential due to a dipole	20
2.3.4	Potential due to a quadrupole	21
2.4	A Dipole in an Electric Field	23
3	MAGNETIC FIELD DUE TO CURRENT	24
3.1	Magnetic Field Produced by Current	24
3.2	Magnetic Field Due to a Moving Charge	24
3.3	The Biot-Savart Law	26
3.4	Applications of Biot-Savart law	27
3.4.1	A straight wire segment	27
3.4.2	A circular current loop	29
3.5	Magnetic Field in a Solenoid	31
3.5.1	Loosely wire wound solenoid	31
3.5.2	Closely wire wound solenoid	31
3.6	Ampere's Law	32
3.7	Application of Ampere's Law	33
3.7.1	A long straight wire	33

3.7.2	A solenoid	35
3.7.3	A toroid	35
3.8	Examples	36
4	FARADAY'S LAW OF INDUCTION	38
4.1	Faraday's Experiment	38
4.2	Faraday's Law	39
4.3	Lenz's Law	40
4.4	Eddy Current	41
4.5	Inductance	42
4.5.1	Self inductance	43
4.5.2	Mutual inductance	43
4.5.3	The inductance of a solenoid	44
4.5.4	The inductance of a toroid	45
4.6	Examples	45
5	MAGNETIC PROPERTIES OF MATER	48
5.1	The Magnetic Dipole	48
5.1.1	The field of a dipole	50
5.1.2	Microscopic view of atomic magnetism	51
5.2	Magnetization	53
5.2.1	Diamagnetism	53
5.2.2	Paramagnetism	53
5.2.3	Ferromagnetism	53
5.3	Three Magnetic Vectors	54
5.4	Permeability & Susceptibility	56
5.5	Magnetic Hysteresis	57
5.6	Examples	58
	Index	60

Chapter 1

GAUSS'S LAW

Coulomb's law can always be used to calculate the electric field $\vec{E}$ for any discrete or continuous distribution of charges at rest. The sums or integrals might sometimes be complicated to calculate the resultant field at any point. In the chapter, we wish to discuss an alternative to Coulomb's law, called Gauss's law that provides a more useful and instructive approach to calculate the electric field in situations having certain symmetries. Although Gauss's law and Coulomb's law give identical results in the cases in which both can be used, Gauss's law is considered a more fundamental equation than Coulomb's law.

1.1 The Electric Field

Every body have an idea about *gravitational field*. Around the earth surface there is a space region with existence of one kind of force field known as *gravitational field*. If a body falls through the earth's gravitational field, it experience one kind of force expressed by the magnitude

$$\vec{F} = m_0 \vec{g}, \quad (1.1)$$

$$\text{where, } \vec{g} = \frac{\vec{F}}{m_0}. \quad (1.2)$$

The field $\vec{g}$ is a vector whose direction gives the direction of the gravitational force at that point and whose magnitude indicates the “**strength**” of the gravitational effect at that point.

Similarly around an electric charged particle or particles, there is a space region where one kind of force field exists. *This force field existing around the electric charged particle or particles is called electric field and denoted by $\vec{E}$* . If a test charge q_0 is placed such kind of electric field $\vec{E}$, in analogy with gravitational field Eq. 1.1, it also experienced a kind of force and can be expressed as

$$\vec{F} = q_0 \vec{E}, \quad (1.3)$$

which gives the electric field strength can be written as

$$\vec{E} = \frac{\vec{F}}{q_0}. \quad (1.4)$$

The direction of the vector $\vec{E}$ is the same as the direction of $\vec{F}$.

1.2 Electric Field Lines

The concept of the electric field was introduced in the early 19th century by **Michael Faraday**. He

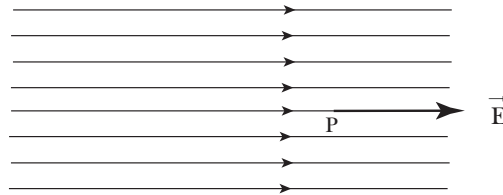


Figure 1.1: Electric field lines for a uniform electric field and direction at any point.

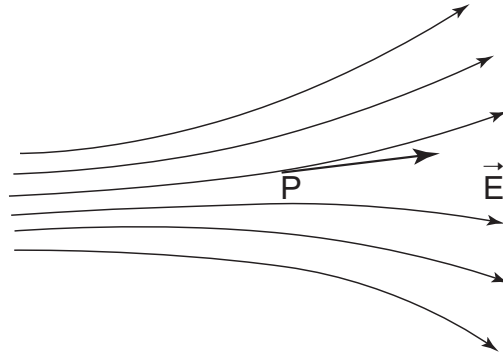


Figure 1.2: Nonuniform electric field lines and direction at any point.

developed a graphical representation, in which he imagined the space around an electric charge to be filled with *lines of force*. According to his concept of force lines we shall try to visualize some pattern of *electric field lines*.

Direction of the field line: The tangent to the electric field line passing through any point in space gives the direction of the electric field at that point, as it is shown in Figs. 1.1, 1.2, 1.3 and 1.4.

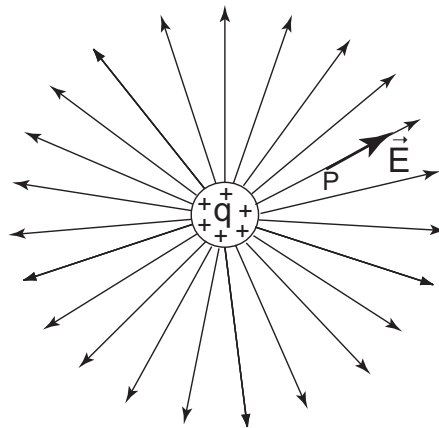


Figure 1.3: Electric field lines surrounding a group of point charges and direction at any point.

Figure 1.1 shows the electric force lines which represents a uniform electric field. Note that the lines are parallel and equally spaced to each other.

Figure 1.2 shows nonuniform electric field lines. At any point P the direction of field $\vec{E}$ has been shown by an arrowhead marker. Figure 1.3 represents the field lines for an isolated group of positive point charges. The electric field lines start on positive charges and point radially outward. If the charge were negative, the field lines would point in the opposite direction i.e. radially inward. Figure 1.4 shows the electric field lines for an electric dipole. The electric field lines start on positive charge and end on negative charge.

Magnitude of the electric field: The magnitude of the electric field at any point is proportional to the number of field lines per unit cross sectional area perpendicular to the lines.

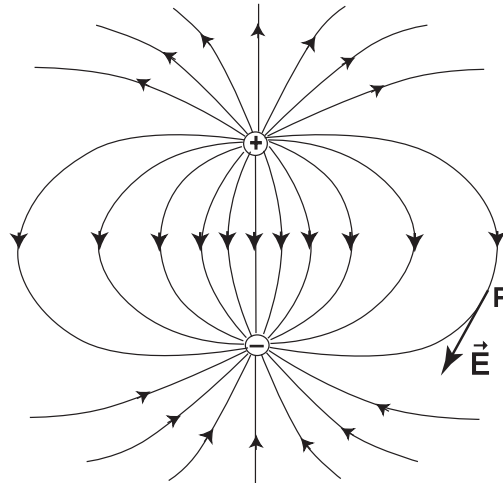


Figure 1.4: Lines of electric field for an electric dipole and direction at any point.

1.3 The Flux of the Electric Field

The 'flux' comes from a Latin word meaning 'to flow.' The symbol of the flux is written by Φ . It is a property of any vector field that refers to a hypothetical surface in the field, which may be closed or open. **The flux of a vector field to be a measure of the flow or penetration of the field vectors through an imaginary surface in the field. For an electric field the flux Φ_E is a measure of the**

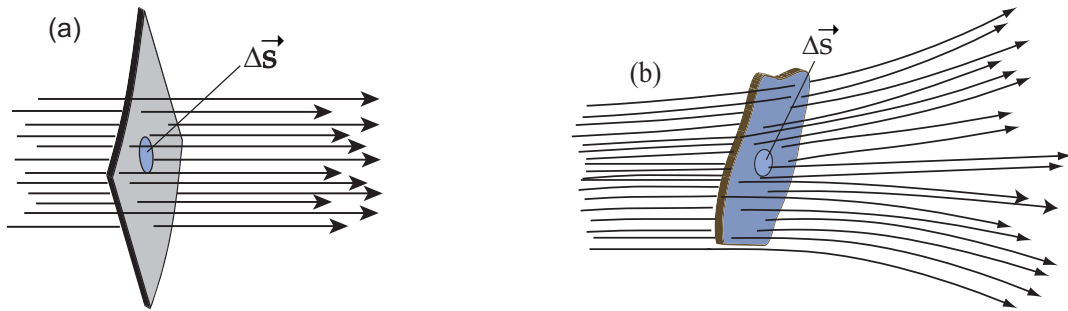


Figure 1.5: (a) Surface in a uniform electric field $\vec{E}$, (b) Surface in a nonuniform electric field $\vec{E}$.

number of lines of the electric field that passes through a surface. Figure 1.5 shows an arbitrary surface immersed in a uniform and nonuniform electric field. Let us divide the surface into small squares of area Δs , each of which is small enough so that it may be considered to be plane. Each element of area can be represented as a vector $\Delta \vec{s}$, whose magnitude to be the area Δs .

The total flux of any electric field over the surface can be calculated by the definition as

$$\Phi_E = \sum \vec{E} \cdot \Delta \vec{s}. \quad (1.5)$$

Replacing the sign of the summation by an integral sign over the surface the Eq. 1.5 yields

$$\Phi_E = \int \vec{E} \cdot d\vec{s}. \quad (1.6)$$

If the angle between the vectors $\vec{E}$ and $d\vec{s}$ be θ then the Eq. 1.6 can be rewritten as

$$\Phi_E = \int E \, ds \cos \theta. \quad (1.7)$$

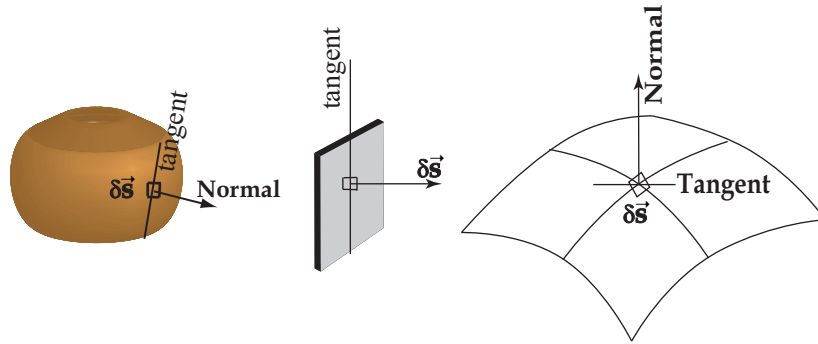


Figure 1.6: Direction of a surface element.

Direction of a surface:

The direction of a small element of surface $\Delta\vec{s}$ is shown in Fig. 1.6. The direction of a surface element can be directed at any point by a line drawn perpendicularly on the tangent of the surface element.

Flux through a closed surface in a uniform field:

Let us consider a hypothetical closed cylindrical surface is immersed in a uniform electric field, as it is shown in Fig. 1.7. The axis of the cylinder is parallel to the electric lines of force. Calculate the flux Φ_E for this closed surface.

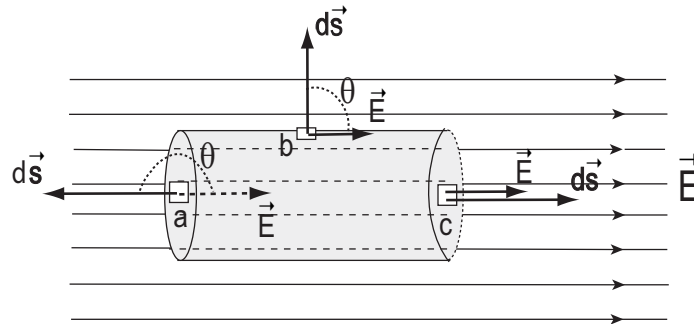


Figure 1.7: A closed cylindrical surface immersed in a uniform electric field.

The flux Φ_E can be calculated by the formula which is applicable for a closed surface,

$$\Phi_E = \oint \vec{E} \cdot d\vec{s}.$$

The integral can be performed by the sum of three parts of integration over the left cylindrical cap 'a,' the curved cylindrical surface 'b' and the right cylindrical cap 'c.'

Therefore,

$$\begin{aligned} \Phi_E &= \oint \vec{E} \cdot d\vec{s}, \\ &= \int_a \vec{E} \cdot d\vec{s} + \int_b \vec{E} \cdot d\vec{s} + \int_c \vec{E} \cdot d\vec{s}, \\ &= \int_a E ds \cos \theta + \int_b E ds \cos \theta + \int_c E ds \cos \theta, \\ &= \int_a E ds \cos 180^\circ + \int_b E ds \cos 90^\circ + \int_c E ds \cos 0^\circ, \end{aligned}$$

$$= -ES_a + 0 + ES_b.$$

Since the surface area S_a is equal to the surface area S_b ,

$$\therefore \Phi_E = 0.$$

We find that the net flux enclosed by the cylindrical surface is zero. Because lines of $\vec{E}$ enter at the left and emerge at the right, and there is no electric charge enclosed by the surface.

1.4 Gauss's Law

Consider an imaginary closed surface which contain some electric charge inside it (see Fig.1.8). Gauss's law gives the relation between the flux that cut through a surface and the net charge enclosed by the surface. The imaginary surface is called **Gaussian surface**. Therefore, Gauss's law which relates the total flux Φ_E through this surface to the net charge q enclosed by the surface, can be written as

$$\Phi_E \propto q,$$

$$\Rightarrow \Phi_E = \frac{1}{\epsilon_0} q$$

$$\text{or} \quad \epsilon_0 \Phi_E = q, \quad (1.8)$$

$$\text{or} \quad \epsilon_0 \oint \vec{E} \cdot d\vec{s} = q. \quad (1.9)$$

The circle on the integral sign indicates that the integral is to be carried out over a closed surface.

Here, ϵ_0 is a constant and its value is $8.854 \times 10^{-12} \text{ C}^2/\text{N} - \text{m}^2$.

In special case, if there is no charge inside the Gaussian surface, from the Eq. 1.8 we have

$$\Phi_E = 0.$$

The same results has already been found in previous section belongs to the the topic of flux calculation in a uniform field, where also no net charge enclosed that surface. The integral in Eq. 1.9 essentially counts the number of field lines passing through the surface. It is entirely reasonable that the number of field lines passing through a surface should be proportional to the net charge enclosed by the surface.

1.5 Coulomb's Law and Gauss's Law

Coulomb's law can be deduced from Gauss's law. Consider a group of positive charge of amount $+q$. We can apply Gauss's law to calculate electric field E at any point situated at a distance r from the charges. To do so, let us consider an imaginary spherical Gaussian surface with radius r by keeping the charges at the center, as it is shown in Fig. 1.8. The electric field lines $\vec{E}$ should certainly be directed radially outward. The direction of electric field $\vec{E}$ and surface element $d\vec{s}$ must be perpendicular to the spherical surface. Therefore, the angle θ between $\vec{E}$ and $d\vec{s}$ is zero everywhere on the surface.

Using Gauss's law

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = \epsilon_0 \int E ds \cos 0^\circ = q,$$

$$\text{or} \quad \epsilon_0 \oint E ds = q.$$

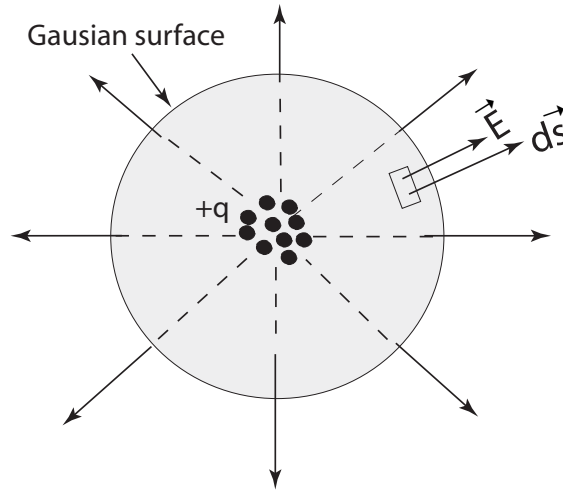


Figure 1.8: A group of positive point charge surrounded by a spherical Gaussian surface.

Since E has the same magnitude for all points on the sphere, it can be termed as constant for integration.

$$\therefore \epsilon_0 E \oint ds = q,$$

where $\oint ds$ is merely the total surface area of the sphere, $4\pi r^2$. We, therefore, obtain after integration

$$\begin{aligned} \epsilon_0 E (4\pi r^2) &= q, \\ \text{or } E &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}. \end{aligned} \quad (1.10)$$

Here, $\frac{1}{4\pi\epsilon_0}$ is a constant and its value is $8.99 \times 10^9 \text{ N-m}^2/\text{C}^2$.

Equation 1.10 gives the magnitude of the electric field $\vec{E}$ at any point a distance r from an isolated point charge q and is identical the result of Coulomb's law. Thus by choosing a Gaussian surface with the proper symmetry, one can obtain Coulomb's law from Gauss's law. The two laws can be regarded as equivalent for our application, *Gauss's law is more generally applicable and is regarded as more fundamental equation of electromagnetism.*

1.6 Applications of Gauss's Law

Gauss's law is suitable to calculate $\vec{E}$ if the symmetry of the charge distribution is high enough. One example of this calculation, the field of a point charge, has already been discussed in section 1.5. Next we will discuss some other examples.

1.6.1 Infinite line of charge

Figure 1.9 shows a part of infinite line of charge of constant linear charge density λ . That is, λ is the charge per unit length. We have to calculate E at a distance r from the line. Since the charge distribution is infinitely long, the electric field always be perpendicular to the charge distribution and having cylindrical symmetry. For this argument we can choose a cylindrical Gaussian surface of radius r and length h .

The field $\vec{E}$ is constant with perpendicular direction of line of charge. Now apply Gauss's law,

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q.$$

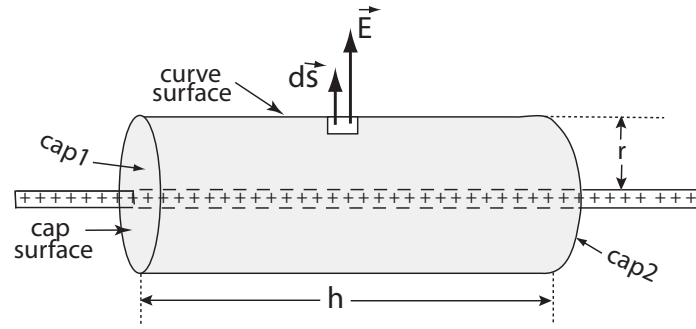


Figure 1.9: Infinite line of charge surrounded by a spherical Gaussian surface.

The left hand side can be subdivided as the integral

$$\begin{aligned} \epsilon_0 \left(\int_{\text{curve}} E ds \cos 0^\circ + \int_{\text{cap1}} E ds \cos 90^\circ + \int_{\text{cap2}} E ds \cos 90^\circ \right) &= q, \\ \text{or } \epsilon_0 E (2\pi r h) + 0 + 0 &= \lambda h, \\ \text{or } E &= \frac{\lambda}{2\pi\epsilon_0 r}. \end{aligned} \quad (1.11)$$

Note that the calculation is much simpler than that of other integration method.

1.6.2 Infinite sheet of charge

Figure 1.10 shows a portion of a thin, nonconducting, infinite sheet of charge of constant positive surface charge density σ (charge per unit area). We have to calculate the electric field at points near the sheet. let us choose a conventional cylindrical surface of cross sectional area S as it is shown in Fig. 1.10.

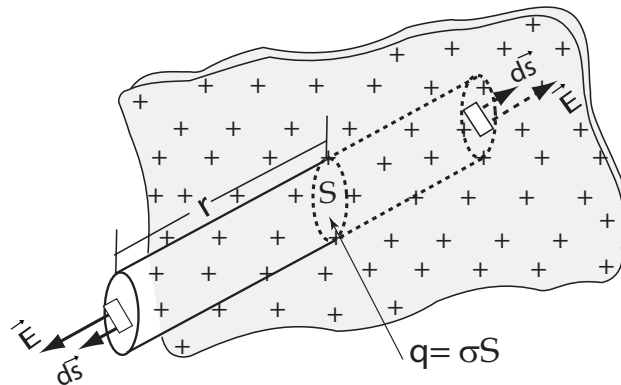


Figure 1.10: An infinite sheet of charge density σ . A conventional cylindrical Gaussian surface is considered here.

Since the sheet is infinitely extended, the electric field should be directed perpendicularly from the plane of the sheet. Only the flux regarding to the two cap surfaces will be effective in the Gaussian surface. Gauss's law gives

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q.$$

Now the surface integral of the L.H.S can be written as

$$\epsilon_0 \left(\int_{\text{curve}} E ds \cos 90^\circ + \int_{\text{cap1}} E ds \cos 0^\circ + \int_{\text{cap2}} E ds \cos 0^\circ \right) = \sigma S,$$

$$\begin{aligned}
\text{or } \epsilon_0(0 + ES + ES) &= \sigma S, \\
\text{or } 2\epsilon_0 ES &= \sigma S, \\
\text{or } E &= \frac{\sigma}{2\epsilon_0}.
\end{aligned} \tag{1.12}$$

Note that E is independent to the distance of the points from the charge sheet. So E is the same for all points on each side of the sheet. Although an infinite sheet of charge cannot exist physically, the derivation of Eq. 1.12 gives approximately correct results for real (not infinite) sheets.

1.6.3 A spherical shell of charge

Figure 1.11 shows a thin spherical shell on which a charge q is uniformly distribution. We calculate E for (i) $r > R$ and (ii) $r < R$.

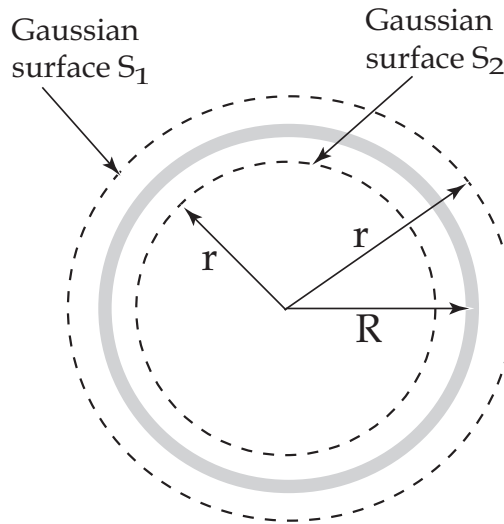


Figure 1.11: A spherical shell of charge q with radius R . Gaussian surface with radius $r > R$ and Gaussian surface with radius $r < R$.

(i) First we calculate E at a distance when $r > R$. In this condition we constitute a Gaussian surface S_1 outside the charge distribution. Applying Gauss's law

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q.$$

For symmetric charge distribution we consider E is also symmetric. For $r > R$ Gauss's law gives

$$\begin{aligned}
\epsilon_0 E (4\pi r^2) &= q \\
\text{or } E_{r>R} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2},
\end{aligned} \tag{1.13}$$

which gives the same result with Eq. 1.10. Thus the uniformly charged shell behaves like a point or group of point charges for all points outside the shell.

(ii) Now we calculate E at a distance when $r < R$. We constitute a Gaussian surface S_2 inside the charge distribution. Applying Gauss's law

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q.$$

Since there is no charge enclosed by the Gaussian surface, that is $q = 0$ for Gaussian surface S_2 . So we have E for inside the shell is as,

$$E_{r<R} = 0. \tag{1.14}$$

The electric field therefore vanishes inside a uniform shell of charge.

1.6.4 Spherical symmetric charge distribution

Figure 1.12 shows a spherical charge distribution with charge density ρ (the charge per unit volume). Radius of the sphere be R . Let us consider the charge density ρ is not constant everywhere of the sphere, but restrict that ρ at any point be radial symmetry. Let us find an expression of E for points (a) outside and (b) inside the charge distribution.

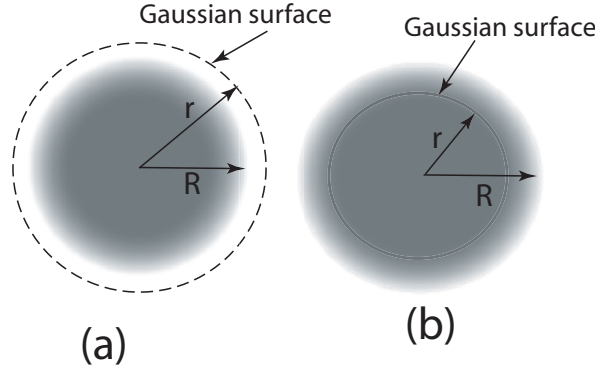


Figure 1.12: A spherical charge distribution of radius R . (i) Gaussian surface with radius $r > R$, (ii) Gaussian surface with radius $r < R$.

(i). Let us consider a spherical Gaussian surface of radius r , where $r > R$ as shown in Fig. 1.12a. Now using Gauss's law we have

$$\begin{aligned}\epsilon_0 \oint \vec{E} \cdot d\vec{s} &= q, \\ \text{or } \epsilon_0 E(4\pi r^2) &= q, \\ \text{or } E_{r>R} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}\end{aligned}\quad (1.15)$$

We calculate q as

$$q = \rho \left(\frac{4}{3} \pi R^3 \right).$$

Then Eq. 1.15 gives the form as

$$E_{r>R} = \frac{\rho R^3}{3\epsilon_0 r^2}. \quad (1.16)$$

(ii). Let us consider a spherical Gaussian surface of radius r , where $r < R$ as shown in Fig. 1.12b. Now applying Gauss's law we have

$$\begin{aligned}\epsilon_0 \oint \vec{E} \cdot d\vec{s} &= q', \\ \text{or } \epsilon_0 E(4\pi r^2) &= q', \\ \text{or } E_{r<R} &= \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2}.\end{aligned}\quad (1.17)$$

We calculate q' as

$$q' = \rho \left(\frac{4}{3} \pi r^3 \right).$$

Then Eq. 1.17 gives the form as

$$E_{r<R} = \frac{\rho r}{3\epsilon_0}. \quad (1.18)$$

We see if $r = 0$, then E vanishes at the center of the sphere. Let us consider a point on the surface of the sphere. That is $r = R$, then Eqs. 1.16 and 1.18 coincide the same result.

1.7 Gauss Law for Dielectric Materials

In this section, it is discussed how an electric field modified in a capacitor and calculated Gauss's law for a dielectric material. This modification is occurred due to polarization which is not the same as in vacuum.

Let consider an external electrical field E_0 is applied in a parallel plate capacitor without dielectric [Fig. 1.13(a)]. Positive charges $+q$ and negative charges $-q$ are accumulated in the two different plates as it is shown in Fig. 1.13 Consider the Gaussian surface of positive charges.

The Gauss's law in vacuum:

$$\text{Differential form} \quad \vec{\nabla} \cdot \vec{E}_0 = \frac{\rho}{\epsilon_0}. \quad (1.19)$$

$$\text{Integration form} \quad \oint \vec{E}_0 \cdot d\vec{S} = \frac{q}{\epsilon_0}. \quad (1.20)$$

From Eq. 1.20

$$E_0 A = \frac{q}{\epsilon_0},$$

The electric field in vacuum

$$E_0 = \frac{q}{\epsilon_0 A}. \quad (1.21)$$

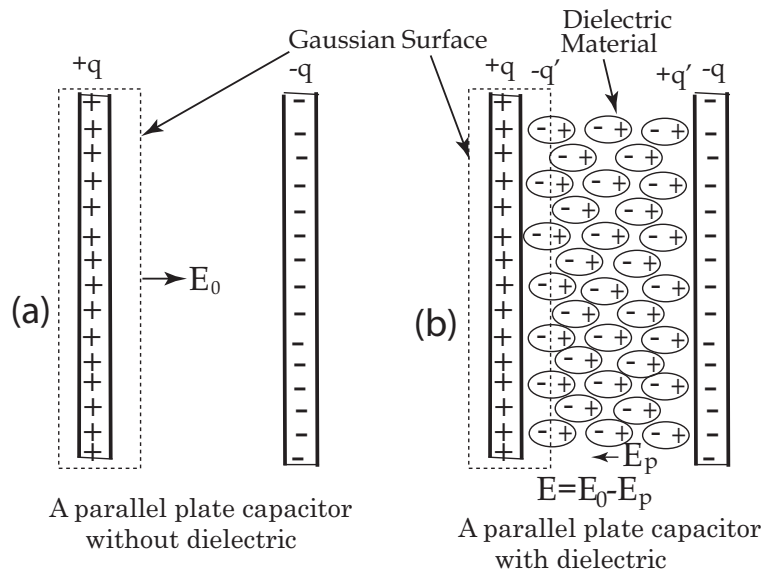


Figure 1.13: A parallel plate capacitor (a) without dielectric, (b) with dielectric.

Consider a dielectric material of dielectric constant κ and permittivity¹ ϵ is inserted between the plates [Fig. 1.13(b)]. There is a polarization of charges occurred in the dielectric medium. As a result induced charges $-q'$ and $+q'$ are accumulated near the surfaces of the plates.

In the present of dielectric, the Gaussian surface encloses two types of charge: Free charge $+q$ and induced charge $-q'$. Let E be the resultant field within the dielectric. According to Gauss's law

$$\begin{aligned} \oint \vec{E} \cdot d\vec{S} &= \frac{q - q'}{\epsilon_0}, \\ EA &= \frac{q - q'}{\epsilon_0}, \end{aligned} \quad (1.22)$$

¹Permittivity is the ability to resist external electric field of a dielectric. This means a substance with high permittivity requires high external electric field to polarize.

$$E = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}. \quad (1.23)$$

The induced charges ($-q'$ and $+q'$ near both the plates) produce their own field which opposes the external field E_0 . It is clear that $E_0 > E$. Now, $E_0/E = \kappa$, where κ is dielectric constant. Therefore, from Eq. 1.21

$$\kappa = \frac{q}{\epsilon_0 A E} \quad \text{or} \quad E = \frac{q}{\epsilon_0 A \kappa}.$$

Now from Eq. 1.23

$$\begin{aligned} \frac{q}{\epsilon_0 A \kappa} &= \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A} \\ \frac{q}{\kappa} &= q - q' \\ q' &= q \left(1 - \frac{1}{\kappa} \right). \end{aligned} \quad (1.24)$$

Equation 1.24 shows that the induced charge q' is less than the free charge q . Substituting the value $q - q' = \frac{q}{\kappa}$ in Eq. 1.22 one get

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0 \kappa} \quad (1.25)$$

Now we known, the dielectric constant is the ratio of permittivity of medium to that of the vacuum i.e, $\kappa = \epsilon/\epsilon_0$. From Eq. 1.25

$$\begin{aligned} \oint \vec{E} \cdot d\vec{S} &= \frac{q}{\epsilon} \\ \text{or} \quad \oint \epsilon \vec{E} \cdot d\vec{S} &= q \\ \text{or} \quad \oint \vec{D} \cdot d\vec{S} &= q \end{aligned} \quad (1.26)$$

where $\epsilon E = D$ and is called the displacement vector. Equation 1.26 is the Gauss's law in the presence of dielectric.

1.7.1 Three electric vectors

Electric displacement denoted by D is the charge per unit area that would be displaced across a layer of conductor placed across an electric field. It is also known as electric flux density. Here total charge density $\rho = \rho_f + \rho_b$, where ρ_f is the free charge density and ρ_b bound charge density due two polarization. Now $\rho_b = -\vec{\nabla} \cdot \vec{P}$, the polarization $\vec{P}$ is defined to be a vector field whose divergence yields the density of bound charges ρ_b in the material.

Differential form of Gauss's law for dielectric

$$\begin{aligned} \vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\epsilon_0}, \\ \vec{\nabla} \cdot \vec{E} &= \frac{\rho_f + \rho_b}{\epsilon_0}, \\ \vec{\nabla} \cdot \vec{E} &= \frac{1}{\epsilon_0}(\rho_f - \vec{\nabla} \cdot \vec{P}), \\ \vec{\nabla} \cdot (\epsilon_0 \vec{E} + \vec{P}) &= \rho_f, \end{aligned} \quad (1.27)$$

$$\therefore \vec{\nabla} \cdot \vec{D} = \rho_f, \quad (1.28)$$

where $\vec{D}$ is called the displacement vector. Therefore,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad (1.29)$$

The relation of Eq. 1.29 is called **three electric vectors**.

Polarization is proportional to the electric field. So, $\vec{P} = \epsilon_0 \chi \vec{E}$, where χ is called the electric susceptibility².

$$\begin{aligned} \vec{D} &= \epsilon_0 \vec{E} + \epsilon_0 \chi \vec{E} \\ \vec{D} &= \epsilon_0 (1 + \chi) \vec{E} \\ \vec{D} &= \epsilon_0 \epsilon_r \vec{E} \\ \vec{D} &= \epsilon \vec{E}, \end{aligned} \quad (1.30)$$

where $\epsilon_r = (1 + \chi)$ is called the relative permittivity³.

1.8 Examples

Example 1.1: Consider charge is distributed uniformly throughout infinitely long solid cylinder of radius R and charge density ρ . (a) Find E at a distance r from the cylinder axis when $r < R$, and (b) E at a distance r from the cylinder axis when $r > R$.

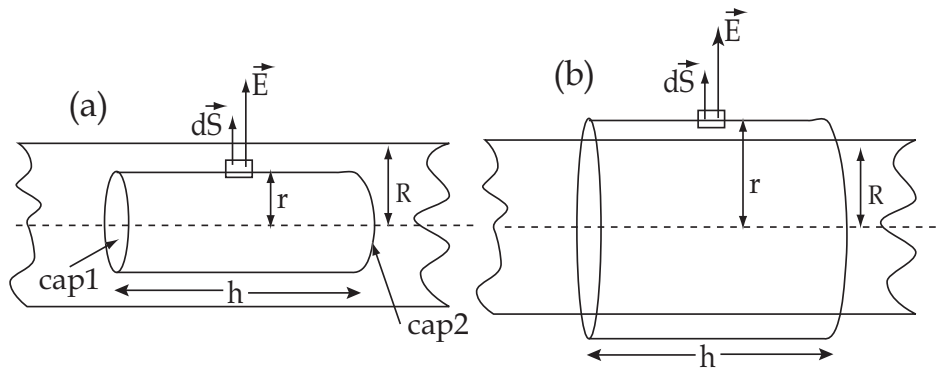


Figure 1.14: A cylindrical charge distribution of radius R . (a) Gaussian surface with radius $r < R$, (b) Gaussian surface with radius $r > R$.

Solution: (a) Consider a cylindrical Gaussian surface of radius r ($r < R$) and length h as it is shown in Fig. 1.14a. Charge enclosed by the Gaussian surface is $q = \pi r^2 h \rho$. Now using Gauss' law

$$\begin{aligned} \epsilon_0 \oint \vec{E} \cdot d\vec{s} &= q, \\ \text{or} \quad \epsilon_0 \int_{\text{cap1}} \vec{E} \cdot d\vec{s} + \epsilon_0 \int_{\text{cap2}} \vec{E} \cdot d\vec{s} + \epsilon_0 \int_{\text{curve}} \vec{E} \cdot d\vec{s} &= \pi r^2 h \rho, \\ \text{or} \quad \epsilon_0 \int_{\text{cap1}} E ds \cos 90^\circ + \epsilon_0 \int_{\text{cap2}} E ds \cos 90^\circ + \epsilon_0 \int_{\text{curve}} E ds \cos 0^\circ &= \pi r^2 h \rho, \\ \text{or} \quad 0 + 0 + \epsilon_0 \int_{\text{curve}} E ds &= \pi r^2 h \rho, \end{aligned}$$

²Susceptibility is defined as the ability to polarize of a dielectric. The greater the electric susceptibility, the greater the ability of a material to polarize in response to the field.

³The permittivity is often represented by the relative permittivity ϵ_r which is the ratio of the permittivity ϵ in the dielectric and permittivity ϵ_0 in the vacuum. i.e., $\epsilon_r = \epsilon/\epsilon_0$

$$\text{or} \quad \epsilon_0 E(2\pi rh) = \pi r^2 h \rho,$$

$$\therefore E = \frac{\pi r^2 h \rho}{\epsilon_0 2\pi rh} = \frac{\rho r}{2\epsilon_0}.$$

(b) Similarly consider a cylindrical Gaussian surface of radius r ($r > R$) and length h as it is shown in Fig. 1.14b. Here charge enclosed by the Gaussian surface is $q = \pi R^2 h \rho$. Now using Gauss' law

$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q = \pi R^2 h \rho$$

we have by the same calculation as before

$$\epsilon_0 \int_{cap1} \vec{E} \cdot d\vec{s} + \epsilon_0 \int_{cap2} \vec{E} \cdot d\vec{s} + \epsilon_0 \int_{curve} \vec{E} \cdot d\vec{s} = \pi R^2 h \rho,$$

$$\text{or} \quad \epsilon_0 \int_{cap1} E ds \cos 90^\circ + \epsilon_0 \int_{cap2} E ds \cos 90^\circ + \epsilon_0 \int_{curve} E ds \cos 0^\circ = \pi R^2 h \rho,$$

$$\text{or} \quad 0 + 0 + \epsilon_0 \int_{curve} E ds = \pi R^2 h \rho,$$

$$\text{or} \quad \epsilon_0 E(2\pi rh) = \pi R^2 h \rho,$$

$$\therefore E = \frac{\pi R^2 h \rho}{\epsilon_0 2\pi rh} = \frac{\rho R^2}{2\epsilon_0 r}.$$

Example 1.2: Gauss' law for gravitation is

$$\frac{1}{4\pi G} \Phi_g = \frac{1}{4\pi G} \oint \vec{g} \cdot d\vec{s} = m,$$

where m is the enclosed mass and G is the universal gravitation constant. Derive Newton's law of gravitation from this. What is the significance of negative sign?

Solution: Consider the mass m enclosed by a spherical Gaussian surface surface of radius r .

$$\text{Then} \quad \frac{1}{4\pi G} \Phi_g = \frac{1}{4\pi G} \oint \vec{g} \cdot d\vec{s} = \frac{1}{4\pi G} \oint g ds \cos 180^\circ = m,$$

$$\text{or} \quad -\frac{g}{4\pi G} \oint ds = m,$$

$$\text{or} \quad -\frac{g \cdot 4\pi r^2}{4\pi G} = m,$$

$$\text{or} \quad g = -\frac{Gm}{r^2},$$

Let a mass M is placed in the gravitational field g , which experienced a force F is found by

$$F = Mg = -\frac{GMm}{r^2},$$

which is Newton's law of gravitation. The negative sign indicates that the force is attractive.

Example 1.3: A proton orbits with a speed $v = 294 \text{ km/s}$ just outside a charged sphere of radius $r = 11.3 \text{ cm}$. Find the charge on the sphere.

Solution: Let the charge on the sphere be Q . Then the Coulomb's force between the proton (with charge q) and the sphere is

$$F_{Coul} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2}.$$

Now the centripetal force

$$F_c = \frac{mv^2}{r},$$

where m is the mass of the proton. For electrostatic equilibrium

$$F_{Coul} = F_c,$$

$$\text{or } \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} = \frac{mv^2}{r},$$

$$\text{or } Q = \frac{4\pi\epsilon_0 mv^2 r}{q},$$

$$= \frac{4\pi(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(1.67 \times 10^{-27} \text{ kg})(294 \times 10^3 \text{ m/s})^2(0.0113 \text{ m})}{1.6 \times 10^{-19} \text{ C}} = 1.1338 \times 10^{-9} \text{ C}.$$

Example 1.4: A solid nonconducting sphere of radius R carries a nonuniform charge distribution, with charge density $\rho = \rho_s r/R$, where ρ_s is a constant and r is distance from the center of the sphere. Show that (a) the total charge of the sphere is $Q = \pi\rho_s R^3$, and (b) the electric field inside the sphere is given by $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^4} r^2$.

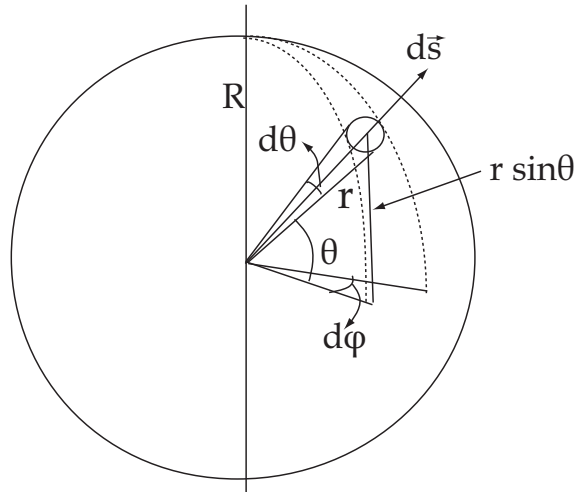


Figure 1.15: A non uniform cylindrical charge distribution with charge density $\rho = \rho_s r/R$, where ρ_s is constant.

Solution: (a) The total charge is the volume integral over the whole sphere, Therefore $Q = \int \rho dV$. From Fig. 1.15 we have a surface element $ds = (r d\theta)(r \sin \theta d\phi)$. So we have a volume element as

$$dV = (dr)(ds) = (dr)(r d\theta)(r \sin \theta d\phi)$$

$$\therefore Q = \int \rho dV$$

$$= \int \left(\frac{\rho_s r}{R} \right) (dr)(r d\theta)(r \sin \theta d\phi)$$

$$\begin{aligned}
&= \frac{\rho_s}{R} \int_0^R r^3 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = \frac{\rho_s}{R} \int_0^R r^3 dr [-\cos \theta]_0^\pi [\phi]_0^{2\pi} \\
&= \frac{4\pi\rho_s}{R} \int_0^R r^3 dr = \frac{4\pi\rho_s}{R} \left[\frac{r^4}{4} \right]_0^R \\
&= \frac{\pi\rho_s R^4}{R} = \pi\rho_s R^3.
\end{aligned}$$

(b) Let us now consider a Gaussian surface of radius r ($r < R$). Therefore, charge enclosed by the Gaussian surface can be calculated

$$\begin{aligned}
q' &= \int \rho dV' \\
&= \frac{\rho_s}{R} \int_0^r r^3 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = \frac{\rho_s}{R} \int_0^r r^3 dr [-\cos \theta]_0^\pi [\phi]_0^{2\pi} \\
&= \frac{4\pi\rho_s}{R} \int_0^r r^3 dr = \frac{4\pi\rho_s}{R} \left[\frac{r^4}{4} \right]_0^r \\
&= \frac{\pi\rho_s r^4}{R}.
\end{aligned}$$

Now using Gauss' law

$$\begin{aligned}
&\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q', \\
&\text{or} \quad \epsilon_0 E (4\pi r^2) = \frac{\pi\rho_s r^4}{R}, \\
&\text{or} \quad E = \frac{\pi\rho_s r^4}{4\pi\epsilon_0 r^2 R} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^4} r^2.
\end{aligned}$$

Chapter 2

ELECTRIC POTENTIAL

Many electrical phenomena are associated with the transfer of large quantities of energy. For example, when a lightning flash strikes the Earth from a cloud, an energy of typically 10^8 J is released in the form of light, sound, heat, and shock wave. Where does this energy comes from and how is it stored in clouds? To understand this question, we must consider the energy associated with electrical force. The electrostatic force law is very similar to the gravitational force law:

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2} \quad (2.1)$$

$$F = G \frac{m_1 m_2}{r^2} \quad (2.2)$$

The difference of the forces is that the gravitational force is always attractive, whereas electrostatic force can be either attractive or repulsive. Both forces depends on the inverse square of the separation distance between the two objects. When an object moves from place to place under the gravitational force of another object, the work done is occurred by the gravitational force. Similarly when an electric charge moves from one place to another place, a work done is performed under the electrical force. The forces are conservative, because the work done does not depends on the path of the travelling. For a conservative force we could define a potential energy. The difference in potential energy ΔU of the system as the object moves from its initial to its final position is equal to the negative of the work done by the force:

$$\Delta U = U_f - U_i = -W_{if} = - \int_i^f \vec{F} \cdot d\vec{l}, \quad (2.3)$$

where W_{if} is the work done by the force $\vec{F}$ when the object moves from place i to f. The electrostatic force is conservative, and therefore there is a potential energy associated with the configuration of a system in which electrostatic forces act.

2.1 Potential Difference and Potential

Imagine a charge q fixed at the origin of a coordinate system, as it is shown in Fig. 2.1. Let a test charge q_0 is to move from point 'a' to point 'b'. The distance of the two points be r_a and r_b , respectively. Consider $q > q_0$, therefore we have to apply a working force which will be equal to Coulomb's force but in opposite direction. The change in potential energy ΔU of the system can be calculated by Eq. 2.3

$$\Delta U = -W_{ab} = - \int_a^b \vec{F} \cdot d\vec{l} = - \int_a^b F dl \cos 180^\circ = + \int_a^b F dl.$$

But $dl = -dr$, and we have

$$\Delta U = -W_{ab} = - \int_{r_a}^{r_b} F dr = - \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} dr = \frac{1}{4\pi\epsilon_0} qq_0 \left(\frac{1}{r_b} - \frac{1}{r_a} \right), \quad (2.4)$$

$$\text{or} \quad \frac{\Delta U}{q_0} = \frac{-W_{ab}}{q_0} = V_b - V_a, \quad (2.5)$$

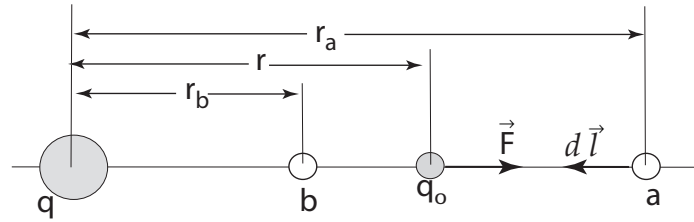


Figure 2.1: A test charged particle q_0 is moving from a to b under the influence of the electrostatic force $\vec{F}$ exerted by q .

$$\text{or} \quad \frac{\Delta U}{q_0} = \Delta V. \quad (2.6)$$

We define the electric potential difference ΔV to be the electrical potential energy difference per unit test charge.

Let us consider the point 'a' is placed at infinite distance. That is $r_a = \infty$. From Eq. 2.4 we have

$$\frac{\Delta U}{q_0} = \frac{-W_{ab}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r_b}, \quad (2.7)$$

which gives the potential difference per unit test charge is equal to the potential at point 'b'. Therefore, taking the magnitude of Eq. 2.4 we can write in general form

$$\frac{W}{q_0} = V. \quad (2.8)$$

$$\text{If } q_0 = 1 \text{ then} \quad W = V. \quad (2.9)$$

Therefore, the potential at any point situated in an electric field is defined as the work done in moving a unit test charge from infinite distance to that point.

2.2 Relation Between E and V

The interactions of a charge can be characterized using four different properties: electric force, electric field, electric potential energy and electric potential. Here we try to relate electric potential to electric field. The connection between V and $\vec{E}$ follows directly from the definition of potential in Eqs. 2.5 and 2.6

$$\begin{aligned} \Delta V &= \frac{-W_{ab}}{q_0} = \frac{-\int_a^b \vec{F} \cdot d\vec{l}}{q_0} = \frac{-\int_a^b q_0 \vec{E} \cdot d\vec{l}}{q_0}, \\ \text{or} \quad \Delta V &= V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{l}. \end{aligned} \quad (2.10)$$

Let the point 'a' to be at infinity, then $V_a = 0$ and the potential at point 'b' is given by

$$V_b = -\int_{\infty}^b \vec{E} \cdot d\vec{l}. \quad (2.11)$$

2.2.1 Potential when E is uniform

Suppose we move a test charge q_0 (Fig. 2.2) through a uniform electric field $\vec{E}$ from a to b along the path acb . We try to find potential difference between a to b . For the path ac we have potential difference from Eq. 2.10,

$$\begin{aligned} V_c - V_a &= - \int_a^c E \, dl \cos(\pi - \theta) \\ &= E \cos \theta \int_a^c dl = E \cos \theta \cdot ac = E \cos \theta \cdot \frac{L}{\cos \theta} = EL. \end{aligned}$$

Potential difference between c to b is

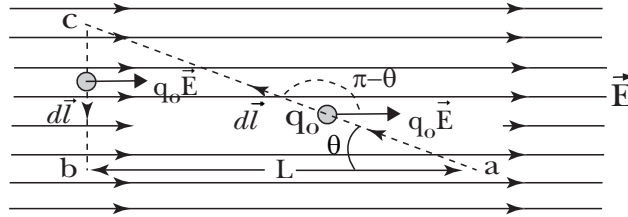


Figure 2.2: Consider a uniform electric field E . A test charge q_0 have to move from point a to point b .

$$V_b - V_c = - \int_c^b E \, dl \cos 90^\circ = 0.$$

Therefore, potential difference between a to b is given by

$$V_b - V_a = (V_c - V_a) + (V_b - V_c) = EL + 0 = EL.$$

This the same value if we follow the direct path a to b , because we know that the potential difference between two points is independent of the path.

2.2.2 Potential when E is not uniform

Let a test charge q_0 is to move from a arbitrary point ' a ' to ' b ' as it is shown in Fig. 2.3. The electric field exerts a force $q_0 \vec{E}$ on the test charge q_0 . To keep the test charge from accelerating, an external agent must be applied a force $\vec{F}$ chosen to be exactly equal to $-q_0 \vec{E}$ for all position of the test

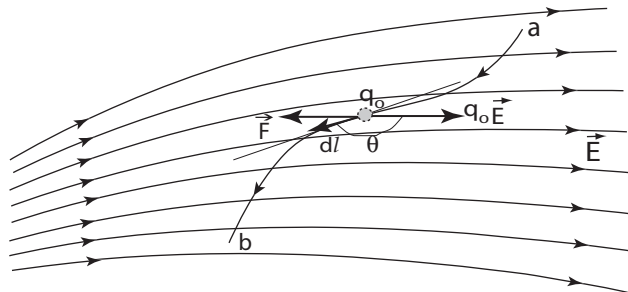


Figure 2.3: A non-uniform electric field E . A test charge q_0 has to move from point ' a ' to point ' b '.

charge. To find the total work done W_{ab} be the external agent in moving the test charge from a to b is given as

$$W_{ab} = \int_a^b -q_0 \vec{E} \cdot d\vec{l}.$$

Potential difference between the points a and b

$$V_b - V_a = \frac{W_{ab}}{q_0} = - \int_a^b \vec{E} \cdot d\vec{l} = - \int_a^b E dl \cos \theta.$$

Consider a is at infinity, then $V_0 = 0$. Hence

$$V_b = V = - \int_{\infty}^b \vec{E} \cdot d\vec{l}.$$

2.3 Calculation of Potential

2.3.1 Potential due to a point charge

A point charge rendered the electric field spherically outward as shown in Fig. 2.4. Let a test charge q_0 is to move from a arbitrary point a to b. The potential difference between a to b can be calculated by using Eq. 2.10 and written as

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l} = - \int_a^b E \cdot dl \cos \theta.$$

Here the angle θ between $\vec{E}$ and $d\vec{s}$ along the line ab is 180° .

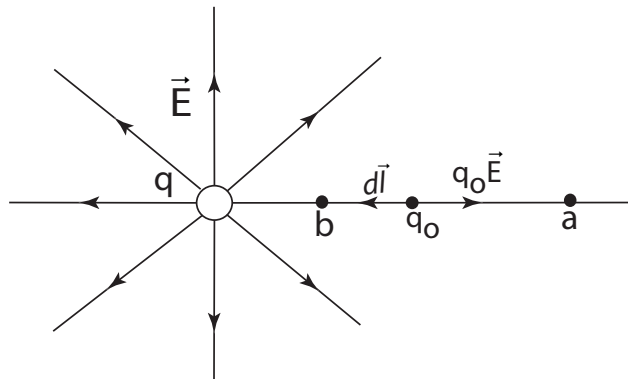


Figure 2.4: Consider a point charge and electric field E is directed radially outward. A test charge q_0 have to move from point a to point b.

$$\therefore V_b - V_a = - \int_a^b E \cdot dl \cos 180^\circ = + \int_a^b E dl.$$

Since the electric field is directed spherically outward,

$$\therefore dl = -dr.$$

$$\begin{aligned} \therefore V_b - V_a &= + \int_a^b E dl = - \int_{r_a}^{r_b} E dr = - \int_{r_a}^{r_b} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr \\ &= - \frac{q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = + \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_b} - \frac{1}{r_a} \right). \end{aligned}$$

Let us consider A is placed at infinity, then $\frac{1}{r_a} = 0$ and $V_a = 0$, and we put $V_b = V$ and $\frac{1}{r_b} = \frac{1}{r}$. Now we have

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}. \quad (2.12)$$

Actually the expressions shown by the Eqs. 2.12 and 2.7 are the same.

2.3.2 Potential due to a collection of point charges

Suppose we have a collection of point charges $q_1, q_2, \dots, q_n$ located at various fixed points. We wish to find the potential at any arbitrary point P due to these collection of charges. We can find the potential at P by the sum of the potentials due to individual point charges.

Thus we have

$$V = V_1 + V_2 + \dots + V_n = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} + \dots + \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_n}, \quad (2.13)$$

which can be written as compact form

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}. \quad (2.14)$$

The expression indicates that q_i is the magnitude of the i th charge and r_i is the distance of the i th charge from the point P .

2.3.3 Potential due to a dipole

The configuration of two equal but opposite charges separated by a finite distance is called an electric dipole. In Fig. 2.5 two equal charges, one positive $+q$ and another negative $-q$ are separated by a distance d . The configuration is said to be an electric dipole.

The potential due to an electric dipole can be calculated in a straightforward way using Eq. 2.14. We place the origin at the center of the dipole, and we seek the potential at point P , which is located a distance r from the center of the dipole. Let the angle between r and y axis be θ . The

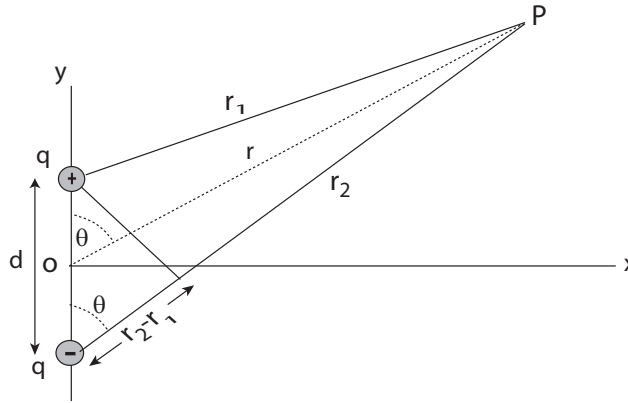


Figure 2.5: A configuration of electric dipole. Two equal charges $+q$ and $-q$ are separated by a distance d . O is the origin of the configuration.

distances from the positive and negative charges to P are r_1 and r_2 respectively. Using Eq. 2.14, we find the potential at point P to be

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{+q}{r_1} + \frac{-q}{r_2} \right). \quad (2.15)$$

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{r_2 - r_1}{r_1 r_2} \right). \quad (2.16)$$

Equation 2.15 is the exact expression for the potential due to a dipole. However, in many applications it is considered that the point P is very far from the dipole, compared with the distance d between the charges; that is $r \gg d$.

In this assumption,
 $r_2 - r_1 \approx d \cos \theta$ and $r_1 r_2 \approx r^2$, Now Eq. 2.16 gives the result

$$V = \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}, \quad (2.17)$$

where $p = qd$ and is called electric dipole moment. Dipole moment p is a vector whose direction is from $-q$ to $+q$. Equation 2.17 gives the potential due to a dipole at any point in space.

Note that potential due to dipole varies as $1/r^2$ which is in contrast to the potential for a single charge which varies (see Eq. 2.12) as $1/r$. If $\theta = 90^\circ$, Eq. 2.17 gives the potential along the x axis as $V = 0$. Therefore, potential along the perpendicular bisector of an electric dipole is zero. For a given r , the potential varies from positive values on the positive y axis (for $\theta = 0$) to zero in the x axis (for $\theta = 90^\circ$) to negative values on the negative y axis (for $\theta = 180^\circ$). Though $V = 0$ in the x axis but actually $E \neq 0$.

We know that there is a relation between Electric field E and Electric potential V . So, if one know the potential at any point of an electric field, he can also calculate the electric field easily from that expression of potential. The relation between Electric field E and Electric potential V is

$$E = -\frac{dV}{dr}.$$

Substituting the value of V of Eq. 2.17 we have

$$\begin{aligned} E &= -\frac{d}{dr} \left(\frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \right), \\ \text{or } E &= -\left(\frac{p \cos \theta}{4\pi\epsilon_0} \right) \frac{d}{dr} (r^{-2}), \\ \text{or } E &= \frac{p \cos \theta}{2\pi\epsilon_0} \frac{1}{r^3}. \end{aligned} \quad (2.18)$$

Note: We see from Eqs. 2.17 and 2.18, for an electric dipole, $V \propto \frac{1}{r^2}$ and $E \propto \frac{1}{r^3}$.

2.3.4 Potential due to a quadrupole

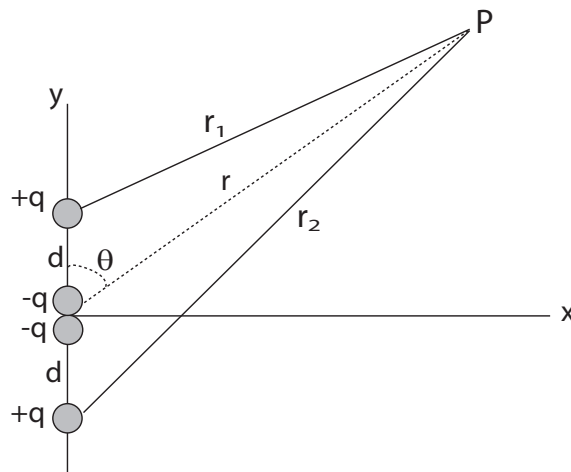


Figure 2.6: A configuration of electric quadrupole.

When two electric dipole are placed closely like as shown in Fig. 2.6, the configuration is called an *electric quadrupole*. We try to find electric potential at point P due to quadrupole. Let we choose the origin at the center of the quadrupole. The potential at P due to this quadrupole is

$$V(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_1} - \frac{2q}{r} + \frac{q}{r_2} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} - \frac{2}{r} \right). \quad (2.19)$$

Using cosine law, we can write with respect to Fig. 2.6,

$$r_1^2 = r^2 + d^2 - 2rd \cos \theta$$

$$\text{Hence, } \frac{1}{r_1} = \frac{1}{(r^2 + d^2 - 2rd \cos \theta)^{1/2}} = \frac{1}{r} \left[1 + \left(\frac{d^2}{r^2} - \frac{2d}{r} \cos \theta \right) \right]^{-1/2}.$$

Since $r \gg d$ we can expand by Binomial theorem, as,

$$(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3x^2}{8} \dots\dots\dots$$

$$\text{Hence, } \frac{1}{r_1} = \frac{1}{r} \left[1 - \frac{1}{2} \left(\frac{d^2}{r^2} - \frac{2d}{r} \cos \theta \right) + \frac{3}{8} \left(\frac{d^2}{r^2} - \frac{2d}{r} \cos \theta \right)^2 \dots\dots\dots \right]$$

Neglecting higher order terms we have

$$\frac{1}{r_1} = \frac{1}{r} \left[1 + \frac{d^2}{2r^2} (3 \cos^2 \theta - 1) + \frac{d}{r} \cos \theta \right],$$

$$\text{and similarly } \frac{1}{r_2} = \frac{1}{r} \left[1 + \frac{d^2}{2r^2} (3 \cos^2 \theta - 1) - \frac{d}{r} \cos \theta \right].$$

After substituting these values in Eq. 2.19, we find

$$V(r) = \frac{q}{4\pi\epsilon_0} \frac{1}{r} \left[1 + \frac{d^2}{2r^2} (3 \cos^2 \theta - 1) + \frac{d}{r} \cos \theta + 1 + \frac{d^2}{2r^2} (3 \cos^2 \theta - 1) - \frac{d}{r} \cos \theta - 2 \right],$$

$$\text{or } V(r) = \frac{qd^2}{4\pi\epsilon_0 r^3} (3 \cos^2 \theta - 1) \quad (2.20)$$

The product $2qd^2$ is called the electric quadrupole moment and can be denoted by Q , then potential due to electric quadrupole is

$$V(r) = \frac{Q}{8\pi\epsilon_0 r^3} (3 \cos^2 \theta - 1) \quad (2.21)$$

Note that the potential due to quadrupole varies as $1/r^3$, whereas the potential due to dipole varies as $1/r^2$ and the potential for a single charge varies as $1/r$.

<i>Note: Potential</i> $V \propto \frac{1}{r}$, for single charge, $V \propto \frac{1}{r^2}$, for dipole, $V \propto \frac{1}{r^3}$, for quadrupole.	<i>Note: Electric field</i> $E \propto \frac{1}{r^2}$, for single charge, $E \propto \frac{1}{r^3}$, for dipole, $E \propto \frac{1}{r^4}$, for quadrupole.
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2.4 A Dipole in an Electric Field

Figure 2.7 shows an electric dipole placed in a uniform electric field $\vec{E}$. The dipole axis makes an angle θ with the field direction E . Therefore, the charge $+q$ will experience a force $F = qE$ along the direction of field E and the charge $-q$ will experience the same force $F = qE$ but opposite to the direction of field E , as it is shown in Fig. 2.7.

These two equal and opposite forces produce a torque due to the couple of moment and is given by

$$\begin{aligned}\tau &= qE \times d \sin \theta, \\ \text{or } \tau &= qd E \sin \theta.\end{aligned}\quad (2.22)$$

Magnitude qd is defined by dipole moment and denoted by $p = qd$. Therefore, from Eq. 2.22 we have,

$$\tau = pE \sin \theta. \quad (2.23)$$

In vector notation this torque can be written as

$$\vec{\tau} = \vec{p} \times \vec{E}. \quad (2.24)$$

This torque tends to align the dipole to the direction of E . Therefore, if a dipole placed in an external electric field experiences a torque which tends to align it with the direction of applied field.

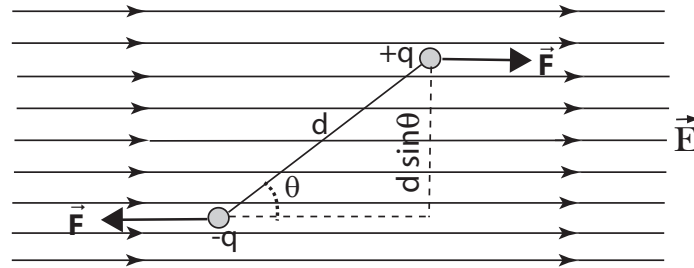


Figure 2.7: Electric dipole in a uniform electric field.

So to keep in position the dipole in the field, an external force equal to qE must be applied to it which can do work. This work done is stored as potential energy U in the system. Let the external force makes an angle $d\theta$ against the torque, then the work done by the force

$$dW = pE \sin \theta d\theta.$$

Therefore, the work done in rotating the dipole with making an angle θ which is stored as potential energy

$$\begin{aligned}U = W &= \int dW = pE \int \sin \theta d\theta, \\ \text{or } U &= -pE \cos \theta,\end{aligned}\quad (2.25)$$

$$\text{or } U = -\vec{p} \cdot \vec{E}. \quad (2.26)$$

Let $\theta = 0$, then from Eq. 2.25 we have

$$U = -pE.$$

Therefore, to keep p and E in parallel, the work done is minimum and stored potential energy is also minimum. Let $\theta = 90$, then from Eq. 2.25 we have potential energy $U = 0$. Let $\theta = 180$, then from Eq. 2.25 we have potential energy $U = pE$, which gives the maximum value.

Chapter 3

MAGNETIC FIELD DUE TO CURRENT

In this chapter we shall study the magnetic field produced by moving charges. We know, usually, a large number of charges move when we say a current is flowing in a wire. We shall investigate in this chapter the magnetic field produced by some simple current distribution say straight wires, circular loops. We shall also see the characteristics of produced field, lines of field and direction of the force lines as well as the amount of magnetic field related to the amount of current.

3.1 Magnetic Field Produced by Current

In 1820 Hans Christian Oersted demonstrated an experiment and observed when a compass is placed near a straight wire carrying a current, like shown in Fig. 3.1, the compass needle aligns so that it is tangent to a circle drawn around the wire. Oersted's discovery provided the first link between electricity and magnetism.

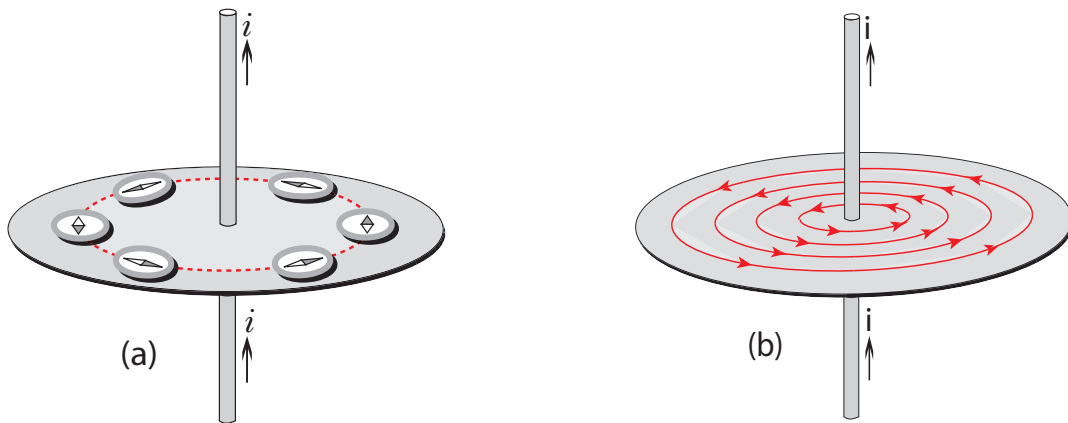


Figure 3.1: (a) Magnetic field produced by current is shown by Oersted's experiment. The direction of the compass is always perpendicular to the direction of the current in the wire. (b) The lines of magnetic field are concentric circles produced by a long straight wire of carrying current i .

3.2 Magnetic Field Due to a Moving Charge

To discuss the physics of produced magnetic field, first we wish to introduce how a single moving charge produces a magnetic field, later we will see why this approach is not practical and why it is easier to produce magnetic field in the laboratory by using moving charges in the form currents in wires.

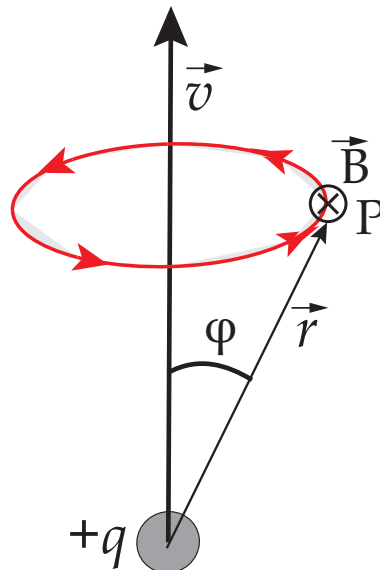


Figure 3.2: The magnetic field at point P due to a moving charge. $\vec{B}$ is directed perpendicularly to the page (going away from the eye sight).

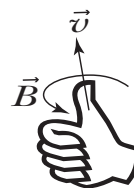
Figure 3.2 shows a moving charge $+q$ sets a magnetic field $\vec{B}$. We can also observe the properties of charge and magnetic field which determine magnitude of B . The properties are

1. The field strength is directly proportional to the magnitude of the velocity v and also to the charge q .
2. If $\vec{v}$ reverses direction or q changes sign, the direction of $\vec{B}$ is reversed.
3. The direction of $\vec{B}$ is tangent to circles drawn on planes perpendicular to the direction of $\vec{v}$, with the direction of $\vec{B}$ determined by the right-hand rule¹, and on any given circle the magnitude of $\vec{B}$ is the same at all points.
4. The magnitude of $\vec{B}$ observed at any point P depends on the distance $\vec{r}$ from the charge q . The magnitude of $B \propto 1/r^2$.
5. The magnitude of $\vec{B}$ depends on $\sin \phi$, where ϕ is the angle between $\vec{v}$ and $\vec{r}$. If $\phi = 0$, then $\sin \phi = 0$. This means the field is zero along the direction of $\vec{v}$. Again if $\phi = 90^\circ$, then $\sin \phi = 1$. This means the field is maximum if $\vec{r}$ is perpendicular to the direction of $\vec{v}$. In Fig. 3.2 the direction of $\vec{B}$ at point P is perpendicular to the plane determined by the vectors $\vec{v}$ and $\vec{r}$. From the above observations we find that the magnitude of $\vec{B}$ is directly proportional to q , v and $\sin \phi$, and inversely proportional to r^2 .

Finally we can write

$$B \propto \frac{qv \sin \phi}{r^2},$$

¹point your thumb in the direction of $\vec{v}$, and your fingers will curl in the direction of $\vec{B}$.



$$\text{or} \quad B = K \frac{qv \sin \varphi}{r^2}, \quad (3.1)$$

where K is a constant of proportionality.

It indicates that φ is the angle between the vectors $\vec{v}$ and $\vec{r}$, and $\vec{B}$ is directed perpendicularly to the plane containing $\vec{v}$ and $\vec{r}$, provided that the direction of B follows the right-hand rule.

The proportionality constant K is defined in SI unit $K = \frac{\mu_0}{4\pi} = 10^{-7}$ tesla.meter/ampere (T.m/A). Hence

$$\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A},$$

where μ_0 is called permeability constant, here simply magnetic constant. The magnetic constant μ_0 plays a role in calculating magnetic field to that of the electric constant ϵ_0 in calculating electric field. The two constant are related through the speed of light as $c = (\epsilon_0 \mu_0)^{-1/2}$. The magnitude of $\vec{B}$ can be written as

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \varphi}{r^2}. \quad (3.2)$$

In vector form Eq. 3.2 can be written as

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \vec{r}}{r^3}. \quad (3.3)$$

where $\hat{r}$ is the unit vector along $\vec{r}$. Equations 3.2 or 3.3 are not practically useful because of difficulties of measuring of q . In practice it is easy to measure i instead of q .

3.3 The Biot-Savart Law

The relations of Eqs. 3.2 or 3.3 are not usually useful. In electrostatics we observed that the electric field produced by charges whose location did not change. Here we are also interested for magnetostatics; the production of steady magnetic fields by charges whose motion does not change.

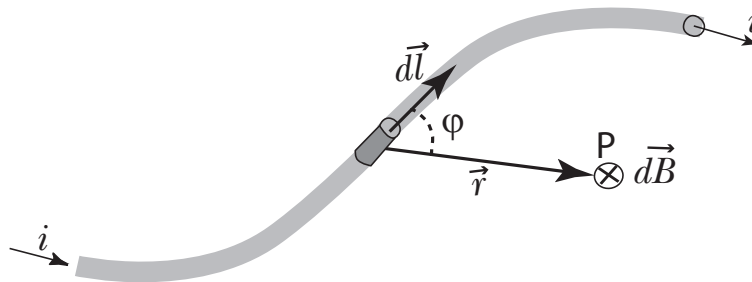


Figure 3.3: The magnetic field $d\vec{B}$ at point P produced by an element $d\vec{l}$ of an wire carrying current i . $d\vec{B}$ is directed perpendicularly to the plane formed by $d\vec{l}$ and $\vec{r}$.

In laboratory we produce magnetic fields using current carrying wires rather than the motion of individual charges, as it is shown in Fig. 3.3. If here we consider a steady-state current in the wire, it is obvious that a certain amount of charge per second passes through any cross sectional area at any instant of time which may create a steady magnetic field at any point near the wire.

Let the current element dl contains an amount of charge dq in the time interval dt , and the steady-state velocity of a charge in the wire be v . Now the relations for dB produced by the moving

charge dq at point P with distance r from the location of charge can be written as

$$\begin{aligned} dB &\propto dq \\ dB &\propto v \\ dB &\propto \sin \phi \\ dB &\propto \frac{1}{r^2} \end{aligned}$$

From the above relations let's find an equation for dB at point P as

$$dB = \frac{\mu_0}{4\pi} \frac{dq v \sin \phi}{r^2}. \quad (3.4)$$

Notice that the Eq. 3.4 is similar to Eq. 3.2.

Now we can write $v = dl/dt$, hence

$$dq v = dq \frac{dl}{dt} = \frac{dq}{dt} dl = i dl.$$

Substituting this in Eq. 3.4, we obtain the Biot-Savart law in scalar form can be rewritten by as follows

$$dB = \frac{\mu_0}{4\pi} \frac{i dl \sin \phi}{r^2}, \quad (3.5)$$

where ϕ is the angle between the vectors dl and r .

The expression of Eq. 3.5 is known as the **Biot-Savart law**. The direction of dB is perpendicular to the plane containing the vectors dl and r . The direction of dB follows the so called right-hand rule. The symbol $\otimes$ shown in Fig. 3.3 indicates the direction of $d\vec{B}$ which is perpendicular to the page going away from the observer.

The Biot-Savart law in vector form

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{l} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{i d\vec{l} \times \vec{r}}{r^3}. \quad (3.6)$$

To calculate the total field at any point P due to the entire current distribution, one has to integrate over all current element $i dl$. Therefore,

$$\text{or} \quad B = \int dB = \frac{\mu_0}{4\pi} \int \frac{i dl \sin \phi}{r^2}. \quad (3.7)$$

3.4 Applications of Biot-Savart law

We now consider how to apply the Biot-Savart law to calculate the magnetic fields of some current-carrying wires of different shapes.

3.4.1 A straight wire segment

Figure 3.4 shows a current segment of length L carrying current i . Consider the wire segment lies along the y axis and current is flowing along $+y$ direction. We have to calculate magnetic field B at point P , a distance d from the current segment. Let us consider a small current element $d\vec{l}$ and suppose d lies along the x axis. The distance between $d\vec{l}$ and point P is r . The contributed magnetic field $d\vec{B}$ will produce at point P due to element $d\vec{l}$. The direction of $d\vec{B}$ involves the vector cross product $d\vec{l} \times \vec{r}$ and should be followed right-hand rule. Therefore, in Fig. 3.4 $d\vec{B}$ can be indicated perpendicularly on the xy plane through $-z$ direction.

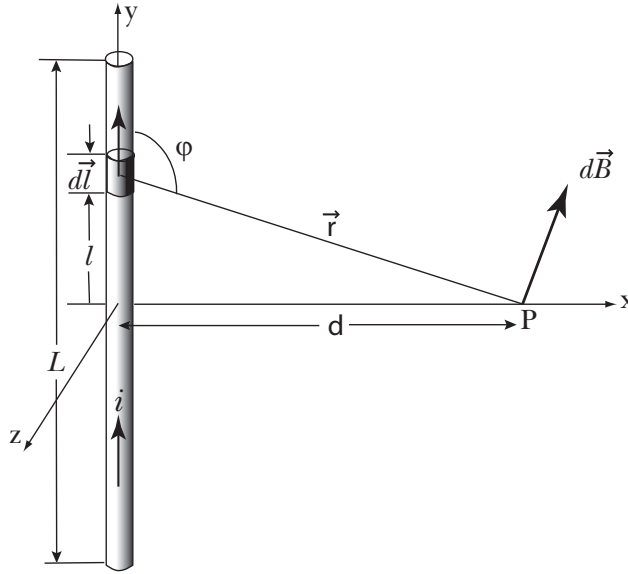


Figure 3.4: The magnetic field $d\vec{B}$ at point P produced by a current segment L of carrying current i . $d\vec{B}$ is directed perpendicularly on the xy plane through $-z$ direction.

Using Biot-Savart law the magnitude of $d\vec{B}$ at point P can be calculated by the formula

$$dB = \frac{\mu_0 i dl \sin \phi}{4\pi r^2}.$$

From figure we can write

$$r = \sqrt{l^2 + d^2}$$

$$\text{and} \quad \sin \phi = \sin(\pi - \phi) = \frac{d}{r} = \frac{d}{\sqrt{l^2 + d^2}}.$$

Substituting the above values of r and $\sin \phi$ we have from the Biot-Savart law

$$dB = \frac{\mu_0 i d}{4\pi} \frac{dl}{(l^2 + d^2)^{3/2}}. \quad (3.8)$$

Using integration limit $-L/2$ to $+L/2$, we find the total field;

$$B = \int dB = \frac{\mu_0 i d}{4\pi} \int_{l=-L/2}^{l=+L/2} \frac{dl}{(l^2 + d^2)^{3/2}},$$

Therefore, the integration is follows as²

$$B = \int dB = \frac{\mu_0 i d}{4\pi} \left[\frac{l}{d^2(l^2 + d^2)^{1/2}} \right]_{l=-L/2}^{l=+L/2},$$

$$\text{or} \quad B = \int dB = \frac{\mu_0 i}{4\pi d} \frac{L}{(L^2/4 + d^2)^{1/2}}. \quad (3.9)$$

$$2 \int \frac{dx}{\sqrt{(x^2 + a^2)^3}} = \frac{x}{a^2 \sqrt{x^2 + a^2}}$$

What happened if the wire is to be considered infinitely:

Then Eq. 3.8 can be written using integration limit $l = -\infty$ to $+\infty$

$$B = \int dB = \frac{\mu_0 i}{4\pi} \int_{l=-\infty}^{l=+\infty} \frac{d}{(l^2 + d^2)^{3/2}} dl. \quad (3.10)$$

To evaluate the integral let us take a trigonometric technic, suppose

$$l = d \tan \theta,$$

$$\therefore dl = d \sec^2 \theta d\theta,$$

when $l = +\infty$ then $\theta = +\pi/2$ and when $l = -\infty$

then $\theta = -\pi/2$.

Now from Eq. 3.10 we obtain,

$$B = \frac{\mu_0 i}{4\pi} \int_{\theta=-\pi/2}^{\theta=+\pi/2} \frac{d^2 \sec^2 \theta d\theta}{\{d^2(\tan^2 \theta + 1)\}^{3/2}},$$

$$\text{or } B = \frac{\mu_0 i}{4\pi d} \int_{-\pi/2}^{+\pi/2} \frac{1}{\sec \theta} d\theta,$$

$$\text{or } B = \frac{\mu_0 i}{4\pi d} \int_{-\pi/2}^{+\pi/2} \cos \theta d\theta = \frac{\mu_0 i}{4\pi d} [\sin \theta]_{-\pi/2}^{+\pi/2},$$

$$\text{or } B = \frac{\mu_0 i}{4\pi d} \{\sin(+\pi/2) - \sin(-\pi/2)\} = \frac{\mu_0 i}{4\pi d} (\sin \pi/2 + \sin \pi/2),$$

$$\text{or } B = \frac{\mu_0 i}{2\pi d}. \quad (3.11)$$

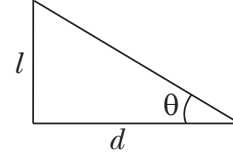
**3.4.2 A circular current loop**

Figure 3.5 shows a circular loop of radius R carrying current i . Let us calculate $\vec{B}$ at a point P on the axis a distance x from the center of the loop. The Biot-Savart law

$$dB = \frac{\mu_0 i}{4\pi} \frac{dl \sin \phi}{r^2}.$$

In this problem, the angle ϕ lies between the vectors $d\vec{l}$ and $\vec{r}$ is 90° . Hence the Biot-Savart law stands here

$$dB = \frac{\mu_0 i}{4\pi} \frac{dl}{r^2}. \quad (3.12)$$

The vector $d\vec{B}$ is at right angles to the plane formed by $d\vec{l}$ and $\vec{r}$.

Let us resolve $d\vec{B}$ into components; one $dB \cos \alpha$ along the axis of the loop and another $dB \sin \alpha$ at right angles to the axis. Let us consider an analogous current element $d\vec{l}$ lies at the opposite position of the loop as it is considered in the previous account. Due to this argument the previous component $dB \sin \alpha$ must be cancelled by an equal and oppositely directed amount of $dB \sin \alpha$. Thus the sum of all $dB \sin \alpha$ for complete loop acting perpendicularly on the axis is zero. Only $dB \cos \alpha$ contributes to the total magnetic field $\vec{B}$ at point P .

Therefore, the the total magnetic field at point P arises

$$B = \int dB \cos \alpha. \quad (3.13)$$

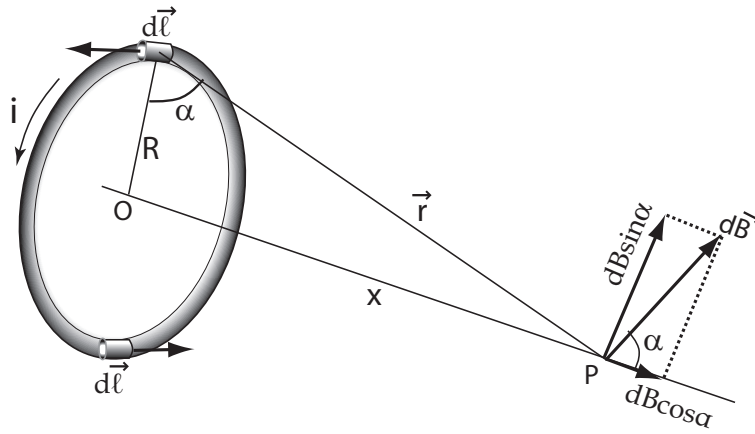


Figure 3.5: The magnetic field $d\vec{B}$ at point P produced by an element $d\vec{l}$ of a wire carrying current i . $d\vec{B}$ is directed perpendicularly to the plane formed by $d\vec{l}$ and $\vec{r}$.

Now from Eqs. 3.12, 3.13 we have

$$B = \frac{\mu_0 i}{4\pi} \int \frac{\cos \alpha \, dl}{r^2}. \quad (3.14)$$

We see in Fig. 3.5 that α and r are not independent of each other. The relationships are

$$r = \sqrt{R^2 + x^2}$$

and $\cos \alpha = \frac{R}{r} = \frac{R}{\sqrt{R^2 + x^2}}.$

Substituting these values into Eq. 3.14 we obtain

$$B = \frac{\mu_0 i R}{4\pi(R^2 + x^2)^{3/2}} \int dl,$$

or $B = \frac{\mu_0 i R}{4\pi(R^2 + x^2)^{3/2}} \times 2\pi R,$

or $B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}, \quad (3.15)$

note that $\int dl$ is simply the circumference of the loop ($= 2\pi R$). If one wants to calculate B at the center of the loop then he has to consider $r = R$ everywhere. The Biot-Savart law gives that

$$B = \int dB = \frac{\mu_0 i}{4\pi r^2} \int dl = \frac{\mu_0 i}{4\pi R^2} \times 2\pi R,$$

or $B = \frac{\mu_0 i}{2R}. \quad (3.16)$

The result could be found also from Eq. 3.15 by putting $x = 0$.

If $x \gg R$ i.e., very far from the current loop B gives from Eq. 3.15 by putting $R \approx 0$

$$B = \frac{\mu_0 i R^2}{2x^3}. \quad (3.17)$$

This dependence of the field on the inverse cube of the distance reminds as of the electric field on an electric dipole (discussed in Section 2.3 on page 20). So it is often convenient to consider a current loop of wire to be a magnetic dipole. Just as the electric behavior of many molecules can be characterized in terms of electric dipole moment, so also the magnetic behavior of atoms can be described in terms of their magnetic dipole moment. In the case of atoms, the current loop is due to the circulation of electrons about the nucleus.

3.5 Magnetic Field in a Solenoid

A solenoid is a helical wire wound on a cylindrical core. The wire windings carry a current i . The use of a solenoid is to get a nearly uniform magnetic field in a system produced by current. A solenoid is a long cylindrical device comparing its breadth or radius. So a large number of windings are needed to form a solenoid and n number of windings are considered per unit length. Inside the solenoid it is considered that the field distribution has the axial symmetry and the magnetic force lines are parallel to the axis of the cylindrical solenoid. How to produce the magnetic field in a solenoid is shown below by the Figs. 3.6 and 3.7.

3.5.1 Loosely wire wound solenoid

Figure 3.6 shows a cross section of a solenoid having loosely wire wound, cutting through its length. The symbol $\otimes$ shown in the figure indicates the direction of current i , which is perpendicular to the page going away from the observer and the symbol $\odot$ ³ indicates the direction of current perpendicular to the page coming to the observer. The circular behavior of the magnetic field close

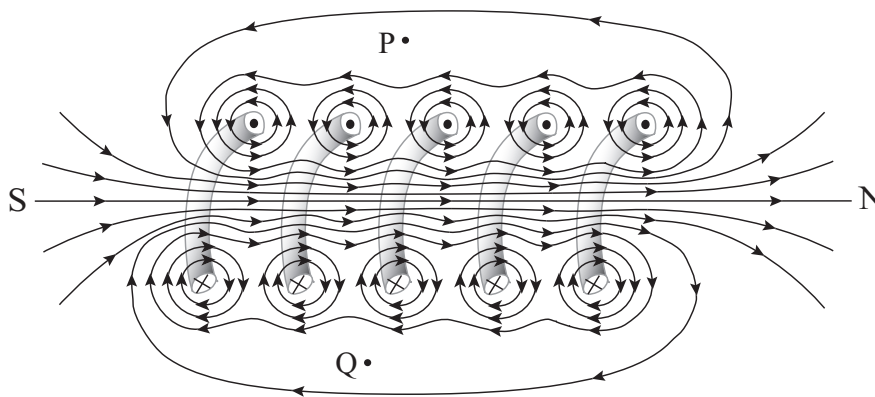


Figure 3.6: Formation of magnetic field in a loosely wound current carrying solenoid. Cross section along the length of the solenoid.

to a single wire is same as that of single straight wire. When two circular force lines regarding two adjacent wires interact each other, they combine together into a single line and exists around the adjacent wires. Thus the total field of the entire solenoid is the vector sum of the fields set up by all the turns. Lines of B tend to form parallel to each other in the interior points of the solenoid. For an ideal solenoid, the field becomes uniform and parallel to the axis and at exterior points, such as P and Q the magnetic field is considered negligible comparing to the interior points. However, interior lines of force of a loosely wire wound solenoid may not be completely parallel as it is shown in Fig. 3.6 and field intensity may also not be uniform at each point.

3.5.2 Closely wire wound solenoid

Figure 3.7 shows a cross section of a solenoid having closely wire wound, cutting through its length. To get an ideal solenoid, the windings of the wire should be very close-packed helix; so that the force lines in interior of the solenoid are almost be parallel to each other. In this case the field intensity along the axial direction is almost be uniform. Taking the external field to be zero if its length is much greater that its radius and if we consider only external points such as P and Q . The solenoid shown in Fig. 3.7 is also not an ideal because its length is not much greater than its radius.

³the symbol $\otimes$ means an observer is just looking the back view and the symbol $\odot$ just looking the front view of an arrow.

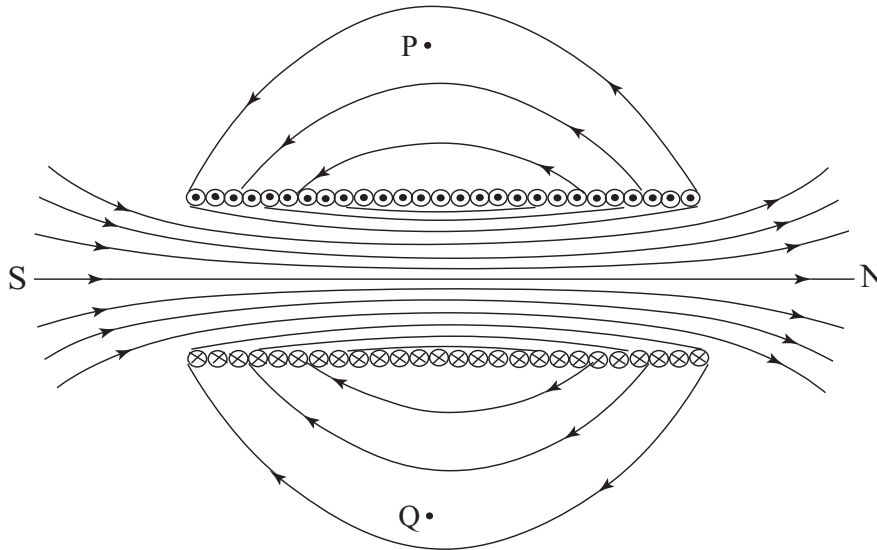


Figure 3.7: Formation of magnetic field in a closely wound current carrying solenoid. Cross section along the length of the solenoid.

A relatively simple device capable of producing a field that is approximately uniform is the parallel plate capacitor if the plate separation is small compared to the dimensions of the plate.

3.6 Ampere's Law

Coulomb's law can use to calculate the electric field associated with any distribution of electric charges. In Chapter 1, however, we have seen that Gauss's law is more suitable to solve a certain class of problems, those containing a high degree of symmetry of field distribution. We consider Gauss's formula is more basic formula than that of Coulomb's formula and Gauss's law is one of the four fundamental of Maxwell's equations of electromagnetism. The situation is similar to magnetism, where using the Biot-Savart law, one can calculate the magnetic field of any distribution of current. But to calculate magnetic field $\vec{B}$, Ampere's law is convenient where the field has symmetry like current in a long straight wire described in the subsection 3.4.1. Ampere's law is considered more fundamental than the Biot-Savart law and leads to another of the Maxwell's law of electromagnetism.

Ampere's law is written by the formula

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i. \quad (3.18)$$

In Chapter 1, using Gauss's law the integration were performed over an imaginary closed surface called Gaussian surface. In using Ampere's law similarly the integration is performed over imaginary closed curve called Amperian loop.

Here i is the net current enclosed by the Amperian loop. The left side of Eq. 3.18, the $d\vec{l}$ is small segment of imaginary loop. The symbol $\oint$ indicates the integration all around the closed loop. The dot product of $\vec{B}$ and $d\vec{l}$ gives the scalar quantity $Bdl \cos \theta$. Where θ is angle between $\vec{B}$ and $d\vec{l}$ at any segment.

Therefore, Ampere's law can be rewritten as

$$\oint \vec{B} \cdot d\vec{l} = \oint Bdl \cos \theta = \mu_0 i. \quad (3.19)$$

The magnetic field $\vec{B}$ at any point is the net effect of the currents in all wires. In Fig. 3.6(a) and 3.6(b) the net currents are $i = i_1 + I_2 + i_3$ and $i = i_1 - I_2 + i_3$, respectively.

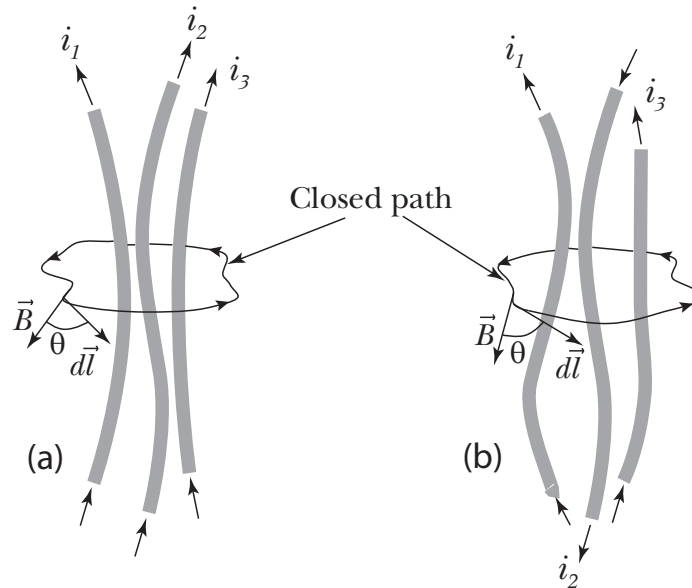


Figure 3.8: In Applying Ampere's law, the integration is performed around a closed contour that enclosed the net current in wires. (a) Net current enclosed by the path is $i = i_1 + i_2 + i_3$. (b) Net current enclosed by the path is $i = i_1 - i_2 + i_3$.

3.7 Application of Ampere's Law

As it is mentioned before that Ampere's law is convenient to calculate magnetic field $\vec{B}$, where the fields have symmetric configurations. Application of Ampere's law in some more general cases are described below.

3.7.1 A long straight wire

B for an external point of a long straight current carrying wire:

In the subsection 3.4.1, we have already calculated the magnetic field B for a straight wire using Biot-Savart law. Let us now use Ampere's law to find B in the same case which is more easier process to calculate it. Figure 3.9 shows a top view of the cross section of a long current carrying wire. We can choose a circle of radius r as Amperian loop for closed path integration. The directions of $\vec{B}$ and $d\vec{l}$ at any point on the Amperian loop will be along the tangent drawn at that point of the loop. B for an external point, the radius r of the loop must be greater than the radius of the wire as

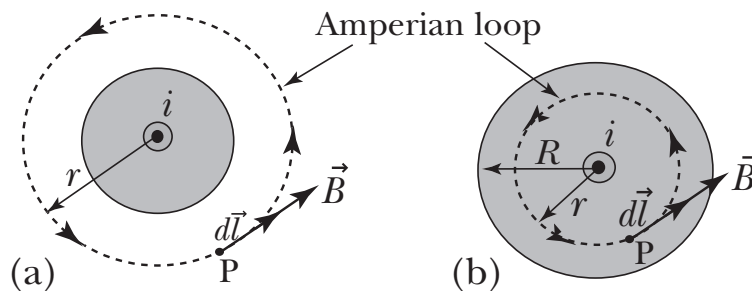


Figure 3.9: $\vec{B}$ for a long straight current carrying wire. (a) $\vec{B}$ for an external point, (b) $\vec{B}$ for an internal point.

it is shown in Fig. 3.9(a). We know Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \oint B dl \cos \theta = \mu_0 i.$$

The magnitude of $\vec{B}$ is constant around the circular path. The value of θ is zero, because $\vec{B}$ and $d\vec{l}$ have the same direction. Hence from above formula we have

$$B \oint dl = \mu_0 i,$$

$\oint dl$ simply gives the circumference of the circle of radius r . Hence

$$B(2\pi r) = \mu_0 i,$$

$$\text{or} \quad B = \frac{\mu_0 i}{2\pi r}. \quad (3.20)$$

The result of Eq. 3.20 is identical with Eq. 3.11, a result we obtained using Biot-Savart law.

B for an internal point of a long straight current carrying wire:

We can also use Ampere's law to find magnetic field inside a wire. B for an external point, we choose the radius r of the Amperian loop less than the radius of the wire ($r < R$) as it is shown in Fig. 3.9(b). Here first we have to know the effective current that passing through the area enclosed by the Amperian loop.

Let the effective current be i' which gives

$$i' = i \frac{\pi r^2}{\pi R^2}.$$

We have from Ampere's law

$$\oint B dl \cos \theta = B \oint dl = \mu_0 i' = \mu_0 i \frac{\pi r^2}{\pi R^2},$$

$$\text{or} \quad B(2\pi r) = \mu_0 i \frac{\pi r^2}{\pi R^2},$$

$$\text{or} \quad B = \frac{\mu_0 i r}{2\pi R^2}. \quad (3.21)$$

If we consider $r = R$, i.e. B on the surface gives the value from Eq. 3.21

$$B = \frac{\mu_0 i}{2\pi R}. \quad (3.22)$$

We see from Eqs. 3.21 and 3.22 that the value of B on the surface of the wire is greater than that of any interior point. We see from Eq. 3.21 $B \propto r$, and from Eq. 3.20 $B \propto r^{-1}$. Therefore, starting from the center of the wire, B at any point increases upto the surface and again decreases from the surface of the wire.

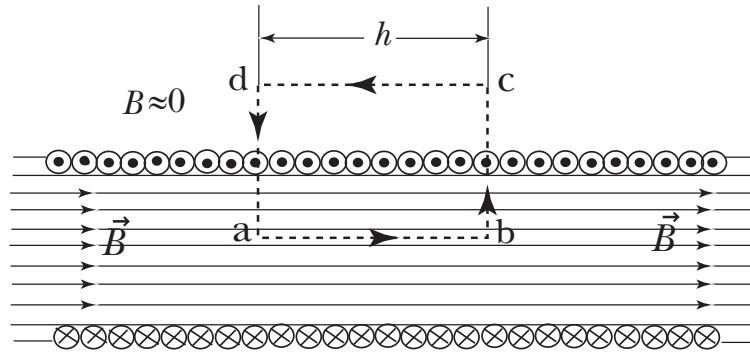


Figure 3.10: $\vec{B}$ for an ideal solenoid. Doted rectangle $abcd$ is the Amperian loop.

3.7.2 A solenoid

Figure 3.10 shows an example of ideal solenoid. In this analysis we assume the magnetic field is parallel to the axis of this ideal solenoid and constant in magnitude along the axis. It also suggest that for an ideal solenoid the field is uniform at any point in the interior point. We have to calculate B for an interior point of the solenoid. For our convenient let us choose an Amperian loop of rectangular shape, where the direction of integration would be perform in the path $abcda$.

Applying Ampere's law we have

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i'. \quad (3.23)$$

The total integration $\oint \vec{B} \cdot d\vec{l}$ can be calculated as the sum of integration of four individual segments along the path $abcda$. Which follows

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \int_a^b B dl \cos 0^\circ + \int_b^c B dl \cos 90^\circ + \int_c^d (0) dl \cos 180^\circ + \int_d^a B dl \cos 90^\circ, \\ \text{or} \quad \oint \vec{B} \cdot d\vec{l} &= Bh + 0 + 0 + 0 = Bh, \end{aligned} \quad (3.24)$$

the integration for the segment c to d where the value of B outside the solenoid is ideally zero.

Now we look the effective current i' enclosed by the Amperian loop. Let the number of turns per unit length of the solenoid be n and the current carrying for a single loop be i . Then the effective current is given by

$$i' = nhi. \quad (3.25)$$

Now from Eqs. 3.23, 3.24 and 3.25 we have

$$\begin{aligned} Bh &= \mu_0 nhi, \\ \text{or} \quad B &= \mu_0 ni. \end{aligned} \quad (3.26)$$

Equation 3.26 concludes that the magnetic field inside an ideal solenoid is uniform over its cross section, and just knowing the number of turns per unit length and amount of current in a winding one can calculate magnetic field at any point inside it.

3.7.3 A toroid

Figure 3.11 shows a cross section of an electrical device named toroid. A toroid may be visualized to be a solenoid bent into a ring shape. The lines of field $\vec{B}$ inside the toroid are formed by a number of concentric circles and have the symmetry with uniform field distribution. We have to calculate B for a point inside it. Let us consider a concentric circle of radius r as an Amperian loop. Applying

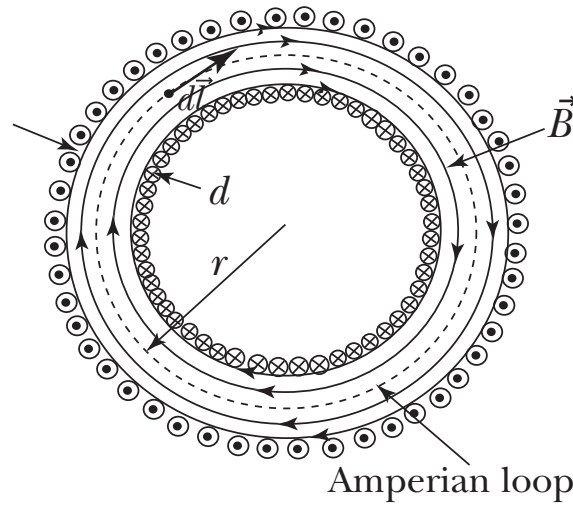


Figure 3.11: $\vec{B}$ for a toroid. The magnetic force lines inside the toroid are concentric circles.

Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \oint B dl \cos 0^\circ = B \oint dl = B(2\pi r) = \mu_0 i N,$$

$$\text{or } B = \frac{\mu_0 i N}{2\pi r}, \quad (3.27)$$

where i is the current in the toroid windings and N is the total number of turns. Note that the Amperian loop encloses the total number of turns N of the toroid. Equation 3.27 discusses that B is inversely proportional to r . Therefore, we can observe that B for an ideal solenoid is constant everywhere in the interior points, in contrast that B for an ideal toroid is not constant everywhere in the interior points. One can able to show that, using Ampere's law, $B = 0$ for points outside and in the central cavity of an ideal toroid.

As we stated earlier that a toroid is a solenoid bent into the shape of a ring. Let us see Eq. 3.27 justifies our earlier statement as setting $n = N/2\pi r$ the number of turns per unit length. Then Eq. 3.27 reduces to $B = \mu_0 n i$ which has analogy of Eq. 3.26.

3.8 Examples

Example 3.1: In the Bohr model of the hydrogen atom, the electron circulates around the nucleus in a circular path of radius 5.29×10^{-11} m at a frequency f of 6.60×10^{15} Hz. What value of B is set up at the center of the orbit?

Solution: The rate of charge passes any point on the orbit gives the current at any point on the orbit and is given by

$$i = ef,$$

$$\text{or } i = (1.60 \times 10^{-19} \text{C})(6.60 \times 10^{15} \text{Hz}) = 1.06 \times 10^{-3} \text{A}.$$

The magnetic field B at the center of the orbit is given by Eq. 3.16

$$B = \frac{\mu_0 i}{2R} = \frac{(4\pi \times 10^{-7} \text{T.m/A})(1.06 \times 10^{-3} \text{A})}{2(5.29 \times 10^{-11} \text{m})} = 12.6 \text{T}.$$

Example 3.2: A solenoid 1.33 m long and 2.60 cm in diameter carries a current of 17.8 A. The magnetic field inside the solenoid is 22.4 mT. Find the length of the wire forming the solenoid.

Solution: The magnetic field inside an ideal solenoid is given by the Eq. 3.26 on page 35,

$$B = \mu_0 n i.$$

Hence the number of turns per unit length of the solenoid,

$$n = \frac{B}{\mu_0 i} = \frac{0.0224 \text{ T}}{(4\pi \times 10^{-7} \text{ T.m/A})(17.8 \text{ A})} = 1.00 \times 10^3 \text{ m}^{-1}.$$

The length of the solenoid is 1.33 m. So the total number of turns of the solenoid be

$$N = nL = (1.00 \times 10^3 \text{ m}^{-1})(1.33 \text{ m}) = 1330.$$

Now radius of the solenoid i.e., each turn of the coil $r = 2.60 \text{ cm} = 0.013 \text{ m}$. Therefore total length wire which forms this solenoid is

$$= N(2\pi r) = (1330)(2\pi)(0.013 \text{ m}) \approx 109 \text{ m}.$$

Example 3.3: Figure 3.12 shows a an arrangement known as **Helmholtz coil**. It consists of two circular coaxial coils each of N turns and radius R , separated by a distance R . They carry equal current i in the same direction. Find the field at P , mid way between the coils.

Solutions: From the Subsection 3.4.2 magnetic field on the axis of a circular loop, we can use the formula 3.15 on page 30 as

$$B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}.$$

For magnetic field at P , here we have to introduce the turns factor N , $x = R/2$ and magnitude will

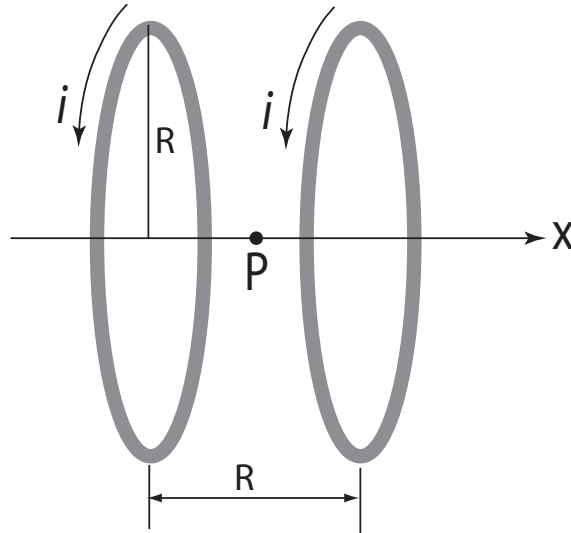


Figure 3.12: Helmholtz coil of radius R . The point P is placed in the middle of the coils on their axis.

be twice for the two coils, and we can replace the above formula as

$$B = \frac{2\mu_0 N i R^2}{2(R^2 + (R/2)^2)^{3/2}} = \frac{8\mu_0 N i}{(5)^{3/2} R}.$$

Chapter 4

FARADAY'S LAW OF INDUCTION

In the previous chapter (Chap. 3), we have studied that how a flow of current induces a magnetic field. In this chapter we shall discuss how a magnetic field induces an emf in a circuit. Faraday's law of induction gives the basic idea of electromagnetism and the equation produces by this law is one of the four Maxwell's equation. Electric and magnetic fields with change of time are related to each other. In the following sections we will relate the electric field with changing of magnetic field and demonstrate some simple experiments that can easily be done in the laboratory.

4.1 Faraday's Experiment

Michael Faraday first discovered in England in 1831 through an experiment that a magnetic field can produce an electric field. He suggested a law how a magnetic field produces current in a circuit and according to his honor this law is named as Faraday's law of induction. Figure 4.1 shows a coil

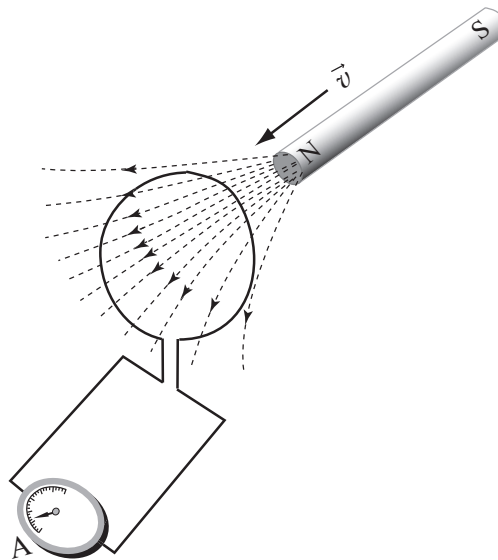


Figure 4.1: When the magnet is moving with respect to the coil, the ammeter 'A' shows a deflection which indicates the appearance of a current in the coil.

of conducting wire makes a circuit connecting an ammeter in series. Normally we would expect the ammeter to show no deflection in the circuit because there is no driving source as connected to the circuit. However, if we push a bar magnet towards the coil with its north pole facing the coil, astonishingly a physical property shows there. While the magnet is moving, the ammeter deflects, showing that a current has been set up in the coil. The ammeter does not show any deflection if

we hold the magnet stationary with respect to the coil. Again, if we move the magnet away from the coil, the ammeter repeats to show the deflection but in the opposite direction, which explains the current in the coil appears in the opposite direction. If we push the south pole instead of north pole in the same manner the ammeter shows also a deflection but in reversed direction as described before. The described results are also observed if we move the coil in keeping the magnet stationary.

The electromotive force that appears in the circuit is called an *induced electromotive force (emf)* and the induced *emf* set up a current in the coil is called an *induced current*.

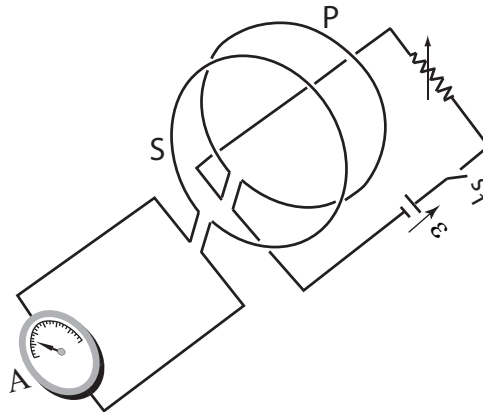


Figure 4.2: When the switch s_1 is operated in the primary coil P as on-off, the ammeter 'A' deflects immediately as the sequence of operation. The coils are not necessary to move.

Figure 4.2 shows an experimental arrangement where a magnet is not necessary to achieve a magnetic field. The two coils (one is called primary P and another is secondary S) are placed close together but at rest with respect to each other. When we close the switch s_1 in the primary coil P, the ammeter in the secondary coil S deflects immediately. Here the physical phenomenon is occurred that when we close the switch the current that flows in the primary coil P produces a magnetic field around that coil. This magnetic field cross the secondary coil S and induces an *emf* which makes the cause of deflection in the ammeter. When we open the switch s_1 the indicator of the ammeter shows again a movement immediately, but in the opposite direction. The experiment shows that there is an induced *emf* in the secondary coil whenever the current in the primary coil is changing. The common feature of these two experiment is motion or change. It is the moving magnet or the changing current that is responsible for the induced *emf*.

4.2 Faraday's Law

Faraday's law is a successful law of induction which can deduce the magnitude and direction of the induced *emf*. In practise this induced *emf* is very important, as for example, the current in a commercial electric generator are operated in the phenomenon of producing induced *emf*.

We see from Figs. 4.1 and 4.2 that an induced *emf* is produced in a coil whenever a number of lines of magnetic field cross that coil with changing their number of force lines. We know that the change of force lines through a cross sectional area means the change of flux in that area. Therefore, *the change of magnetic flux passing through the cross sectional area of a circuit of wire induces an emf in that circuit. Actually, it is the rate of change in the number of field lines passing through a circuital loop that determines the induced emf in that process.*

The verbal statement of Faraday's law of induction is said to be that the induced *emf* in a circuit is equal to the rate at which the magnetic flux through the circuit is changing with time.

Mathematically the definition of Faraday's law is written by

$$\varepsilon = -\frac{d\Phi_B}{dt}, \quad (4.1)$$

where ε is the induced *emf* and Φ_B is the magnetic flux that passing through the circuit. The negative sign indicates the direction of induced *emf* and the physical significance of it is described the next section. If the coil consists of N turns, then an induced *emf* appears in every turn, and the total induced *emf* in the circuit is the sum of the individual values.

Hence, the total induced *emf* is given by

$$\varepsilon = -N\frac{d\Phi_B}{dt}. \quad (4.2)$$

Note: The magnetic flux Φ_B to be a measure of the number of magnetic force lines crossing through a surface area. The magnetic flux through a surface is defined by the similar formula of an electric flux stated in Eq. 1.6 and given by

$$\Phi_B = \int \vec{B} \cdot d\vec{A}. \quad (4.3)$$

The SI units of induced *emf* and magnetic flux are *Volt* and *Weber*, respectively. 1 *Weber* = 1 *Tesla.meter*².

4.3 Lenz's Law

Significance of negative sign of Faraday's law:

In Eq. 4.1 we see that a negative sign has been used which indicates the direction of induced *emf*. In this section we shall discuss the physical significance, that why a negative sign has been chosen in Faraday's law of induction. In 1834 Heinrich Lenz proposed a rule for the direction of induced current that produced in the coil and it is known as Lenz's law.

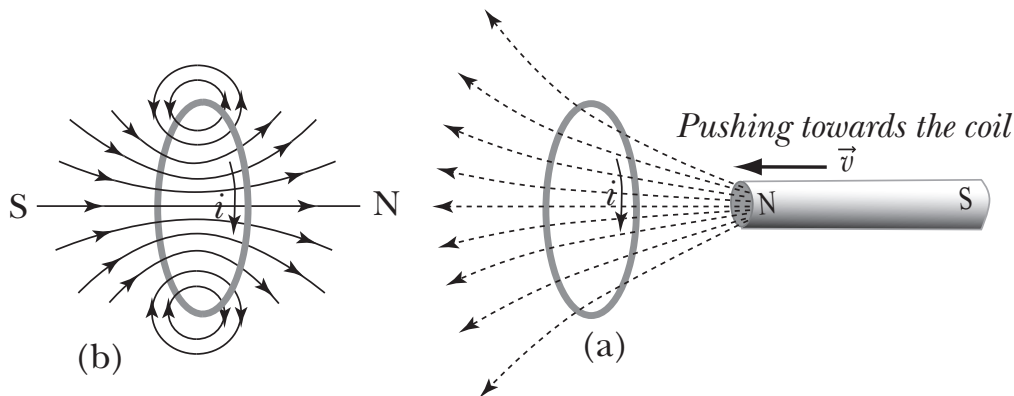


Figure 4.3: Indication of direction of induced current in the loop. (a) Due to pushing the magnet towards the loop, the magnetic flux through the loop is increasing. (b) Due to the change of flux through the loop, the induced current i is produced in such a direction that the magnetic field produced by induced current i opposes the pushing of the bar magnet.

Figure 4.3(a) shows the N pole of a bar magnet is pushing towards a conducting loop. Due to pushing the magnet, it is increasing the magnetic flux with time. According to Faraday's law, the change of magnetic flux through the loop with time will induce an *emf* that creates a current in the loop. Practically, it is experienced that there is a appearance of force which opposes the

pushing of the bar magnet toward the loop. According to Lenz's law this opposition force comes into appearance due to the direction of induced current i . The direction of i is shown by arrow markers in figures. Figure 4.3(b) shows that the induced current i sets up a magnetic field in the loop by the system of right-hand rule. It is clear that the direction of i appears in such that it creates an N pole facing towards the N pole of the bar magnet. By the process of repulsion of N-N poles, the creating situation finally opposes the pushing of the bar magnet towards the loop.

Again let us now imagine the Fig 4.4(a). It describes that the N pole of a bar magnet is pulling away from the conducting loop. Due to pulling away the magnet, it is decreasing the magnetic flux with time. Practically, the experienced force opposes the pulling away of the bar magnet from the loop. Lenz proposed regarding this opposition force comes into appearance due to the opposite direction of induced current i as it is stated earlier. Figure 4.4(b) shows that the induced current i sets up a magnetic field also by the system of right-hand rule. Here, the direction of i is such that it creates an S pole facing towards the N pole of the bar magnet. By the process of attraction of N-S poles, the creating situation finally opposes the pulling away of the bar magnet from the loop.

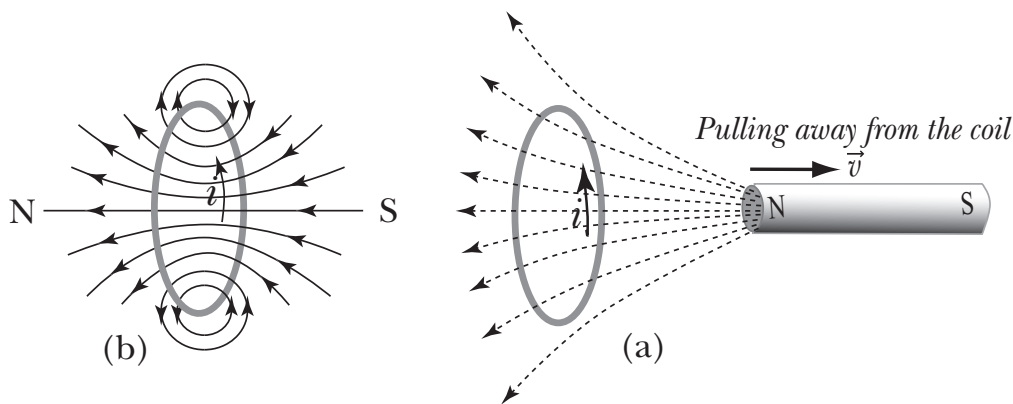


Figure 4.4: Indication of direction induced current in the loop. (a) Due to pulling the magnet away from the loop, the magnetic flux through the loop is decreasing. (b) Due to the change of flux through the loop, the induced current i is produced in such a direction that the magnetic field produced by induced current i opposes the pulling of the bar magnet.

However, Fig 4.3 describe the direction of induced current in the loop which opposes the increasing of the change of flux and Fig. 4.4 describe the direction of induced current which opposes the decreasing of the change of flux through the loop. We see that in both cases *the flux of the magnetic field produced by the induced current opposes the change of flux that causes the induced current*. This unavoidable opposite factor of appearing induced current in the loop suggests the acceptance of negative sign in Faraday's law of induction.

4.4 Eddy Current

An induced current produces not only in a loop or coil, the induced effect is also observed if a metal sheet or plate is to be moved in a magnetic field. *When a metal sheet or plate is made to move in a magnetic field, due to the change of magnetic flux through the metal, an induced current appears in that metal sheet or plate. This induced current is called eddy current.* In most cases the eddy current is undesirable effect due to the lose of energy in the system. For example, they increase the internal energy and thus can increase the temperature of the material. For this reason, materials that are subject to changing magnetic field are often laminated or insulated. In some cases eddy-current heating effect can be used to advantage, as in an induction furnace, in which a sample of material can be heated using a rapidly changing magnetic field. Figure 4.5 shows when a conducting plate is taking away from the magnetic field, an eddy current is produced can be shown by a current loop.

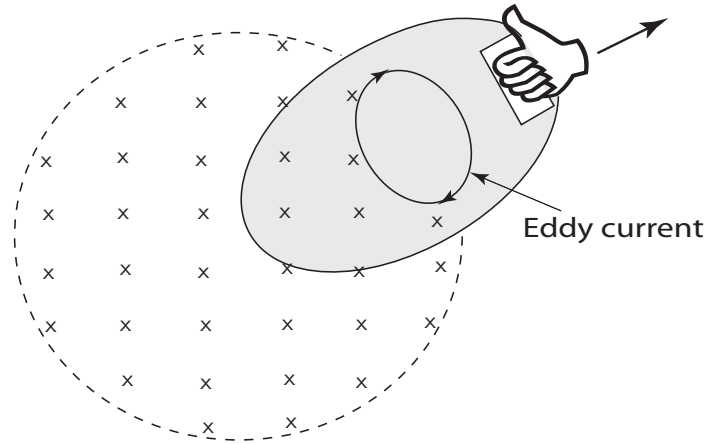


Figure 4.5: Eddy current is produced in a conducting material when it moves in magnetic field.

The effective eddy current path experience a magnetic force $\vec{F}_B = i\vec{L} \times \vec{B}$, where L is the effective length of the current path. This force is transmitted to the material and according to Lenz's law it opposes the motion of the conductor. This effect gives rise a deceleration in a rotating wheel or a moving track operated in magnetic field.

4.5 Inductance

An inductor is an ideal solenoid, carrying a current i that produces an magnetic field $\vec{B}$ in its interior position. An inductor stores energy in the form of magnetic field surrounding its current carrying wires, just like as a capacitor stores energy in the form of electric field between its charged plates. Inductance is the property of an inductor depends on its geometry, like the capacitance, the property of a capacitor also depends on its geometry. In an electrical circuit diagram, an inductor is represented by the symbol $\text{---}\text{|||||}\text{---}$, which pictures the shape of a solenoid.

Mathematical representation:

Whenever we change the current in a solenoid, it changes the magnetic field. Faraday's law shows that there is an emf generated in the solenoid directly proportional to the rate of change of current.

$$\begin{aligned} \epsilon_L &\propto \frac{di}{dt}, \\ \text{or} \quad \epsilon_L &= L \frac{di}{dt}, \end{aligned} \quad (4.4)$$

where L is a proportionality constant and called inductance. If the solenoid has N number of turns, it sets a flux $N\Phi_B$ and is known as the number of *flux linkages* of the inductor. According to Faraday's law, the emf is given by

$$\epsilon_L = -\frac{d(N\Phi_B)}{dt}. \quad (4.5)$$

From the Eqs. 4.4 and 4.5 (taking the magnitude) we have

$$L \frac{di}{dt} = \frac{d(N\Phi_B)}{dt}.$$

Integrating respect to time

$$Li = N\Phi_B + C,$$

where C is the integrating constant. Assuming that when $i = 0$, then $\Phi_B = 0$ and $C = 0$. Therefore, we find

$$Li = N\Phi_B,$$

$$\text{or } L = \frac{N\Phi_B}{i}. \quad (4.6)$$

Note that, since Φ_B is proportional to the current i , the Eq. 4.6 indicates the inductance (like the capacitance) depends only on the geometry of the device.

4.5.1 Self inductance

Suppose that a steady current i , supplied by a battery, exists in a coil. Let us suddenly the current in the circuit is reduced to zero. This will cause the change of magnetic field and an induced emf will appear in such a direction, according to the Lenz's law it will oppose the decrease in current. Again if the current i in the circuit is to be increased, the induced emf will oppose to increase the current in that circuit. This effect in that circuit created itself is called self-inductance.

However, when we talk about inductance, it refers the self-inductance and is also denoted by L .

Unit of inductance:

Inductance is written as with SI unit

$$L = \frac{\mathcal{E}}{\frac{di}{dt}} \cdot \frac{\text{volt}}{\frac{\text{ampere}}{\text{second}}} = \frac{\mathcal{E}}{di/dt} \text{ henry}.$$

$\therefore 1 \text{ henry} = 1 \text{ volt.second/ampere}.$

4.5.2 Mutual inductance

If current i is set up in primary coil P from zero to a maximum value, the change of flux sets up an induced emf in the secondary coil S to maximum. This induced emf sets a current in S. Again if current in P is cut-off suddenly, then current in S will change, which causes the change in flux linkage in P and sets up an induced emf in P. In this case the induced emf in P sets up a current in opposite direction of the former current in S. The presence of the secondary coil in the neighborhood of the primary coil makes a delay of the growth of current. Again when the current is made to decrease in P, due to the presence of secondary coil S, it makes also delay to decrease the current in P. This mutual action of this two coil is called mutual induction between them.

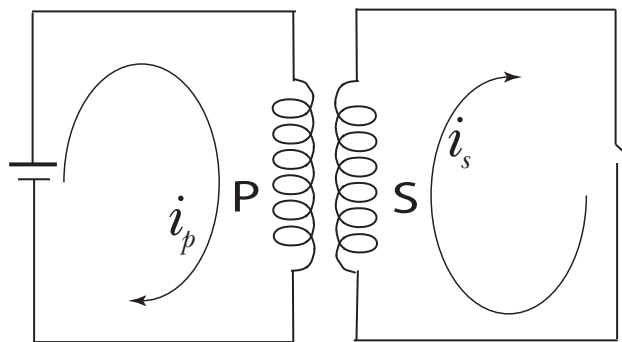


Figure 4.6: Mutual inductance creates between two coils situated close together. P and S are called primary and secondary coils, respectively.

Let a current i_p is sets up in P, which creates a magnetic field B_p in it. This $\vec{B}_p$ produces a flux in S. Therefore, the flux in secondary coil S due to primary coil P can be calculated by

$$\Phi_{sp} = \int \vec{B}_p \cdot d\vec{A}_s,$$

where A_s is the area of single turn of S. If there is N_s turns of S, then

$$N_s \Phi_{sp} \propto i_p,$$

$$\text{or} \quad N_s \Phi_{sp} = M_{sp} i_p,$$

where M_{sp} is the mutual inductance induces in S coil due to the P coil.

This flux produces an *emf* in S due to P, is given by

$$\epsilon_{sp} = -N_s \frac{d\Phi_B}{dt} = -M_{sp} \frac{di_p}{dt} \quad (4.7)$$

This induced *emf* ϵ_{sp} in secondary coil S, causes a flow of current i_s in that coil, and similarly produces a mutual inductance M_{ps} in P.

Therefore, the induced *emf* in P due to coil S is to be calculated as

$$\epsilon_{ps} = -M_{ps} \frac{di_s}{dt} \quad (4.8)$$

For any two circuits close together, in fact

$$M_{sp} = M_{ps} = M,$$

where M is generally called the mutual inductance. *The unit of mutual inductance is also denoted by henry.*

4.5.3 The inductance of a solenoid

Consider a cross section of a solenoid of length l , cross sectional area A , number of turns per unit length be n and carrying a current i . We have seen, in Subsection 3.7.2, the magnetic field B inside a solenoid carrying a current i has been shown

$$B = \mu_0 n i.$$

The number of flux linkage in the length l is

$$N\Phi_B = (nl)(BA).$$

Now substituting the value of B , we have

$$N\Phi_B = \mu_0 n^2 l i A \quad (4.9)$$

Now using Eq. 4.6, the inductance for a solenoid can be calculated

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 n^2 l i A}{i} = \mu_0 n^2 l A, \quad (4.10)$$

and the inductance per unit length of the solenoid can written by

$$\frac{L}{l} = \mu_0 n^2 A. \quad (4.11)$$

Equation 4.10 shows that the inductance of a solenoid does not depends on the current or the magnetic field. It depends on geometrical factors number of turns n , length l and cross sectional area A .

4.5.4 The inductance of a toroid

Figure 4.7 shows a rectangular cross section of a toroid. The toroid has outer radius 'b', inner radius 'a' and height of the rectangular cross section 'h'. The magnetic field in a toroid is given by Eq. 3.27

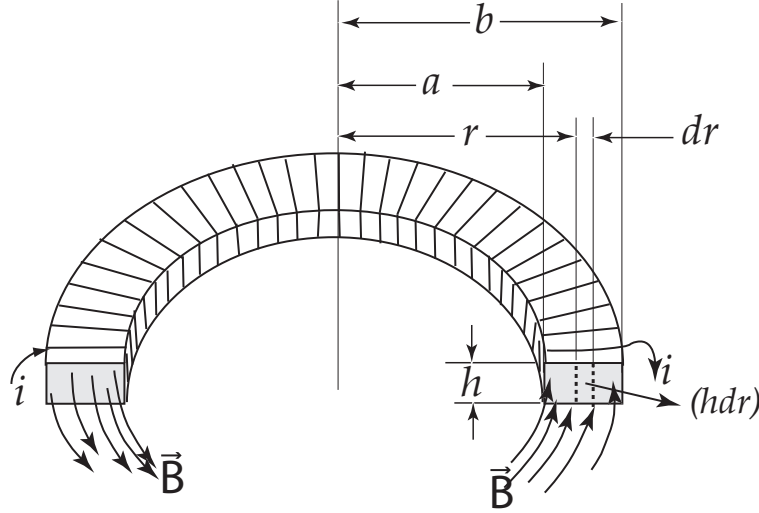


Figure 4.7: A cross section of a toroid. $\vec{B}$ concentric magnetic field, 'a' and 'b' are inner and outer radius respectively. The thickness of the toroid is 'h'.

$$B = \frac{\mu_0 i N}{2\pi r},$$

where we see that the magnetic field inside the toroid is not constant and varies with the radius r .

Let us take a small cross sectional area of the toroid be $h dr$. The total flux Φ_B through the cross section is

$$\begin{aligned}\Phi_B &= \int \vec{B} \cdot d\vec{A} = \int_a^b B(h dr) = \int_a^b \frac{\mu_0 i N}{2\pi r} h dr \\ &= \frac{\mu_0 i N h}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu_0 i N h}{2\pi} \ln \frac{b}{a}.\end{aligned}$$

The inductance of the toroid can be found by the equation

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a}. \quad (4.12)$$

Here we also see the inductance L depends only on geometrical factors.

4.6 Examples

Example 4.1 The long solenoid S of Fig. 4.8 has 220 turns/cm; its diameter d is 3.2 cm. At its center we place a 130-turn close-packed coil C of diameter $d_c=2.1$ cm. The current in the solenoid is increased from zero to 1.5 A at a steady rate over a period of 0.16 s. What is the magnitude of the induced *emf* that appears in the central coil while the current in the solenoid is being changed?

Solution: Since the current is changing with time, the magnetic field produced by the current is also changing in the interior points of the solenoid. The field is uniform and perpendicular to the area of each turn of coil C.

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = BA,$$

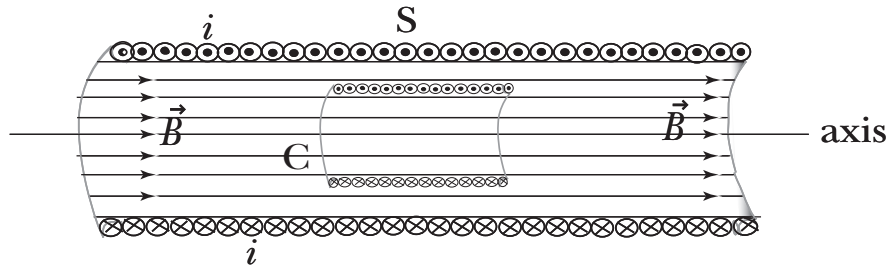


Figure 4.8: A coil 'C' is placed inside a solenoid 'S'. The solenoid carries a current that emerges from the page at the top and enters at the bottom, as indicated by the dots and crosses. When the current is changing in the solenoid, an induced emf appears in the coil.

where A is the area of each turn of the coil C.

Now from Faraday's law of induction followed by Eq. 4.2 with absolute value

$$|\varepsilon| = N \frac{d\Phi_B}{dt} = N \frac{d(BA)}{dt} = NA \frac{d(B)}{dt} = NA \frac{d(\mu_0 ni)}{dt} = NA \mu_0 n \frac{di}{dt}.$$

Area of the single turn of the central coil,

$$A = \frac{\pi d_c^2}{4} = \frac{\pi (0.021)^2}{4} \text{ m}^2 = 3.4636 \times 10^{-4} \text{ m}^2.$$

$N = 130$, $\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A}$, $n = 2.2 \times 10^4 \text{ turns/m}$. $di = (1.5 - 0) = 1.5 \text{ A}$ and $dt = 0.16 \text{ s}$. Therefore,

$$|\varepsilon| = (130)(3.4636 \times 10^{-4} \text{ m}^2)(4\pi \times 10^{-7} \text{ T.m/A})(2.2 \times 10^4 \text{ turns/m})(1.5 \text{ A}/0.16 \text{ s}) \\ = 1.2 \times 10^{-2} \text{ V} = 12 \text{ mV}.$$

Example 4.2 A long solenoid of diameter 5 cm has 100 turns/cm of its length. At the center of the solenoid, a small coil of 50 turns and 2 cm diameter is placed such that the magnetic field produced by the solenoid is parallel to the axis of the coil. The current is changed from -2.0 A to +2.0 A over a period of 0.1 second. Calculate the emf induced in the coil during the change.

Solution: Faraday's law of induction

$$|\varepsilon| = N \frac{d\Phi_B}{dt} = N \frac{d(BA)}{dt} = NA \frac{d(B)}{dt} = NA \frac{d(\mu_0 ni)}{dt} = NA \mu_0 n \frac{di}{dt}.$$

Area of the single turn of the central coil,

$$A = \frac{\pi d_c^2}{4} = \frac{\pi (0.02)^2}{4} \text{ m}^2 = 3.1416 \times 10^{-4} \text{ m}^2.$$

$N = 50$, $\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A}$, $n = 1 \times 10^4 \text{ turns/m}$.

$di = \{2 - (-2)\} \text{ A} = 4 \text{ A}$ and $dt = 0.1 \text{ s}$.

Therefore,

$$|\varepsilon| = (50)(3.1416 \times 10^{-4} \text{ m}^2)(4\pi \times 10^{-7} \text{ T.m/A})(1 \times 10^4 \text{ turns/m})(4 \text{ A}/0.1 \text{ s}) \\ = 7.88 \times 10^{-3} \text{ V} = 7.88 \text{ mV}.$$

Example 4.3 A section of a solenoid of length $l = 12 \text{ cm}$ and having a circular cross section of diameter $d = 1.6 \text{ cm}$ carries a steady current of $i = 3.80 \text{ A}$. The section contains 75 turns along the length. (a) What is the inductance of the solenoid when the core is empty? (b) the current is reduced at a constant rate to 3.20 A in a time of 15 s. What is the resulting emf developed by the solenoid?

Solution: (a) From Eq. 4.10, we have the inductance of a solenoid is

$$L = \mu_0 n^2 l A.$$

Here, $n=75$ turns/0.12 m and $A=\pi(0.008)^2$.

Therefore,

$$L = (4\pi \times 10^{-7} H/m)(75 \text{ turns}/0.12 \text{ m})^2 (0.12 \text{ m}) \pi (0.008 \text{ m})^2 = 1.184 \times 10^{-5} H = 11.84 \mu H.$$

(b) The rate at which the current changes is

$$\frac{di}{dt} = \frac{3.20 \text{ A} - 3.80 \text{ A}}{15 \text{ s}} = -0.04 \text{ A/s}.$$

The induced emf in magnitude

$$\varepsilon = \left| L \frac{di}{dt} \right| = (11.84 \mu H)(0.04 \text{ A/s}) = 0.4737 \mu V.$$

Example 4.4 A toroid having square cross section with a side 5 cm. It contains 536 turns, inner radius 15.3 cm and carries a current 810 mA. Calculate magnetic flux, total flux-linkage and inductance.

Solution: We have inner radius $a=15.3$ cm, $h=50$ cm and outer radius $b=(15.3+5)$ cm $=20.3$ cm. The magnetic flux

$$\begin{aligned} \Phi_B &= \frac{\mu_0 i N h}{2\pi} \ln \frac{b}{a} \\ &= \frac{(4\pi \times 10^{-7} H/m)(0.810 \text{ A})(536 \text{ turns})(0.05 \text{ m})}{2\pi} \ln \frac{0.203 \text{ m}}{0.153 \text{ m}} = 1.2277 \times 10^{-6} \text{ weber.} \end{aligned}$$

Total flux linkage

$$N\Phi_B = \frac{\mu_0 i N^2 h}{2\pi} \ln \frac{b}{a} = 6.5803 \times 10^{-4} \text{ weber.}$$

Finally the inductance of the toroid is to be

$$L = \frac{N\Phi_B}{i} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a} = 8.1238 \times 10^{-4} H = 0.81238 \text{ mH}.$$

Example 4.5: A circular loop made of a stretched conducting elastic material has a 1.23 m radius. It is placed with its plane at right angles to a uniform 785 mT magnetic field. When released, the radius of the loop starts to decrease at an instantaneous rate of 7.50 cm/s. Calculate the emf induced in the loop at that instant.

Solution: The magnetic field is perpendicular to the surface area. So, $\vec{B} \cdot d\vec{A} = B dA \cos 0^\circ = B dA$.

$$\therefore \Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA = BA = B(\pi r^2),$$

where r is the radius of the loop. When the loop is released then the radius starts to decrease, hence area A starts to decrease which produce the change of magnetic flux.

$$\therefore d\Phi_B = B dA = B d(\pi r^2) = 2\pi r B dr.$$

Now the induced emf is

$$\varepsilon = -\frac{d\Phi_B}{dt} = -2\pi r B \frac{dr}{dt}.$$

It is given that $B = 0.785 \text{ T}$, $r = 1.23 \text{ m}$, and $\frac{dr}{dt} = -7.50 \times 10^{-2} \text{ m/s}$ (here negative sign indicates the decreasing of the radius).

$$\begin{aligned} \text{Then } \varepsilon &= -2\pi r B \frac{dr}{dt} = 2\pi(1.23 \text{ m})(0.785 \text{ T})(-7.50 \times 10^{-2} \text{ m/s}) \\ &= 0.455 \text{ V.} \end{aligned}$$

Chapter 5

MAGNETIC PROPERTIES OF MATER

Magnetic materials play important roles in our daily lives. Materials such as iron, which can be permanent magnets at ordinary temperatures, are commonly used in electric motors and generators as well as loudspeakers. Some materials are analogous to dielectric in that they acquire an induced magnetic field in response to an external magnetic field and the induced field vanishes when the external field is removed. The materials which can be magnetized and demagnetized easily are widely used for storing information in such applications as magnetic recording tape (audio and video tape recorders), computer disks, bank and credit cards etc.

In the starting of this chapter, we shall introduce the internal structure of materials that is responsible for their magnetic properties. We shall show that the behavior of different magnetic materials can be understood in terms of the magnetic dipole moments of individual atoms. A vast understanding of magnetic properties of mater requires quantum mechanical discussions. But in this chapter we shall not enter to the interior of quantum mechanics. We wish to discuss qualitative understanding of magnetic properties of mater in the chapter.

5.1 The Magnetic Dipole

In the Sec. 1.2 of Chapter 1, we have seen the configuration of two static charges as *electric dipole* and how it can produce an electric field (Fig. 1.4). For static electric fields, the single isolated charge is the fundamental quantity. Individual charges produce an electric field, and in turn the electric field set up by one group of charges can influence the behavior of other charges. The common phenomena regarding this are the interaction between nucleus and electrons in an atom, the force exerted by one atom on another atom in ionic molecules and solids etc.

For steady magnetic field, the fundamental quantity is the moving electric charges in a current element, which can produce produce a magnetic field and can also influenced by other magnetic field. However, in trying to magnetic properties of materials that how moving charges in the atoms can produce magnetic dipoles. Actually the magnetic properties of materials to be the collection of atoms with individual magnetic dipole moments.

Let us begin by considering the magnetic field due to a circular loop (Subsection 3.4.2 Eq. 3.15) at a point on the axis 'x' is

$$B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}.$$

We have far from the loop (if, $x \gg R$)

$$B = \frac{\mu_0 i R^2}{2x^3},$$

or $B = \frac{\mu_0 i \pi R^2}{2\pi x^3}.$ (5.1)

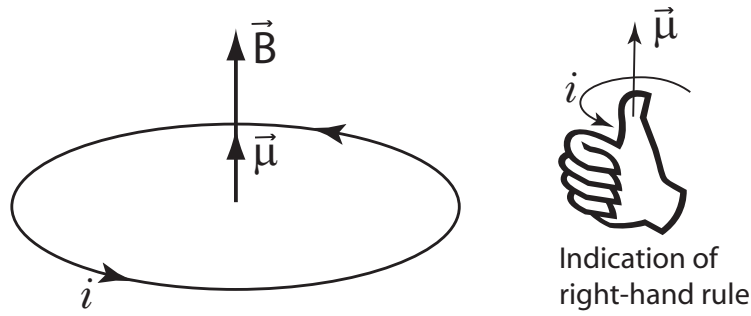


Figure 5.1: The magnetic dipole moment $\vec{\mu}$ and field vector $\vec{B}$ along the axis which is perpendicular to the plane of the current loop.

The quantity $i\pi R^2$ in Eq. 5.1 can be written as iA , where $A = \pi R^2$ is the area of the circular loop. We define this quantity to be the magnitude of the *magnetic dipole moment* μ of the loop:

$$\mu = iA. \quad (5.2)$$

The magnetic dipole moment of a current loop is the product of the current and the area of the loop. If the loop has N turns, the magnetic dipole moment be

$$\mu = NiA. \quad (5.3)$$

Like the electric dipole moment, the magnetic dipole moment is a vector quantity. The direction of $\vec{\mu}$ is perpendicular to the plane of the current loop, determined by the right-hand rule as it is shown in Fig. 5.1. Therefore, Eq. 5.1 can be rewritten in vector notation as

$$\vec{B} = \frac{\mu_0 \vec{\mu}}{2\pi x^3}. \quad (5.4)$$

Note that $\vec{B}$ and $\vec{\mu}$ are vectors in the same direction and it is also shown in Fig. 5.1.

Let us now see the effect of a magnetic dipole placed in a magnetic field. Figure 5.2 shows a current loop placed in a uniform magnetic field $\vec{B}$ produced by some external agent. We know that

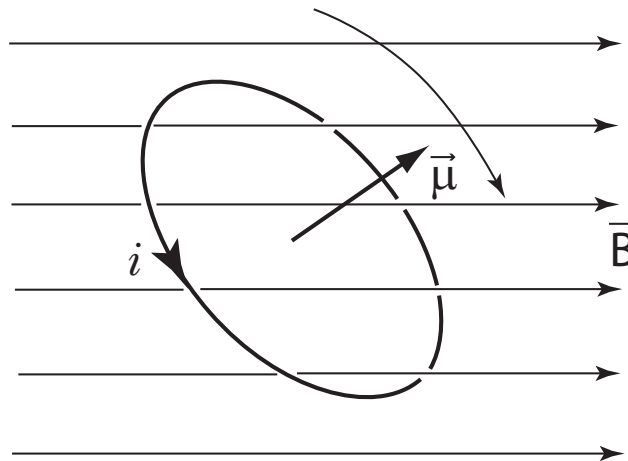


Figure 5.2: A magnetic dipole moment experiences a torque that rotates $\vec{\mu}$ into alignment with $\vec{B}$. $\vec{B}$ is applied by an external agent.

a current loop in a uniform magnetic field experiences no net force but does experience a net torque given by

$$\vec{\tau} = iA\hat{n} \times \vec{B},$$

where $\hat{n}$ is a unit vector perpendicular to the loop. Since $\hat{n}$ and $\vec{\mu}$ have the same direction, we can write

$$\vec{\mu} = iA\hat{n}.$$

Therefore, the current loop experiences a net torque and can be expressed by the formula

$$\vec{\tau} = \vec{\mu} \times \vec{B}. \quad (5.5)$$

This torque tends to rotate the loop so that $\vec{\mu}$ lines up with the magnetic field vector $\vec{B}$. Equation 5.5 has the similar characteristics for the torque that an electric dipole in an electric field as given $\vec{\tau} = \vec{p} \times \vec{E}$.

Equation 5.4 and 5.5 give us two basic idea of magnetic properties of matter: Eq. 5.4 indicates how a magnetic field is produced by a magnetic dipole, and Eq. 5.5 shows how a magnetic dipole is influenced by an applied magnetic field. These two concepts will help us to understand the magnetic behavior of materials.

[If the magnetic dipole moment $\vec{\mu}$ makes an angle θ with $\vec{B}$, then we can write the potential energy as

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \theta. \quad (5.6)$$

The result is similar for an electric dipole moment, energy $U = -\vec{p} \cdot \vec{E}$. In Eq. 5.6 $U = 0$ when $\theta = 90^\circ$ ($\vec{\mu}$ is perpendicular to $\vec{B}$). U has its smallest value ($= -\mu B$) when $\vec{\mu}$ and $\vec{B}$ are parallel, and has largest value ($= \mu B$) when $\vec{\mu}$ and $\vec{B}$ are antiparallel.]

Many physical system have magnetic dipole moments: the earth, bar magnets, current loops, atoms, nuclei, and elementary particles. Nuclear magnetic dipole moments are typically three to six order smaller in magnitude than atomic dipole moment. Ordinary magnetic effects in materials are determined by atomic magnetism. To exert a particular torque necessary to align nuclear dipoles requires a magnetic field about three to six orders larger that that necessary to align atomic dipoles.

5.1.1 The field of a dipole

Here we show the magnetic field pattern of magnetic dipole of a current loop and compare the analogy with an electric dipole and a dipole of a bar magnet. We have shown the complete field pattern of an electric dipole with the Fig. 1.4 discussed on page 3. A few lines for an electric dipole are shown in Fig. 5.3(a) and can be compared with the field lines for a current loop shown in Fig. 5.3(b). We see the great similarity between the patterns of field lines outside the loop. But we see a significant difference between electric and magnetic field lines is that electric field lines start on positive charges and end on negative charges, whereas magnetic field lines start on the plane of the closed loop and ends on the same plane. Actually the magnetic field lines always form closed loops.

Figure 5.3(c) shows the field lines of a bar magnet. It shows the same pattern of the field lines as the current loop, so a bar magnet can also be considered to be a magnetic dipole. From the outside view of the bar magnet, it seems that the force lines are leaving the N pole and converging on the S pole. The assuming N and S poles are to behave like the positive and negative charges of an electric dipole. But the actual situation is different instead of N and S poles concept of a bar magnet. The interior inspection of Fig. 5.3(c) shows that the field lines do not start and end on the poles but continue through the interior of the magnet, again forming closed loops. Therefore, N and S poles do not behave like the charges in an electric dipole and there is no real existence of isolated N and S poles in a bar magnet. We know that one of the Maxwell's equation $\vec{\nabla} \cdot \vec{B} = 0$ established by the conception as there is no isolated magnetic pole can be found in nature.

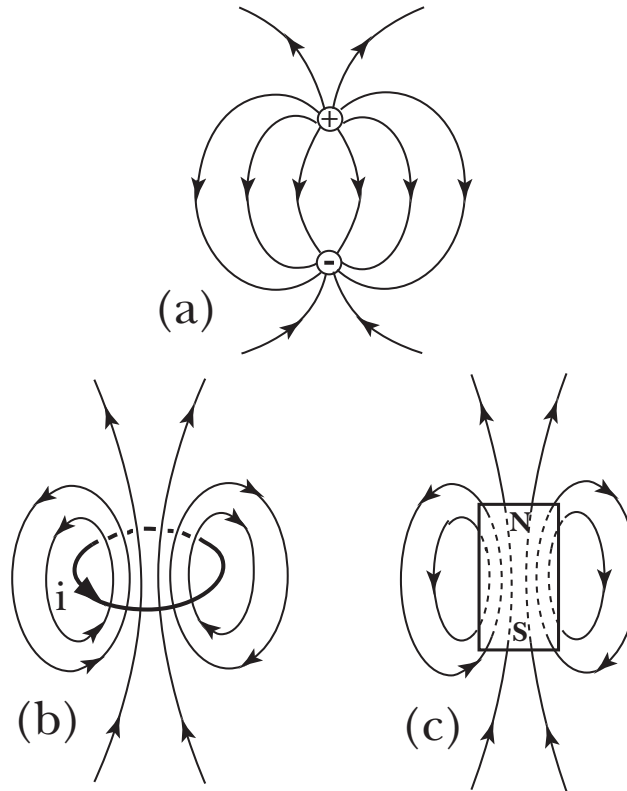


Figure 5.3: (a) Electric field configuration of an electric dipole. (b) The dipole magnetic field of a current loop. (c) The dipole magnetic field of a bar magnet.

5.1.2 Microscopic view of atomic magnetism

The magnetic properties of materials depend on the magnetic dipole moments of individual atoms, and we can consider magnetic materials to be composed of a collection of atomic dipoles, which might align when an external magnetic field is applied, as it is shown in Fig. 5.4. In practise we

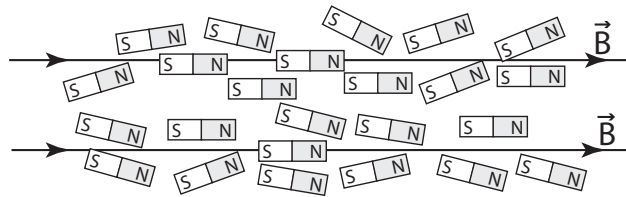


Figure 5.4: A magnetic materials can be considered as a collection of magnetic dipole moments. Each dipole moment consists by a schematic north and a south pole. In microscopic view, each dipole is actually a current loop that can not produce individual magnetic poles.

can not break apart the atoms into separate N and S magnetic poles, like the individual charges of an electric dipole.

We consider the magnetic moments to be tiny current loops, caused for example by the circulation of electrons in orbits in the atom. Figure 5.5(a) shows the current loop and magnetic field pattern of an atomic magnetism. Figure 5.5(b) shows the schematic drawing of an atomic magnet which represents the analogy of Fig. 5.5(a). Now we are going to discuss the magnetic dipole moment associated with a circulating electron.

We consider a simple model of an atom in which an electron moves in a circular orbit of radius r and speed v about the nucleus. The circulating electron can be considered as a current loop. If T

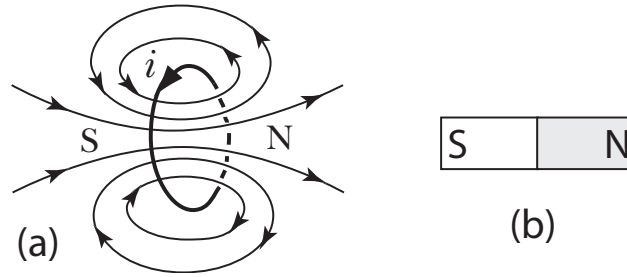


Figure 5.5: (a) Magnetic dipole shown by a tiny current loop, the current loop is formed by the circulation of orbital electrons in the atom. (b) Schematic drawing of magnetic dipole of an atom constitutes with N and S poles.

be the revolving time of the electron then current

$$i = \frac{e}{T} = \frac{e}{2\pi r/v}. \quad (5.7)$$

Now the magnetic dipole moment can be found

$$\mu = iA = \left(\frac{ev}{2\pi r} \right) (\pi r^2) = \frac{evr}{2}. \quad (5.8)$$

Here the magnetic dipole moment is known as the *orbital magnetic dipole moment*. Because it is due to the orbital motion of electrons about the nucleus. The Eq. 5.8 can be rewritten as

$$\mu_l = \frac{evr}{2} = \frac{e}{2m} (mvr) = \frac{e}{2m} l, \quad (5.9)$$

where m is the mass of electron, l is the angular momentum of the electron moving in a circular orbit about the nucleus of the atom and μ_l dipole moment due to angular momentum l . According to the theory of quantum mechanics, the angular momentum is quantized and defined as $l = h/2\pi$, where h is the Planck constant. Substituting this fundamental value of angular momentum into Eq. 5.9, we obtain a basic unit of the dipole magnetic moment called the *Bohr magnetron* μ_B :

$$\mu_B = \frac{e}{2m} \frac{h}{2\pi} = \frac{eh}{4\pi m} = 9.27 \times 10^{-24} \text{ J/T}, \quad (5.10)$$

where the numerical value is obtained by substituting the numerical value of e , h and m into Eq. 5.10. Atomic magnetic moments are usually measured in Bohr magnetron and are typically on the order of $1\mu_B$ in magnitude.

Besides the orbital dipole moment, electrons themselves have dipole moments called the *intrinsic or spin* magnetic moment. Electrons in different states of motion have different orbital magnetic dipole moments, but all electrons have exactly the same intrinsic magnetic dipole moment. The intrinsic magnetic moment of an electron is near equal to $1\mu_B$.

The orbital and spin magnetic dipole moments of electrons are about the same size (of $1\mu_B$), and thus both are important in determining the magnetic properties of atoms. The total magnetic moment of an atom is obtained from the vector sum of the orbital and spin magnetic moments of all its electrons. In an atom with many electrons, these vector sums may be very complicated. In some atoms, however, the total orbital and spin magnetic moments turn to be zero. Materials made from these atoms are virtually nonmagnetic and show only a weak induced effect called *diamagnetism*. In other atoms, the total orbital or spin or both magnetic moment may be nonzero, so that atoms align in the presence of an external magnetic field. These materials are called *paramagnetism*. The most familiar type of magnetic behavior is *ferromagnetism*. The total magnetic moments of ferromagnetic materials are nonzero but have a significant magnitude. The alignment of the atoms remains even when the external field is removed.

5.2 Magnetization

When a substance is placed in an external magnetic field, all atoms of the substance may acquire an induced magnetic dipole moment due to alignment of the individual dipoles. The process of alignment of the dipoles to acquire an induced magnetic dipole moment is known as magnetization. Consider a specimen of volume V consisting n atoms having magnetic dipole moments $\vec{\mu}_n$. The net dipole moment $\vec{\mu}$ can be computed as

$$\vec{\mu} = \sum \vec{\mu}_n.$$

We then define the magnetization $\vec{M}$ in vector form to be the net dipole moment per unit volume,

$$\vec{M} = \frac{\vec{\mu}}{V} = \frac{\sum \vec{\mu}_n}{V}. \quad (5.11)$$

If the magnetization to be uniform then the substance has a continuous dipole moment. In this condition the magnetization to be considered a microscopic quantity and written as

$$\vec{M} = \lim_{\Delta V \rightarrow 0} \frac{\sum \vec{\mu}_n}{V}. \quad (5.12)$$

5.2.1 Diamagnetism

In 1847, Michael Faraday discovered that a specimen of bismuth was repelled by a strong magnet. (Whereas paramagnetic substances are always attracted by a magnet.) Diamagnetic substances become weakly magnetized when placed in a magnetic field and resultant magnetization is opposed to the applied field. The atomic dipole moments of such materials are zero. Perhaps atoms having several electrons with their orbital and spin magnetic moments adding vectorially to zero. Diamagnetic substances have no permanent dipole moments and only acquire dipole moments when it is placed in an external magnetic field. Examples of diamagnetic materials are: bismuth, Zn, Ag, Cu, Sb, Pb, Au.

5.2.2 Paramagnetism

Paramagnetic substances are those which, when placed in a magnetic field, become weakly magnetized in the direction of the applied field. Paramagnetic substances have permanent atomic dipole moments and initially are randomly oriented in space. Paramagnetic materials are weakly attracted by the external magnetic field. When the external magnetic field is removed from a paramagnetic sample, the thermal motion causes the directions of the magnetic dipole moments to become random again. So that the paramagnetic materials do not become permanent magnets. Examples of paramagnetic materials are: Mn, Pt (Platinum), Pd (Palladium) etc.

5.2.3 Ferromagnetism

Ferromagnetic substances are those which, when placed in a magnetic field, become strongly magnetized in the direction of the applied field. The atoms of ferromagnetic materials have permanent dipole moments. Ferromagnetic materials are sometimes classed as strongly paramagnetic. All ferromagnetic materials are paramagnetic but all paramagnetic substances are not ferromagnetic. When an external field is applied in ferromagnetic materials, a strong interaction occurs that the materials keep their atomic dipole moments aligned even when the external field is removed. The ferromagnetism disappears if the temperature is increased above a certain critical value called *Curie temperature* and substances become paramagnetic. The temperature at which a ferromagnetic material becomes paramagnetic is called its Curie temperature. Familiar ferromagnetic materials at

room temperature are iron, cobalt and nickel. The Curie temperature of iron is 700°C ; above this temperature iron becomes paramagnetic.

5.3 Three Magnetic Vectors

When a ferromagnetic material is placed in an external magnetic field, it is to be strongly magnetized. This external field (say $\vec{H}$) magnetizes the magnetic material and makes alignment the atomic dipole moments to the direction of applied field. Let the magnetization vector of the ferromagnetic materials be $\vec{M}$. The two vectors, one is applied field vector $\vec{H}$ and another is $\vec{M}$, interact to each other in the material and gives a resultant magnetic field vector, can be denoted by $\vec{B}$. We now try to make a relation of these three magnetic vectors.

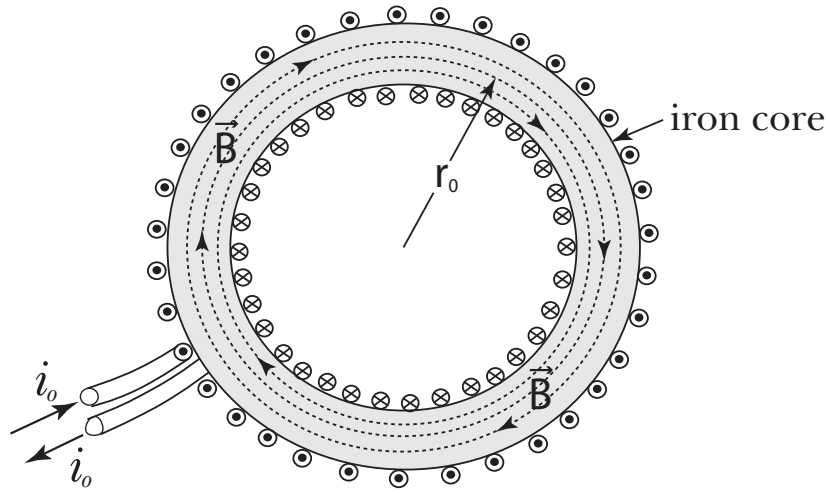


Figure 5.6: A toroid with core of ferromagnetic (iron) material. Current in each winding is i_o .

Let us consider a toroidal coil carrying a current i_o in it. Let a small length of the toroid be $d\vec{l}$ at a radial distance r_o . First consider the toroid without iron core, and according to Ampere's law we have

$$\oint \vec{B} \cdot d\vec{l} = \mu_o i. \quad (5.13)$$

If the total turns of the toroid be N_o , then the Eq. 5.13 can be written as

$$B (2\pi r_o) = \mu_o i_o N_o. \quad (5.14)$$

Let us consider the existence of iron core. The value of $\vec{B}$ determined experimentally is different when an iron core is placed than without it. The net current in the toroid must be changed due to in placing the iron core. An additional amount of magnetized current i_m will be introduced in the system.

In the presence of magnetic material, the Ampere's law is modified by

$$\oint \vec{B} \cdot d\vec{l} = \mu_o (i + i_m) \quad (5.15)$$

$$\text{or} \quad B(2\pi r_o) = \mu_o i_o N_o + \mu_o i_{mo} N_o. \quad (5.16)$$

In fact i_{mo} is related to the magnetization $\vec{M}$. The current i_{mo} is the magnetization current due to contributions of individual atomic dipoles of the material.

Let the rectangular cross sectional area of the toroid be A . Then the volume of the element dl be Adl . The amount of dipole moment due to element dl can be found by the help of Eq. 5.11

$$d\mu = M (Adl). \quad (5.17)$$

But magnetic moment in a current loop is (see Eq. 5.3)

$$\mu = NiA. \quad (5.18)$$

For element dl the number of turns stands

$$\frac{N_o}{2\pi r_o} dl.$$

So considering the Eq. 5.18 we get another relation of dipole moment due to magnetization for element dl

$$d\mu = \frac{N_o}{2\pi r_o} dl.i_{mo} A. \quad (5.19)$$

Now comparing the Eqs. 5.17 & 5.19 we have

$$\begin{aligned} M (Adl) &= \frac{N_o}{2\pi r_o} dl.i_{mo} A, \\ \text{or} \quad N_o i_{mo} &= M (2\pi r_o). \end{aligned} \quad (5.20)$$

Substituting the value of $N_o i_{mo}$ in Eq. 5.16, one can get

$$B (2\pi r_o) = \mu_o i_o N_o + \mu_o M (2\pi r_o).$$

or in general

$$\begin{aligned} \oint \vec{B}.d\vec{l} &= \mu_o i + \mu_o \oint \vec{M}.d\vec{l}, \\ \text{or} \quad \oint \left(\frac{\vec{B} - \mu_o \vec{M}}{\mu_o} \right).d\vec{l} &= i. \end{aligned} \quad (5.21)$$

Now we try to give the special name of the quantity $\frac{\vec{B} - \mu_o \vec{M}}{\mu_o}$, which is responsible for the magnetization of the material. We define it as the applied field intensity and denoted by $\vec{H}$. Sometimes $\vec{H}$ is called the *magnetizing force* or *magnetizing intensity*. So, $\vec{H}$ can be written as

$$\vec{H} = \frac{\vec{B} - \mu_o \vec{M}}{\mu_o}, \quad (5.22)$$

Which follows

$$\vec{B} = \mu_o \vec{H} + \mu_o \vec{M}, \quad (5.23)$$

$$\text{or} \quad \vec{B} = \mu_o \left(1 + \frac{\vec{M}}{\vec{H}} \right) \vec{H}. \quad (5.24)$$

Equations 5.23 or 5.24 gives the relation between three magnetic vectors in a material placed in a magnetic field. $\vec{B}$ is the net magnetic field vector is equal to the sum of the applied field and magnetization vectors multiplied by μ_o . Now from Eqs. 5.21 and 5.22, we find Ampere's law in presence of magnetic material as

$$\oint \vec{H}.d\vec{l} = i, \quad (5.25)$$

where i is the current applied to the circuit and does not include the magnetizing current. The line integral of field intensity $\vec{H}$ around any closed path equals the total conduction current in the presence of magnetized material. Since in Eq. 5.25 i indicates the current applied by an external source, hence the field $\vec{H}$ is said to be magnetizing field.

5.4 Permeability & Susceptibility

Relative permeability:

For non-ferromagnetic materials, it is found experimentally that $\vec{B}$ is directly proportional to $\vec{H}$. That is, the proportionality is applicable only for diamagnetic and paramagnetic materials. Therefore,

$$\begin{aligned}\vec{B} &\propto \vec{H}, \\ \text{or} \quad \vec{B} &= \mu_o \kappa_r \vec{H},\end{aligned}\tag{5.26}$$

where κ_r is called relative permeability and μ_o is called simply permeability in vacuum. Vectors $\vec{B}$ and $\vec{H}$ have the same direction. Putting the value $\vec{B}$ of Eq. 5.26 in Eq. 5.23 we get

$$\begin{aligned}\mu_o \kappa_r \vec{H} &= \mu_o \vec{H} + \mu_o \vec{M}, \\ \text{or} \quad \vec{M} &= (\kappa_r - 1) \vec{H}.\end{aligned}\tag{5.27}$$

Now in vacuum, the magnetization vector $\vec{M}$ is zero, because there is no magnetic dipole moment present. Hence putting $\vec{M} = 0$ in Eq. 5.23 we have

$$\vec{B} = \mu_o \vec{H},\tag{5.28}$$

and in Eq. 5.27 it is found

$$\begin{aligned}(\kappa_r - 1) &= 0, \\ \text{or} \quad \kappa_r &= 1.\end{aligned}$$

Therefore, we see in vacuum the relative permeability κ_r is unity. For paramagnetic materials κ_r is slightly greater than unity and for diamagnetic materials κ_r is slightly less than unity.

Magnetic Susceptibility:

For para and diamagnetic substances the magnetization is proportional to the field intensity $\vec{H}$,

$$\begin{aligned}\text{or} \quad \vec{M} &\propto \vec{H}, \\ \text{or} \quad \vec{M} &= \chi_m \vec{H},\end{aligned}\tag{5.29}$$

where χ_m is a proportionality constant and known as magnetic susceptibility. Since $\chi_m = M/H$, it may be defined as the ratio of magnetization to the magnetizing field. Comparing Eqs. 5.27 and 5.29 we get

$$\begin{aligned}\chi_m &= \kappa_r - 1, \\ \text{or} \quad \kappa_r &= 1 + \chi_m.\end{aligned}\tag{5.30}$$

Permeability:

Substituting κ_r of Eq. 5.30 in Eq. 5.26, we have

$$\vec{B} = (1 + \chi_m) \mu_o \vec{H},$$

or simply

$$\vec{B} = \kappa_m \vec{H},\tag{5.31}$$

where $\kappa_m = \mu_o(1 + \chi_m)$ is the magnetic permeability of the medium. We see

$$\kappa_m = \mu_o(1 + \chi_m),$$

$$\text{or} \quad \frac{\kappa_m}{\mu_o} = (1 + \chi_m) = \kappa_r.$$

Therefore, the permeability of the medium is the measure of the total capacity of the medium to take up magnetization and the relative permeability κ_r of the medium is the measure of the ratio of permeability in the presence of materials to the absence of materials (free space). In other words, the relative permeability of the medium is defined as the ratio of the number of field per square meter crossing perpendicularly to that in air or free space.

5.5 Magnetic Hysteresis

We know that for diamagnetic and paramagnetic materials $\vec{B}$ is directly proportional to $\vec{H}$. That is, $\kappa_m = B/H$. But the permeability κ_m of the ratio B/H is not a constant for ferromagnetic materials. It means $\vec{B}$ is not proportional to $\vec{H}$ for ferromagnetic materials. In order to know the characteristics

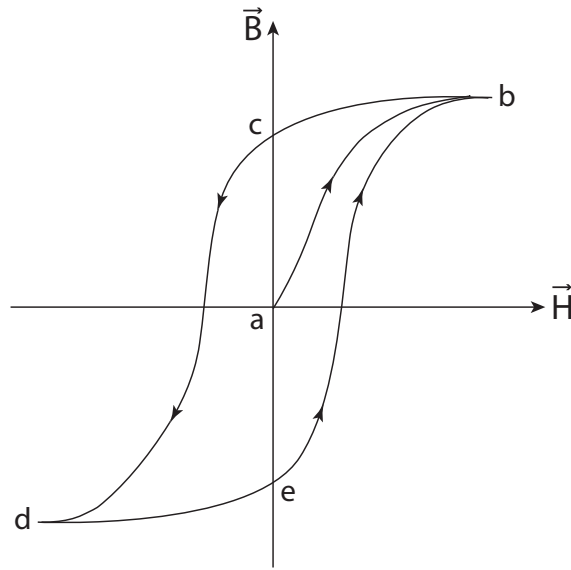


Figure 5.7: The hysteresis curve, shows the variation of magnetization of a specimen of ferromagnetic material with the variation of applied field.

between $\vec{B}$ and $\vec{H}$ for a ferromagnetic material, let us insert an iron core in the toroid like as Fig. 5.6. A curve can be plotted by putting $\vec{B}$ as ordinate and $\vec{H}$ as abscissa. In the beginning, the initial current is zero in the coil and that the iron core is demagnetized. So, initially both H and B are zero. We increase H by increasing increasing current in the solenoid. The induction field B will increase rapidly following a path ab , until it reaches a saturation value as it is shown in Fig. 5.7. Now we decrease the current to zero, see the net magnetized field B does not retrace its original path ba , but it follows the path bc . Notice that the iron remains magnetized (indicated value ac) even when the applied field H is zero. We then reverse direction of the current in the toroid, we reach a saturated magnetization in the opposite direction (point d). Again we reduce the current to zero, we see that the magnetization does not retrace to da , but it follows the path dc and the sample retains a permanent magnetization (indicated value ae). Now we reverse the current in the opposite direction until the path reaches at the saturation value (point b). If we supply current in the toroid to make a contour, the path is followed the trace $bcdeb$. The lack of retractibility of the net magnetization field B is called *hysteresis* and the loop $bcdeb$ is called *hysteresis loop*.

At point c and e the specimen is magnetized even there is no current in the toroid. The value ac or ae is called *residual magnetization*. Furthermore, the iron ‘remembers’ how it became magnetized, a negative current produces a magnetization different from a positive one. This ‘memory’

is essential to the operation of magnetic storage of information such as on cassette tapes, computer disks, credit cards etc.

Domains:

A material such as iron is composed of a large number of microscopic crystals. The regions of roughly 0.01 mm in size in which the coupling of atomic magnetic dipoles produces essentially perfect alignment to all the atoms within each crystal. This particular region is called *magnetic domains*. There are many domains, each with its dipoles pointing in a different direction, and the net result of adding these dipole moments in an unmagnetized ferromagnetic substance gives a magnetization of zero. When the ferromagnet is placed in an external field, two effects may occur: (1) dipoles outside the walls of domains that are aligned with the field can rotate into alignment, in effect allowing such domains to grow at the expense of neighboring domains: and (2) the dipoles of nonaligned domains may swing entirely into alignment, with the applied field. In either case, there are now more dipoles aligned with the field, and material has a large magnetization. When the field is removed, the domain walls do not move completely back to their former positions, and the material retains a magnetization in the direction of the applied field.

5.6 Examples

Example 5.1 A proton is in a magnetic field of strength $B = 1.5 \text{ T}$. The magnetic dipole moment of the proton is initially antiparallel to the direction of $\vec{B}$. How much external work must be done to reverse the direction of the magnetic dipole moment of the proton? Dipole moment of a proton is $1.41 \times 10^{-26} \text{ J/T}$.

Solution: The energy of interaction of a magnetic dipole with a magnetic field is given by Eq. 5.6

$$U = -\vec{\mu} \cdot \vec{B},$$

Since $\vec{\mu}$ and $\vec{B}$ are antiparallel to each other (angle between them is 180°), the initial energy U_i is

$$U_i = -\vec{\mu} \cdot \vec{B} = -\mu B \cos 180^\circ = +\mu B.$$

When magnetic dipole moments become parallel to B , then final energy U_f is

$$U_f = -\vec{\mu} \cdot \vec{B} = -\mu B \cos 0^\circ = -\mu B.$$

The external work done on the system is equal to the change in energy:

$$W = U_f - U_i = -\mu B - \mu B = -2\mu B = -2(1.41 \times 10^{-26} \text{ J/T})(1.5 \text{ T}) = -4.23 \times 10^{-26} \text{ J} = -0.26 \text{ } \mu\text{eV}.$$

(Note: Here the environment does negative work on the system, the system does positive work on the environment. The energy might be transmitted to the environment in the form of electromagnetic radiation and would have a frequency of 64 MHz.)

Example 5.2 A 300 turns of circular coil of radius 3 cm carries a current of $85 \text{ } \mu\text{A}$. (a) What is the magnetic dipole moment of this coil? (b) The magnetic dipole moment of the coil is lined up with an external magnetic field whose strength is 0.85 T . How much work would be done by an external agent to rotate the coil through 180° .

Solution: (a) The magnitude of the magnetic dipole moment is

$$\mu = NiA = Ni(\pi r^2) = 300(85 \times 10^{-6} \text{ A})(\pi)(0.03 \text{ m}^2) = 7.21 \times 10^{-5} \text{ A}\cdot\text{m}^2 = 7.21 \times 10^{-5} \text{ J/T}.$$

(b) The external work done is equal to the increase of potential energy

$$W = \Delta U = -\mu B \cos 180^\circ - (-\mu B \cos 0^\circ) = 2\mu B = 2(7.21 \times 10^{-6} \text{ J/T})(0.85 \text{ T}) = 1.23 \times 10^{-4} \text{ J}.$$

Example 5.3 Calculate the change in magnetic moment of a circulating electron in an applied field B_0 of 2.0 T acting perpendicular to the plane of the orbit. Take $r = 5.29 \times 10^{-11} \text{ m}$ for the radius of the orbit, corresponding to the normal state of an atom of hydrogen.

Solution: We have from Eq. 5.8

$$\mu = \frac{erv}{2} = \frac{er^2\omega}{2},$$

where $v = \omega r$. The change in magnetic moment corresponding to a change in the angular frequency ($\Delta\omega = \pm \frac{eB_0}{2m}$) is given by

$$\begin{aligned} \Delta\mu &= \frac{er^2\Delta\omega}{2} = \frac{1}{2}er^2 \left(\pm \frac{eB_0}{2m} \right) = \pm \frac{e^2B_0r^2}{4m} \\ &= \pm \frac{(1.6 \times 10^{-19} \text{ C})^2(2.0 \text{ T})(5.29 \times 10^{-11} \text{ m})^2}{4(9.1 \times 10^{-31} \text{ kg})} \\ &= \pm 3.9 \times 10^{-29} \text{ J/T} \\ &= \pm \frac{3.9 \times 10^{-29} \text{ J/T}}{9.27 \times 10^{-24} \text{ J/T}} \mu_B \\ &= 4.2 \times 10^{-6} \mu_B. \end{aligned}$$

We see that the effect amounts to only about 4.2×10^{-6} of the magnetic moment.

Example 5.4 The magnetic susceptibility of a medium is 948×10^{-11} . Calculate the permeability (or absolute permeability) and relative permeability.

Solution: Here the susceptibility $\chi_m = 948 \times 10^{-11}$.

The permeability,

$$\begin{aligned} \kappa_m &= \mu_o(1 + \chi_m) = 4\pi \times 10^{-7}(1 + 948 \times 10^{-11}) = 4\pi \times 10^{-7}(949 \times 10^{-11}) \\ &= 1.186 \times 10^{-14} \text{ h/m}. \end{aligned}$$

The relative permeability,

$$\begin{aligned} \kappa_r &= \frac{\kappa_m}{\mu_o} = \frac{4\pi \times 10^{-7}(949 \times 10^{-11})}{4\pi \times 10^{-7}} \\ &= 949 \times 10^{-11}. \end{aligned}$$

Index

Ampere's law, [32](#)
Biot-Savart law, [26](#)
Bohr magnetron, [52](#)
Curie temperature, [53](#)
Diamagnetism, [52](#), [53](#)
Dipole, [20](#)
Dipole moment, [21](#)
Eddy current, [41](#)
Electric dipole, [20](#)
Electric field lines, [1](#)
Electric potential, [16](#)
Electric quadrupole, [22](#)
Faraday's law, [39](#)
Ferromagnetism, [52](#), [53](#)
Flux, [3](#)
Gauss's law, [5](#)
Helmholtz coil, [37](#)
Hysteresis loop, [57](#)
Inductance, [42](#)
Induction, [38](#)
Lenz's law, [40](#)
Magnetic dipole, [30](#), [48](#)
Magnetic domains, [58](#)
Magnetic hysteresis, [57](#)
Magnetic susceptibility, [56](#)
Magnetization, [53](#)
Mutual inductance, [43](#)
Paramagnetism, [52](#), [53](#)
Permeability, [56](#)
Quadrupole, [21](#)
Relative permeability, [56](#)
Self inductance, [43](#)
Solenoid, [31](#), [35](#)
Toroid, [35](#)